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Contents

1 Vector Spaces

1.1 Motivation (Vectors in 3-space)

Problem 1

Given the point $P = (2, -1, 3)$ and the vector $u = [-3, 4, 5]$, find the point Q such that $\overrightarrow{PQ} = u$.

Proof. We require $\vec{u} = [-3, 4, 5] = (q_1 - 2, q_2 - (-1), q_3 - 3)$. Solving componentwise, we see $(q_1, q_2, q_3) = (-1, 3, 8)$. Thus, $Q = (-1, 3, 8)$. ∎

Problem 2

Given the points $P(2,-1,3)$, $Q(3,4,1)$, $R(4,-3,4)$, $S(5,2,2)$. True or false (explain:) $PQSR$ is a parallelogram.

Proof. For $PQSR$ to be a parallelogram, we require $\vec{PQ} = \vec{SR}$.

$$\vec{PQ} = [3-2,\ 4-(-1),\ 1-3] = [1,5,-2]$$

$$\vec{SR} = [4-5,\ -3-2,\ 4-2] = [-1,-5,2]$$

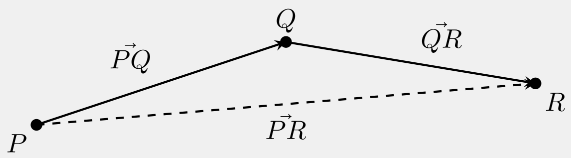
Since

$$[1,5,-2] \neq [-1,-5,2],$$

it follows that $PQSR$ is not a parallelogram. ∎

Problem 4

Show that for every triple points P, Q, R, $\overrightarrow{PQ} + \overrightarrow{QR} = \overrightarrow{PR}$. [Method 1: Calculate components. Method 2: Draw a picture.]



1.2 $\mathbb{R}^n$ and $\mathbb{C}^n$

Problem 2

Consider the vectors $u = (2,1)$, $v = (-5,3)$, $w = (3,4)$ in $\mathbb{R}^2$. Do there exists real numbers a, b such that $au + bv = w$? What if $v = (6,3)$.

Proof. We require $au + bv = w \iff a(2,1) + b(-5,3) = (3,4) \iff (2a,a) + (-5b,3b) = (3,4) \iff (2a-5b, a+3b) = (3,4)$. Thus

$$\begin{cases} 2a - 5b = 3 \\ a + 3b = 4 \end{cases}$$

which has the solution $a = \frac{29}{11},\ b = \frac{5}{11}$. If $v = (6,3)$, we require

$$a(2,1) + b(6,3) = (3,4) \iff (2a+6b, a+3b) = (3,4).$$

Thus

$$\begin{cases} 2a + 6b = 3 \\ a + 3b = 4 \end{cases}$$

which has no solution. To see this has no solution, note

$$a + 3b = 4 \implies 6b = 8 - 2a.$$

Plugging into the first equation gives

$$2a + (8-2a) = 3 \implies 8 = 3,$$

which is a contradiction. ∎

Problem 3

Give a detailed proof of (8) of Theorem 1.2.3 (just checking!). [Write out all steps of the proof and give a reason for each step.]

Proof. Let $x = [x_1, x_2, \dots, x_n] \in F^n$ be a vector and let $c, d \in F$ where F is a field. Then

$$(cd)x = [(cd)x_1, (cd)x_2, \dots, (cd)x_n].$$

Since F is a field, scalar multiplication in F is associative thus

$$(cd)x_i = c(dx_i) \quad \text{for each } i = 1, 2, \dots, n.$$

Therefore,

$$(cd)x = [c(dx_1), c(dx_2), \dots, c(dx_n)] = c(dx),$$

∎

Problem 4

In Definition 1.2.1, $n = 1$ is not ruled out. What do the elements of $\mathbb{R}^1$ look like? Describe sums and scalar multiples in $\mathbb{R}^1$.

Proof. The elements of $\mathbb{R}^1$ look like $[x_1]$ where $x_1 \in \mathbb{R}$. Sums and scalar multiples behave identically to the operations in the field $\mathbb{R}$. ∎

1.3 Vectors Spaces: The Axioms some Examples

Problem 1

Let V be a vector space over a field F. Let T be a nonempty set and let $W = \mathcal{F}(T, V)$ be the set of all functions $x : T \longrightarrow V$. Show that W can be made into a vector space over F in a natural way. [Hint: Use the definitions in Example 1.3.4 as a guide.]

Proof. For $x, y \in W$, $x = y$ means that $x(t) = y(t)$ for all $t \in T$. If $x, y \in W$ and $c \in F$, define functions $x + y$ and cx by the formulas $(x + y)(t) = x(t) + y(t)$ and $(cx)(t) = cx(t)$ for all $t \in T$. Let θ be the function defined by $\theta(t) = 0$ for all $t \in T$, and for $x \in W$, let $-x$ be the function defined by $(-x)(t) = -x(t)$ for all $t \in T$. ∎

Problem 2

The definition of a vector space (1.3.1) can be formulated as follows. A vector space over F is a nonempty set V together with a pair of mappings $\sigma : V \longrightarrow V$ and $\mu : F \times V \longrightarrow V$. ($\sigma$ suggests 'sum' and μ suggests 'multiple') having the following properties $\sigma(x, y) = \sigma(y, x)$ for all $x, y \in V$; $\sigma(\sigma(x, y), z) = \sigma(x, \sigma(y, z))$ for all $x, y, z \in V$ etc. The exercise: Write out the 'etc.' in detail.

Proof. We list the 9 axioms below:

1. $x, y \in V \implies \sigma(x, y) \in V$
2. $x \in V, c \in F \implies \mu(c, x) \in V$
3. $x, y \in V \implies \sigma(x, y) = \sigma(y, x)$
4. $x, y, z \in V \implies \sigma(x, \sigma(y, z)) = \sigma(\sigma(x, y), z)$
5. $\exists 0 \in V$ such that $\sigma(0, v) = v = \sigma(v, 0)$

6. $x \in V \implies -x \in V$ **such that** $\sigma(x, -x) = 0 = \sigma(-x, x)$
7. $x, y \in V$ **and** $c \in F \implies \mu(c, \sigma(x, y)) = \sigma(\mu(c, x), \mu(c, y))$
8. $\exists 1 \in F$ **such that** $x \in V \implies \mu(1, x) = x$
9. $a, b \in F$ **and** $x \in V \implies \mu(a, \mu(b, x)) = \mu(ab, x)$

∎

Problem 3

Let V be a complex vector space (1.3.2). Show that V is also a real vector space (sums as usual, scalar multiplication restricted to real scalars). These two ways of looking at V may be indicated by writing V_c and V_r.

Proof. **Clearly V_r is closed under scalar multiplication and vector addiction and is a vector space.** ∎

Problem 4

Every real vector space V can be 'embedded' in a complex vector space W in the following way: let $W = V \times V$ be the real vector space constructed as in example 1.3.10 and define multiplication by complex scalars by the formula $(a + bi)(x, y) = (ax - by, bx + ay)$ for $a, b \in \mathbb{R}$ and $(x, y) \in W$. [In particular, $i(x, y) = (-y, x)$. Think of (x, y) as '$x + iy$'] Show that W satisfies the axioms for a complex vector space (W is called the complexification of V.)

Proof. **Below are the 9 vector space axioms.**

1. **Let** $x, y \in W$. **Then** $x = (v_1, v_2), y = (w_1, w_2) \in V \times V$, **and** $(v_1, v_2) + (w_1, w_2) = (v_1 + w_1, v_2 + w_2)$. **Now,** $v_1 + w_1 \in V$ **and** $v_2 + w_2 \in V$, **thus** $(v_1 + w_1, v_2 + w_2) \in V \times V$.
2. **Let** $x \in W$ **and** $c \in \mathbb{C}$. **Then** $x = (v_1, v_2) \in V \times V$, $c = a+bi \in \mathbb{C}$, **and** $(a+bi)(v_1, v_2) = (av_1 - bv_2, bv_1 + av_2)$. **Clearly** $av_1 - bv_2 \in V$ **and** $bv_1 + av_2 \in V$, **thus** $(av_1 - bv_2, bv_1 + av_2) \in V \times V$.
3. **Let** $x, y \in W$. **Then** $x = (v_1, v_2), y = (w_1, w_2) \in V \times V$. **Then** $x + y = (v_1, v_2) + (w_1, w_2) = (v_1 + w_1, v_2 + w_2) = (w_1 + v_1, w_2 + v_2) = (w_1, w_2) + (v_1, v_2) = y + x$.
4. **Let** $x, y, z \in W$. **Then** $x = (v_1, v_2), y = (w_1, w_2), z = (s_1, s_2) \in V \times V$. **Then** $(x + y) + z = [(v_1, v_2) + (w_1, w_2)] + (s_1, s_2) = (v_1 + w_1, v_2 + w_2) + (s_1, s_2) = ((v_1 + w_1) + s_1, (v_2 + w_2) + s_2) = (v_1 + (w_1 + s_1), v_2 + (w_2 + s_2)) = (v_1, v_2) + (w_1 + s_1, w_2 + s_2) = (v_1, v_2) + [(w_1, w_2) + (s_1, s_2)] = x + (y + z)$.
5. **Let** $x \in W$. **Then** $x = (v_1, v_2) \in V \times V$ **and** $x + 0 = (v_1, v_2) + (0, 0) = (v_1 + 0, v_2 + 0) = (v_1, v_2)$.
6. **Let** $x \in W$. **Then** $x = (v_1, v_2) \in V \times V$ **and** $x + (-x) = (v_1, v_2) + (-v_1, -v_2) = (v_1 - v_1, v_2 - v_2) = (0, 0) = 0 = (-v_1, -v_2) + (v_1, v_2) = (-x) + x$.
7. **Let** $a + bi \in \mathbb{C}$ **and** $x, y \in W$. **Then** $(a + bi)((v_1, v_2) + (w_1, w_2)) = (a + bi)(v_1 + w_1, v_2 + w_2) = (a(v_1 + w_1) - b(v_2 + w_2), b(v_1 + w_1) + a(v_2 + w_2)) = (av_1 - bv_2, bv_1 + av_2) + (aw_1 - bw_2, bw_1 + aw_2) = (a + bi)(v_1, v_2) + (a + bi)(w_1, w_2)$.
8. **Let** $a+bi, c+di \in \mathbb{C}$ **and** $x = (v_1, v_2) \in W$. **Then** $(a+bi)((c+di)(v_1, v_2)) = (a+bi)(cv_1 - dv_2, dv_1 + cv_2) = ((ac - bd)v_1 - (ad + bc)v_2, (bc + ad)v_1 + (bd + ac)v_2) = ((a + bi)(c + di))(v_1, v_2)$.
9. **Let** $x = (v_1, v_2) \in W$. **Then** $1 \cdot (v_1, v_2) = (1v_1 - 0v_2, 0v_1 + 1v_2) = (v_1, v_2)$.

∎

1.4 Vector Spaces: First Consequences of the Axioms

Problem 1

In a vector space, ① $x - y = \theta$ if and only if ② $x = y$; ③ $x + y = z$ if and only if ④ $x = z - y$.

Proof. (① $\longrightarrow$ ②) Suppose $x - y = \theta \iff x + (-y) = \theta$. It follows that $-y = -x \iff y = x$.

(② $\longrightarrow$ ①) Suppose $x = y \iff -x = -y$. Then $x + (-x) = x + (-y) = x - y = \theta$.

(③ $\longrightarrow$ ④) Suppose $x + y = z \iff x + y + (-y) = z + (-y) \iff x = z - y$.

(④ $\longrightarrow$ ③) Suppose $x = z - y \iff x + y = z - y + y \iff x + y = z$. ∎

Problem 2

Let V be a vector space over a field F. If x is a fixed nonzero vector, then the mapping $f : F \longrightarrow V$ defined by $f(c) = cx$ is injective. If c is a fixed nonzero scalar, then the mapping $g : V \longrightarrow V$ defined by $g(x) = cx$ is bijective.

Proof. Suppose x is a fixed nonzero vector. Let $c, c' \in F$ such that $f(c) = f(c') \iff cx = c'x \iff c = c'$. The last step follows from Corollary 1.4.7 (ii). Thus f is injective. ∎

Proof. Suppose c is a fixed nonzero scalar. The inverse function to $g(x) = cx$ is clearly $g^{-1}(x) = \frac{x}{c}$. Thus g is bijective. ∎

Problem 3

If V is a vector space and y is a fixed vector, then the mapping $\tau : V \longrightarrow V$ defined by $\tau(x) = x + y$ is bijective (it is called translation by the vector y).

Proof. Suppose V is a vector space and y is a fixed vector. The inverse function to $\tau(x) = x + y$ is clearly $\tau^{-1}(x) = x - y$. Thus τ is bijective. ∎

Problem 4

If a is a nonzero scalar and b is vector in the space V, then the equation $ax + b = \theta$ has a unique solution x in V.

Proof. Suppose a is a nonzero scalar and b is a vector in V. The equation $ax + b = \theta$ is equivalent to $ax = -b$. Multiplying both sides by a^{-1} gives $a^{-1}(ax) = a^{-1}(-b) \iff (a^{-1}a)x = -a^{-1}b \iff x = -a^{-1}b$. ∎

Problem 5

If, in a vector space, θ' is a vector such that $0' + x = x$ for even a single vector xS, then $\theta' = \theta$. [Hint: Add $-x$ to both sides of the equation.]

Proof. Suppose in a vector space θ' is a vector such that $\theta' + x = \theta$. Adding $-x$ to both sides gives $\theta' + x + (-x) = x + (-x) \iff \theta' + \theta = \theta \iff \theta' = \theta$. ∎

1.5 Linear Combinations of Vectors

Problem 1

Express the function $f(t) = \cos(t - \pi/3)$ as a linear combination of the functions $\sin t$ and $\cos t$.

Proof. We use the standard trig identity to find $f(t) = \cos(t - \pi/3) = \cos(\pi/3)\cos(t) + \sin(\pi/3)\sin(t)$. ∎

Problem 2

Express the function e^t as a linear combination of the hyperbolic functions $\cosh t$ and $\sinh t$.

Proof. We have the standard identity $e^t = \cosh(t) + \sinh(t)$. ∎

Problem 3

Convince yourself of the correctness of the following formulas pertaining to linear combinations in a vector space:

1. $c\sum_{i=1}^n x_i = \sum_{i=1}^n cx_i$.
2. $\sum_{i=1}^n a_i x_i + \sum_{i=1}^n b_i x_i = \sum_{i=1}^n (a_i + b_i)x_i$.
3. $\sum_{i=1}^m a_i x_I + \sum_{j=1}^n b_j y_j = \sum_{k=1}^{m+n} c_k z_k$. where $c_k = a_k$ and $z_k = x_k$ for $k = 1, 2, \ldots, m$, while $c_k = b_{k-m}$ and $z_k = y_{k-m}$ for $k = m+1, m+2, \ldots, m+n$.
4. $$\begin{aligned}\left(\sum_{i=1}^m c_i\right)\left(\sum_{j=1}^n x_j\right) &= \sum_{i=1}^m \left(\sum_{j=1}^n c_i x_j\right)\\ &= \sum_{j=1}^n \left(\sum_{i=1}^m c_i x_j\right)\\ &= \sum_{i,j} c_i x_j\end{aligned}$$

The last expression signifying with sum, with mn terms, of the vectors $c_i x_j$ for all possible combinations of i and j.

Proof. I'm convinced. ∎

Problem 5

Show that in a vector space, $x - y$ is a linear combination of x and y.

Proof. Let $\alpha = 1$ and $\beta = -1$ and it follows that $x - y = x + (-y) = \alpha x + \beta y$. ∎

Problem 6

True or false (explain): In $\mathbb{R}^3$,

1. the vector $x = (0, 6, 3)$ is a linear combination of $(2, 1, 0)$ and $(-3, 2, 0)$;
2. the vector $x = (2, 1, -2)$ is a linear combination of $(2, 1, 0)$ and $(-3, 2, 0)$.

Proof. We must have

$$(0,6,3) = \alpha(2,1,0) + \beta(-3,2,0) = (2\alpha - 3\beta, \alpha + 2\beta, 0)$$

for $\alpha, \beta \in \mathbb{R}$. We obtain

1. $0 = 2\alpha - 3\beta$
2. $6 = \alpha + 2\beta$
3. $3 = 0$

Since $3 \neq 0$, no solution exists. Now suppose

$$(2,1,-2) = \alpha(2,1,0) + \beta(-3,2,0) = (2\alpha - 3\beta, \alpha + 2\beta, 0).$$

We obtain

1. $2 = 2\alpha - 3\beta$
2. $1 = \alpha + 2\beta$
3. $-2 = 0$

Since $-2 \neq 0$, no solution exists. ∎

1.6 Linear Subspaces

Problem 1

Let M and N be the subsets of $\mathbb{R}^2$ defined as follows:

$$M = \{(a,0) \mid a \in \mathbb{R}\}, N = \{(0,b) \mid b \in \mathbb{R}\}$$

Show that M and N are linear subspaces of $\mathbb{R}^2$ and describe the linear subspaces $M \cap N$ and $M + N$.

Proof. First note $\theta = (0,0) \in M, N$. Let $x, y \in M$ such that $x = (a_1, 0), y = (a_2, 0)$. Then $x + y = (a_1, 0) + (a_2, 0) = (a_1 + a_2, 0) \in M$. Let $c \in \mathbb{R}$. Then $cx = c(a_1, 0) = (ca_1, 0) \in M$. The argument for N is the same. The linear subspace $M + N$ is all of $\mathbb{R}^2$, and the linear subspace $M \cap N$ is $\{(0,0)\}$. ∎

Problem 2

Let $\mathcal{F} = \mathcal{R}$ as in Example 1.6.5, fix a nonempty subset T of $\mathbb{R}$, and let

$$M = \{x \in \mathcal{F} \mid x(t) = 0 \text{ for all } t \in T\}$$

Show that M is a linear subspace of $\mathcal{F}$.

Proof. Let $x, y \in M$. Then for all $t \in T$, $(x+y)(t) = x(t) + y(t) = 0 + 0 = 0$, so $x + y \in M$. Let $x \in M$ and $c \in \mathbb{R}$. Then for all $t \in T$, $(cx)(t) = c \cdot x(t) = c \cdot 0 = 0$, so $cx \in M$. Thus M is a linear subspace of $\mathcal{F}$. ∎

Problem 3

With $\mathcal{F} = \mathcal{F}(\mathbb{R})$ as in 1.6.5, let

$$M = \{x \in \mathcal{F} \mid x = 0 \text{ on } (-\infty, 0]\}$$

$$N = \{x \in \mathcal{F} \mid x = 0 \text{ on } (0, +\infty)\}$$

Prove that $M + N = \mathcal{F}$ and $M \cap N = \{\theta\}$.

Proof. Let $f \in \mathcal{F}$. Define

$$x(t) = \begin{cases} f(t), & t > 0 \\ 0, & t \leq 0 \end{cases} \in M, \quad y(t) = \begin{cases} f(t), & t \leq 0 \\ 0, & t > 0 \end{cases} \in N.$$

Then $f = x + y$, so $M + N = \mathcal{F}$. Now, let $z \in M \cap N$. Then $z(t) = 0$ for $t \leq 0$ and $z(t) = 0$ for $t > 0$. Thus $z = \theta$ so $M \cap N = \{\theta\}$. ∎

Problem 4

Let M and N be linear subspaces of a vector space V, and let

$$M \cup N = \{x \in V \mid x \in M \text{ or } x \in N\}$$

be the union of M and N. True or false (explain): $M \cup N$ is a linear subspace of V. [If true, give a proof. If false, give a counterexample (an example of V, M, N for which $M \cup N$ is not a linear subspace of V).]

Proof. Let M and N be the subsets of $\mathbb{R}^2$ defined as follows:

$$M = \{(a, 0) \mid a \in \mathbb{R}\}, \quad N = \{(0, b) \mid b \in \mathbb{R}\}.$$

We showed in Problem 1 that M and N are linear subspaces of $\mathbb{R}^2$. But notice that $(1, 0) \in M$ and $(0, 1) \in N$, yet $(1, 0) + (0, 1) = (1, 1) \notin M \cup N$. ∎

Problem 6

Let V be a vector space and let A be any subset of V. There exist linear subspaces of V that contain A (for instance, V itself is such a subspace). Let $n = \{N \mid N \text{ is a linear subspace of } V \text{ containing } A\}$. Show that n contains a smallest element. [Try $\cap n$.] Thus, there exists a smallest linear subpsace of V containing A.

Proof. Let P be an arbitrary linear subspace in n. Clearly, all elements in $\cap n$ are in P, thus $\cap n \subseteq P$. Now, since all linear subspaces in n contain 0, we have $0 \in \cap n$. Let $x, y \in \cap n$ and $c \in F$, where F is the field of scalars of V. Let V be an arbitrary element in n. Since $x, y \in \cap n$, it follows that $x, y \in V$, thus $x + y \in V$. Therefore, $\cap n$ is a linear subspace of V. ∎

Problem 7

Let V be a vector space and let n be any set of linear subspaces of V. Let A be the union of the subspaces in n, that is,

$$A = \cup n = \{x \in V \mid x \in M \text{ for some } M \in n\}$$

Apply Exercise 6 to conclude that there exists a linear subspace of V that contains every $M \in n$.

Proof. Clearly $\cup n \subseteq V$. Let $A = \cup n$. By Problem 6, there exists a smallest linear subspace of V containing A. Since for every $M \in n$, $M \subseteq A$, this linear subspace contains every $M \in n$. ∎

Problem 8

A linear subspace M of a vector of a vector space V is closed under subtraction: if $x, y \in M$ then $x - y \in M$.

Proof. Suppose $x, y \in M$, where M is a linear subspace of a vector space V. Let $c = -1$. Then $cy = -1 \cdot y = -y \in M$. Thus $x + (-y) = x - y \in M$. ∎

Problem 9

Let V be a vector space and let n be a set of linear subspaces of V, having the following property: if $M, N \in n$ then there exists $P \in n$ such that $M \subset P$. (That is, any two elements of n are contained in a third.) Prove that $\cup n$ is a linear subspace of V.

Proof. Let $x, y \in \cup n$ and let c be a scalar of the vector space V. Then $x \in M$ for some $M \in n$. Since M is a linear subspace, $cx \in M$, thus $cx \in \cup n$. Now $y \in N$ for some $N \in n$. There exists $P \in n$ such that $M \subseteq P$ and $N \subseteq P$. Thus $x, y \in P$. Since P is a linear subspace, $x + y \in P$, thus $x + y \in \cup n$. It follows that $\cup n$ is a linear subspace of V. ∎

Problem 11

Let M and N be linear subspaces of V. Prove that the following conditions are equivalent:

1. $M \cap N = \{\theta\}$;
2. if $y \in M$, $z \in N$ and $y + z = \theta$, then $y = z = \theta$;
3. if $y + z = y' + z'$, where $y, y' \in M$ and $z, z' \in N$, then $y = y'$ and $z = z'$.

Proof. ($1 \longrightarrow 2$) Suppose $M \cap N = \{\theta\}$. Furthermore, suppose $y \in M$, $z \in N$ and $y + z = \theta$. Then $y = -z \in M$, so $z \in M \cap N$. Since $M \cap N = \{\theta\}$, it follows that $z = \theta$. Then $y = -z = -\theta = \theta$.

($2 \longrightarrow 3$) Suppose that if $y \in M$, $z \in N$ and $y + z = \theta$, then $y = z = \theta$. Suppose $y + z = y' + z'$, where $y, y' \in M$ and $z, z' \in N$. Then $(y - y') + (z - z') = \theta$. By 2 it follows that $y - y' = \theta$ and $z - z' = \theta$, so $y = y'$ and $z = z'$.

($3 \longrightarrow 1$) Suppose that if $y + z = y' + z'$, where $y, y' \in M$ and $z, z' \in N$, then $y = y'$ and $z = z'$. Let $x \in M \cap N$. Then $x \in M$ and $x \in N$. Consider $x + \theta = \theta + x$. By 3 it follows that$x = \theta$ and $\theta = x$. Therefore $M \cap N = \{\theta\}$. ∎

Problem 12

Let M and N be linear subspaces of V. If $M + N = V$ and $M \cap N = \{\theta\}$ (cf. Exercise 11), V is said to be the direct sum of M and N, written $V = M \oplus N$. Prove that the following conditions are equivalent:

1. $V = M \oplus N$;
2. for each $x \in V$, there exist unique elements $y \in M$ and $z \in N$ such that $x = y + z$.

Proof. ($\longrightarrow$) Suppose $V = M \oplus N$. Let x be an arbitrary element in V. Then there exists $m \in M$ and $n \in N$ such that $x = m + n$. Suppose there also exists $m' \in M$ and $n' \in N$ such that $x = m' + n'$. Then $m + n = m' + n'$. By Problem 11, $m = m'$ and $n = n'$, thus m and n are unique.

($\longleftarrow$) Suppose for each $x \in V$, there exist unique elements $y \in M$ and $z \in N$ such that $x = y + z$. Let x be an arbitrary element in V. Clearly $x = y + z$ where $y \in M$ and $z \in N$, thus $x \in M + N$. Let x be an arbitrary element in $M + N$. Thus $x = m + n$ for some $m \in M$ and $n \in N$. Therefore $V = M + N$.

Let $x \in M \cap N$. Then

$$x = x + \theta \quad \text{with } x \in M,\ \theta \in N,$$

and

$$x = \theta + x \quad \text{with } \theta \in M,\ x \in N.$$

Thus $x = \theta$. Therefore $M \cap N = \{\theta\}$. ∎

Problem 13

Let V be the real vector space of all functions $x : \mathbb{R} \longrightarrow \mathbb{R}$ (1.3.4). Call $y \in V$ even if $y(-t) = y(t)$ for all $t \in \mathbb{R}$, and call $z \in V$ odd if $z(-t) = -z(t)$ for all $t \in \mathbb{R}$. Let

$$M = \{y \in V \mid y \text{ is even}\}, \quad N = \{z \in V \mid z \text{ is odd}\}.$$

1. Prove that M and N are linear subspaces of V and that $V = M \oplus N$ in the sense of Exercise 12. [Hint: If $x \in V$ consider the functions of y and z defined by $y(t) = \frac{1}{2}[x(t) + x(-t)]$, $z(t) = \frac{1}{2}[x(t) - x(-t)]$].
2. What does (i) say for $x(t) = e^t$?
3. What does (i) say for a polynomial function x?

Proof. Clearly the zero function is in both M and N. Let $x, y \in M$ and let $c \in \mathbb{R}$. Then $x(-t) = x(t)$ and it follows that $cx(-t) = cx(t)$, thus $cx \in M$. Also,

$$(x + y)(-t) = x(-t) + y(-t) = x(t) + y(t) = (x + y)(t),$$

thus $x+y \in M$. Let $a, b \in N$ and let $c \in \mathbb{R}$. Then $a(-t) = -a(t)$ and it follows that $ca(-t) = -ca(t)$, thus $ca \in N$. Also,

$$(a + b)(-t) = a(-t) + b(-t) = -a(t) - b(t) = -(a + b)(t),$$

thus $a + b \in N$. It follows that M and N are linear subspaces of V. ∎

Proof. Let $x \in V$ be arbitrary. Define functions y and z by

$$y(t) = \frac{1}{2}[x(t) + x(-t)], \qquad z(t) = \frac{1}{2}[x(t) - x(-t)].$$

Notice

$$y(-t) = \frac{1}{2}[x(-t) + x(t)] = y(t),$$

so y is even and thus $y \in M$. Also

$$z(-t) = \frac{1}{2}[x(-t) - x(t)] = -\frac{1}{2}[x(t) - x(-t)] = -z(t),$$

so z is odd and thus $z \in N$. Now, for all $t \in \mathbb{R}$,

$$y(t) + z(t) = \frac{1}{2}[x(t) + x(-t)] + \frac{1}{2}[x(t) - x(-t)] = x(t).$$

Thus $x = y + z$ and therefore $V = M + N$. Suppose $x \in M \cap N$. Then x is both even and odd and

$$x(t) = x(-t) \quad \text{and} \quad x(t) = -x(-t).$$

Thus $x(t) = -x(t)$ for all t, $x(t) = 0$ for all t. Thus $M \cap N = \{\theta\}$. Therefore $V = M \oplus N$. ∎

Solution (2): It says that e^t can be written uniquely as the sum of an even function and an odd function.

Solution (3): It says that any polynomial function can be written uniquely as the sum of an even polynomial and an odd polynomial.

Problem 16

Let V and W be vector spaces over the same field F and let $V \times W$ be the product vector space (1.3.11). If M is a linear subspace of V, and N is a linear subspace of W, show that $M \times N$ is a linear subspace of $V \times W$.

Proof. Suppose M is a linear subspace of V and N is a linear subspace of W. Let $x, y \in M \times N$ and let c be a scalar from the field F. Then $x = (m_1, n_1)$ and $y = (m_2, n_2)$ are elements of $M \times N$, and $cx = c(m_1, n_1) = (cm_1, cn_1)$. Now, $cm_1 \in M$ and $cn_1 \in N$, thus $cx \in M \times N$. Also $x + y = (m_1, n_1) + (m_2, n_2) = (m_1 + m_2, n_1 + n_2)$. Since $m_1 + m_2 \in M$ and $n_1 + n_2 \in N$, it follows that $x + y \in M \times N$. Since $(\theta_V, \theta_W) \in M \times N$, it follows that $M \times N$ is a linear subspace of $V \times W$. ∎

Problem 17

If $V = V_1 \times V_2$ (1.3.11), $M_1 = \{(x_1, \theta) \mid x_1 \in V_1\}$ and $M_2 = \{(\theta, x_2) \mid x_2 \in V_2\}$, then $V = M_1 + M_2$ and $M_1 \cap M_2 = \{\theta\}$ (where θ stands for the zero vector of all vector spaces in sight). In the terminology of Exercise 12, V is the direct sum of its subspaces in M_1 and M_2, that is, $V = M_1 \oplus M_2$. (Generalization?)

Definition 1. If $V = V_1 \times V_2 \times \cdots \times V_n$, and

1. $M_1 = \{(x_1, \theta_2, \dots, \theta_n) \mid x_1 \in V_1\}$
2. $M_2 = \{(\theta_1, x_2, \dots, \theta_n) \mid x_2 \in V_2\}$
3. ...
4. $M_n = \{(\theta_1, \theta_2, \dots, x_n) \mid x_n \in V_n\}$

then V is the direct sum of its subspaces $M_1, M_2, \dots, M_n$, or

$$V = M_1 \oplus M_2 \oplus \cdots \oplus M_n.$$

Problem 18

Let M and N be linear subspaces of V whose union $M \cup N$ is also a linear subspace. Prove that either $M \subset N$ or $N \subset M$. [Hint: Assume to the contrary that neither of M, N is contained in the other; choose vectors $y \in M$, $z \in N$ with $y \notin N$, $z \notin M$ and try to find a home for $y + z$.]

Proof. Let $y \in M$ and $z \in N$ such that $y \notin N$ and $z \notin M$. Then $y \in M \cup N$ and $z \in M \cup N$. Since $M \cup N$ is a linear subspace, $y + z \in M \cup N$. Without loss of generality, suppose $y + z \in M$. Then $-y \in M$ by Problem 8. Since M is closed under addition $y + z + (-y) = z \in M$ which is a contradiction. ∎

2 Linear Mappings

2.1 Linear Mappings

Problem 2

Let T be a set, $V = \mathcal{F}(T, \mathbb{R})$ the vector space of all functions $x : T \longrightarrow \mathbb{R}$ (Example 1.3.4). Fix a point $t \in T$ and define $f : V \longrightarrow \mathbb{R}$ by the formula $f(x) = x(t)$. Then f is a linear form on V.

Proof. Let $x, y \in V$ and $c \in \mathbb{R}$. Then $f(x + y) = (x + y)(t) = x(t) + y(t) = f(x) + f(y)$. and $f(cx) = (cx)(t) = c\,x(t) = cf(x)$. Thus f is a linear mapping. Since it maps functions in V to scalars in $\mathbb{R}$ it is a linear form. ∎

Problem 3

If $T : V \longrightarrow W$ is a linear mapping, then $T(x - y) = Tx - Ty$ for all $x, y \in V$.

Proof. Suppose $T : V \longrightarrow W$ is a linear mapping. Let x, y be arbitrary elements in V. Then $T(x - y) = T(x + (-y)) = Tx + T(-y) = Tx + (-T(y)) = Tx - Ty$. ∎

Problem 4

If V is a vector space, prove that the mapping $T : V \times V \longrightarrow V$ defined by $T(x, y) = x - y$ is linear. (It is understood that $V \times V$ has the product vector space structure described in Example 1.3.10.)

Proof. Suppse V is a vector space. Let $(x,y),(a,b)$ be arbitrary elements in $V \times V$. Let $c \in \mathbb{R}$ be a scalar. Then $T((x,y)+(a,b)) = T(x+a,y+b) = (x+a)-(y+b) = (x-y)+(a-b) = T(x,y)+T(a,b)$. Also $T(c(x,y)) = T(cx,cy) = cx-cy = c(x-y) = c\,T(x,y)$. Thus T is linear. ∎

Problem 5

With notations as in Theorem 2.1.3, an element $(a_1,\dots,a_n)$ of F^n such that $T(a_1,\dots,a_n) = \theta$ is called a (linear) relation among the vectors $x_1,\dots,x_n$. For example, suppose that $V = F^2, n = 3$ and $x_1 = (2,-3), x_2 = (4,1), x_3 = (8,9)$.

1. Find a formula for the linear mapping $F^3 \longrightarrow F^2$ defined in Theorem 2.1.3.
2. Show that $(-2,3,1)$ is a relation among x_1,x_2,x_3.

Solution (a):

$$f(a_1,a_2,a_3) = a_1x_1 + a_2x_2 + a_3x_3$$

Solution (b):

$$f(-2,3,-1) = -2x_1+3x_2-x_3 = -2(2,-3)+3(4,1)-(8,9) = (-4,6)+(12,3)-(8,9) = (8,9)-(8,9) = (0,0) = \theta$$

Problem 6

The proof of Theorem 2.1.2 uses only the additivity of the mapping T. Give a proof using only its homogeneity. [Hint: Corollaries 1.4.3, 1.4.5]

Theorem 1. If $T : V \longrightarrow W$ is a linear mapping, then $T\theta = \theta$ and $T(-x) = -(Tx)$ for all $x \in V$.

Proof. Suppose $T : V \longrightarrow W$ is a linear mapping. Then $T(\theta) = T(0 \cdot x) = 0 \cdot T(x)$. Then by Corollary 1.4.3, $0 \cdot T(x) = \theta$, hence $T(\theta) = \theta$. Also $T(-x) = T((-1)\cdot x) = (-1)T(x)$. Then by Corollary 1.4.5, $(-1)T(x) = -T(x)$. Thus $T(-x) = -Tx$. ∎

Problem 7

If $\mathcal{D}$ is the vector space of real polynomial functions (Example 1.3.5) and $f : \mathcal{D} \longrightarrow \mathbb{R}$ is the mapping defined by $f(p) = p'(1)$ (the value of the derivative of p at 1), then f is a linear form on $\mathcal{D}$. What is the geometric meaning of $f(p) = 0$.

Proof. Let $x,y \in \mathcal{D}$ and $c \in \mathbb{R}$. Then $f(x+y) = (x+y)'(1) = x'(1)+y'(1) = f(x)+f(y)$. and $f(cx) = (cx)'(1) = c\,x'(1) = cf(x)$. It follows that f is a linear form on $\mathcal{D}$. ∎

Solution: $f(p) = 0$ is a critical point at the value $x = 1$ of the function p.

Problem 8

Let $S : \mathcal{D} \longrightarrow \mathcal{D}$ be the linear mapping of Example 2.1.5. Define $R : \mathcal{D} \longrightarrow \mathcal{D}$ by $Rp = Sp + p(5)1$, where $p(5)1$ is the constant function defined by the real number $p(5)$. Then R is linear and $(Rp)' = p$. [So to speak, the constant of integration can be tailor-made for p in a linear fashion.] More generally, look at the mapping $Rp = Sp + f(p)1$, where f is any linear form on $\mathcal{D}$.

Proof. Let $x,y \in \mathcal{D}$ and $c \in \mathbb{R}$. Then $R(x+y) = S(x+y)+(x+y)(5)1 = S(x)+S(y)+x(5)1+y(5)1 = R(x)+R(y)$. Also $R(cx) = S(cx)+(cx)(5)1 = cS(x)+cx(5)1 = cR(x)$. Thus R is linear.

For the general form notice $R(x+y) = S(x+y)+f(x+y)1 = S(x)+S(y)+f(x)1+f(y)1 = R(x)+R(y)$. Also $R(cx) = S(cx)+f(cx)1 = cS(x)+cf(x)1 = cR(x)$. ∎

2.2 Linear Mappings and Linear Subspaces: Kernel and Range

Problem 2

Let V be a vector space over F, $f : V \longrightarrow F$ a linear form on V (Definition 2.1.11). Assume that f is not identically zero and choose a vector y such that $f(y) \neq 0$. Let N be the kernel of f. Prove that every vector x in V may be written $x = z + cy$ with $z \in N$ and $c \in F$, and that z and c are uniquely determined by x. [Hint: If $x \in V$, compute the value of f at the vector $x - [f(x)/f(y)]y$.]

Proof. Suppose $x, y \in V$. Notice $f\left(x - \frac{f(x)}{f(y)}y\right) = f(x) - f\left(\frac{f(x)}{f(y)}y\right) = f(x) - \frac{f(x)}{f(y)}f(y) = f(x) - f(x) = 0$. We've just shown $x - \frac{f(x)}{f(y)}y \in \mathrm{ker}(f)$. Thus $x = \left(x - \frac{f(x)}{f(y)}y\right) + \frac{f(x)}{f(y)}y$. Let $z = x - \frac{f(x)}{f(y)}y$ and $c = \frac{f(x)}{f(y)}$, and we see $x = z + cy$ as required.

Suppose $x = z' + c'y$ with $z' \in \mathrm{ker}(f)$ and $c' \in F$. Then $z + cy = z' + c'y \implies z - z' = (c' - c)y$. Applying f gives $0 = f(z - z') = (c' - c)f(y)$, so $c' - c = 0$ since $f(y) \neq 0$. Thus $c' = c$ and $z' = z$. ∎

Problem 3

Let $S : V \longrightarrow W$ and $T : V \longrightarrow W$ be a linear mappings and let $M = \{x \in V \mid Sx \in T(V)\}$. Prove that M is a linear subspace of V.

Proof. Let $x, y \in M$ and let c be a scalar of the vector space V. Since $x, y \in M$, we have $Sx \in T(V)$ and $Sy \in T(V)$. Since $T(V)$ is a linear subspace of W it follows that $S(x + y) = Sx + Sy \in T(V)$. Thus $x + y \in M$. Similarly $S(cx) = cS(x) \in T(V)$ thus $cx \in M$. Therefore M is a linear subspace of V. ∎

Problem 4

If $T : V \times V \longrightarrow V$ is the linear mapping $T(x, y) = x - y$ (2.1, Exercise 4), determine the kernel and range of T.

Solution: The kernel is $(x, x) \in V \times V$ and the range is V.

Problem 5

Let V be a real vector space, W its complexification (1.3, Exercise 4); write W_R for W regarded as a real vector space (1.3, Exercise 3). For $(x, y) \in W$, we have

$$(x, y) = (x, \theta) + (\theta, y) = (x, \theta) + i(y, \theta).$$

Prove that $x \mapsto (x, \theta)$ is an injective mapping $V \longrightarrow W_R$. [Identifying $x \in V$ with $(x, \theta) \in W$, one can suggestively write $W = V + iV$; so to speak, V is the 'real part' of its complexification.]

Proof.

$$x(x_1) = x(x_2) \iff (x_1, 0) = (x_2, 0) \iff x_1 = x_2$$

∎

Problem 6

Let $\mathcal{D}$ be the vector space of real polynomial functions (1.3.5) and let $T : \mathcal{D} \longrightarrow \mathcal{D}$ be the linear mapping defined by $Tp = p - p'$, where p' is the derivative of p. Prove that T is injective.

Proof. Suppose $x, y \in \mathcal{D}$ such that $T(x) = T(y)$. Then $T(x) = T(y) \iff x - x' = y - y' \iff (x-y)-(x'-y') = 0$. Let $q = x - y$ then $q - q' = 0 \iff q = q'$. Thus $q = 0$ and it follows that $x = y$ and $x' = y'$. ∎

2.3 Spaces of Linear Mappings: $\mathcal{L}(V, W)$ and $\mathcal{L}(V)$

Problem 1

Complete the details in the proof of Theorem 2.3.11.

Theorem 2. *If V and W are vector spaces over F, then $\mathcal{L}(V, W)$ is also a vector space over F for the operations defined in 2.3.9.*

Proof. Let $S, T \in \mathcal{L}(V, W)$ and let $a, b \in F$. For all $x \in V$, $(a(S+T))(x) = a((S+T)(x)) = a(S(x)+T(x)) = aS(x) + aT(x) = (aS + aT)(x)$. For all $x \in V$, $((a+b)T)(x) = (a+b)T(x) = aT(x) + bT(x) = (aT + bT)(x)$. For all $x \in V$, $((ab)T)(x) = ab\,T(x) = a(bT(x)) = (a(bT))(x)$. For all $x \in V$, $(1T)(x) = 1 \cdot T(x) = T(x)$. ∎

Problem 2

Let V be a vector space. If $R, S, T \in \mathcal{L}(V)$ and c is a scalar, the following equalities are true.

1. $(RS)T = R(ST)$;
2. $(R+S)T = RT + ST$;
3. $R(S+T) = RS + RT$;
4. $(cS)T = c(ST) = S(cT)$;
5. $TI = T = IT$; (T is the identity mapping)

Proof. For all $x \in V$.

1. $(RS)T(x) = R(S(T(x))) = R(ST(x)) = (R(ST))(x)$.
2. $(R+S)T(x) = (R+S)(T(x)) = R(T(x)) + S(T(x)) = RT(x) + ST(x) = (RT + ST)(x)$.
3. $R(S+T)(x) = R(S(x)+T(x)) = R(S(x)) + R(T(x)) = RS(x) + RT(x) = (RS + RT)(x)$.
4. $(cS)T(x) = cS(T(x)) = c(ST(x)) = (c(ST))(x) = S(cT)(x)$.
5. $TI(x) = T(I(x)) = T(x)$ and $IT(x) = I(T(x)) = T(x)$.

∎

Problem 4

If $T : V \longrightarrow W$ is a linear mapping and g is a linear form on W, show that the formula $f(x) = g(Tx)$ defines a linear form f on V. [Shortcut: $f = gT = g \circ T$.] The formula $T'g = f$ defines a mapping $T' : \mathcal{L}(W, F) \longrightarrow \mathcal{L}(V, F)$, where F is the field of scalars. Show that T' is linear. [T' is called the transpose (or 'adjoint' of T.)]

Proof. By Theorem 2.3.13 it directly follows that f is linear.

Let $x, y \in \mathcal{L}(W, F)$ and c be a scalar in F. Then for all $v \in V$, $T'(x+y)(v) = (x+y)(T(v)) = x(T(v)) + y(T(v)) = T'(x)(v) + T'(y)(v)$. Similarly, $T'(cx)(v) = (cx)(T(v)) = c\,x(T(v)) = c\,T'(x)(v)$. Therefore, T' is linear. ∎

Problem 5

If V is a vector space over F, the vector space $\mathcal{L}(V, F)$ of linear forms on V is called the dual space of V and is denoted V'. Thus, the correspondence $T \mapsto T'$ described in Exercise 4 defines a mapping $\mathcal{L}(V, W) \longrightarrow \mathcal{L}(W', V')$. Show that this mapping is linear, that is, $(S+T)' = S' + T'$ and $(cT)' = cT'$ for all S, T in

$\mathcal{L}(V,W)$ and all scalars c.

Proof. Let $S,T \in \mathcal{L}(V,W)$ and let c be a scalar. Let v be a vector in V. Then $(S+T)'g(v) = g(S+T)(v) = g(S(v)+T(v)) = gS(v)+gT(v) = S'g(v)+T'g(v)$. Similarly $(cT)'g(v) = g(cT(v)) = cgT(v) = c(T'g)(v)$. Thus the correspondence $T \mapsto T'$ is linear. ∎

Problem 6

With notations as in Exercise 4, prove that if T is surjective then T' is injective; in other words, show that if $T(V) = W$ then $T'g = 0 \implies g = 0$.

Proof. Suppose $T(V) = W$ and $T'g = 0$. Let $v \in V$ then $(T'g)(v) = g(T(v)) = 0$ for all $v \in V$. Thus $g = 0$ and T' is injective. ∎

Problem 7

As in Example 2.1.4, let $V = \mathcal{D}$ be the real vector space of all polynomial functions $p : \mathbb{R} \longrightarrow \mathbb{R}$, $D : V \longrightarrow V$ the linear mapping defined by $Dp = p'$ (the derivative of p); let $D' : V' \longrightarrow V'$ be the transpose of D as defined in Exericise 4. (Caution: The prime is being used with three different meanings.)
For each $t \in \mathbb{R}$ let ρ_i be the linear form on V defined by $\rho_t(p) = p(t)$. Let $[a,b]$ be a closed interval on $\mathbb{R}$ and let ρ be the linear form on V defined by

$$\rho(p) = \int_a^b p(t)dt$$

Prove : $D'\rho = \rho_b - \rho_a$. [Hint: Fundamental Theorem of Calculus.]

Proof. Let $p \in V$. Notice $(D'\rho)(p) = \rho(Dp) = \int_a^b p'(t)dt = p(b) - p(a) = \rho_b(p) - \rho_a(p)$. ∎

Problem 9

Let U, V, W be vector spaces over F. Prove:

1. For fixed $T \in \mathcal{L}(U,V)$, $\psi = S \mapsto ST$ is a linear mapping $\mathcal{L}(V,W) \longrightarrow \mathcal{L}(U,W)$ (see Figure 8 2.3.1 2).
2. For fixed $S \in \mathcal{L}(V,W)$, $\phi = T \mapsto ST$ is a mapping $\mathcal{L}(U,V) \longrightarrow \mathcal{L}(U,W)$.

Proof. Let $x,y \in \mathcal{L}(V,W)$ and $v \in U$. Let c be a scalar. Then $(\psi(x+y))(v) = ((x+y)T)(v) = (x+y)(T(v)) = xT(v)+yT(v) = \psi(x)(v)+\psi(y)(v)$. Also $(\psi(cx))(v) = ((cx)T)(v) = (cx)(T(v)) = c(xT(v)) = c\psi(x)(v)$. Thus ψ is a linear mapping. ∎

Proof. Let $x,y \in \mathcal{L}(U,V)$ and $v \in U$. Let c be a scalar. Then $(\phi(x+y))(v) = S(x+y)(v) = S(x(v)+y(v)) = S(x(v))+S(y(v)) = \phi(x)(v)+\phi(y)(v)$. Also $(\phi(cx))(v) = S(cx(v)) = c\,S(x(v)) = c\,\phi(x)(v)$. Thus ϕ is a linear mapping. ∎

Problem 10

If $T : U \longrightarrow V$ and $S : V \longrightarrow W$ are linear mappings, prove that $Im(ST) \subset ImS$ and $Ker(ST) \supset KerT$.

Proof. Clearly T can only restrict the domain of S thus $Im(ST) \subset ImS$. Clearly any x in $KerT$ will be in $Ker(ST)$ since $S(0) = 0$. ∎

Problem 11

Let $T : V \longrightarrow V$ be a linear mapping such that $ImT \subset Ker(T - I)$, where I is the identity mapping (2.3.3). Prove that $T^2 = T$. [Recall $T^2 = TT$.]

Proof. First note that $Ker(T-I) = \{v \in V \mid (T-I)(v) = 0\}$. For any $v \in V$, $(T-I)(v) = 0 \iff T(v)-I(v) = 0 \iff T(v) - v = 0 \iff T(v) = v$. Let $v \in V$ be arbitrary. Then $T(v) \in ImT$. Since $ImT \subset Ker(T - I)$, it follows that $T(v) \in Ker(T - I)$, and thus $T(T(v)) = T(v)$. Thus $T^2 = T$. ∎

Problem 12

Supose $V = M \oplus N$ in the sense of 1.6, Exercise 12. For each $x \in V$, let $x = y + z$ be its unique decomposition with $y \in M, z \in N$ and define $Px = y$, $Qx = z$. Prove that $P, Q \in \mathcal{L}(V)$, $P^2 = P$, $Q^2 = Q$, $P + Q = I$ and $PQ = QP = 0$.

Proof. Let $a_1, a_2 \in M$ and $b_1, b_2 \in N$. Let $x = a_1+b_1, y = a_2+b_2 \in V$. Then $P(x+y) = P((a_1+a_2)+(b_1+b_2)) = a_1 + a_2 = Px + Py$. Also $P(cx) = P(c(a_1 + b_1)) = P(ca_1 + cb_1) = ca_1 = c(Px)$. Thus $P \in \mathcal{L}(V)$. The argument that $Q \in \mathcal{L}(V)$ is similar. ∎

Proof. Simply note that $ImP \subset Ker(P - I)$ thus $P^2 = P$. The argument that $Q^2 = Q$ is similar. ∎

Proof. Let $a \in M$ and $b \in N$. Let $x = a + b$. Then $(P + Q)(x) = P(x) + Q(x) = a + b = x$. Thus $P + Q = I$. ∎

Proof. Let $a \in M$ and $b \in N$. Let $x = a + b$. Then $(PQ)(x) = P(Q(x)) = P(Q(a + b)) = P(0 + Q(b)) = 0$. The argument for QP is similar. ∎

Problem 13

Let $T : V \longrightarrow V$ be a linear mapping such that $T^2 = T$, and let $M = ImT$ and $N = KerT$. Prove that $M = \{x \in V \mid Tx = x\} = Ker(T - I)$ and that $V = M \oplus N$ in the sense of 1.6, Exercise 12.

Proof. Notice $x \in M \iff Tx = x \iff Tx-x = 0 \iff Tx-Ix = 0 \iff (T-I)(x) = 0 \iff x \in \ker(T-I)$.

Let $x \in V$. Let $y = Tx \in M$ and $z = x - Tx \in N$. Notice

$$T(z) = T(x - Tx) = Tx - T^2x = Tx - Tx = 0$$

Thus $z \in \ker T = N$. Therefore $x = y + z \in M + N$. Suppose $x = y_1 + z_1 = y_2 + z_2$ with $y_i \in M, z_i \in N$. Then

$$y_1 - y_2 = z_2 - z_1 \in M \cap N.$$

But then $M \cap N = \{0\}$ because suppose $v \in M \cap N$, then $Tv = v$ and $Tv = 0$, so $v = 0$. Thus $V = M \oplus N$. ∎

Problem 14

Find $S, T \in \mathcal{L}(\mathbb{R}^2)$ such that $ST \neq TS$.

Proof. Consider the linear maps $S, T \in \mathcal{L}(\mathbb{R}^2)$

$$S(x, y) = (y, 0), \quad T(x, y) = (0, x).$$

Then for any $(x, y) \in \mathbb{R}^2$

$$ST(x, y) = S(T(x, y)) = S(0, x) = (x, 0),$$
$$TS(x, y) = T(S(x, y)) = T(y, 0) = (0, y).$$

Thus $ST \neq TS$. ∎

Problem 15

Let $T : U \longrightarrow V$ and $S : V \longrightarrow W$ be linear mappings. Prove

1. If S is injective, then $Ker(ST) = Ker(T)$.
2. If T is surjective, then $Im(ST) = ImS$.

Proof. Suppose S is injective. Let x be an arbitrary element in U. Suppose $x \in Ker(T)$ thus $T(x) = 0$. Then $(ST)(x) = S(T(x)) = S(0) = 0$. Thus $x \in Ker(ST)$. Suppose $x \in Ker(ST)$ thus $(ST)(x) = 0$. Now $S(0) = 0$ and since $S(T(x)) = 0$ and S is injective, $T(x) = 0$ thus $x \in KerT$. It follows that $Ker(ST) = Ker(T)$. ∎

Proof. Suppose T is surjective. It follows that T maps onto all elements in the domain of S. Therefore $Im(ST) = S$. ∎

Problem 16

Let V be a real or complex vector space and let $t \in \mathcal{V}$ be such that $T^2 = I$. Define

$$M = \{x \in V \mid Tx = x\}, \quad N = \{x \in V \mid Tx = -x\}$$

Prove that M and N are linear subspaces of V such that $V = M \oplus N$. [Hint: For every vector x, $x = \frac{1}{2}(x + Tx) + \frac{1}{2}(x - Tx)$.]

Proof. Let $x \in V$ and consider $x = \frac{1}{2}(x + Tx) + \frac{1}{2}(x - Tx)$. Let $m = \frac{1}{2}(x + Tx)$ and $n = \frac{1}{2}(x - Tx)$. Then $Tm = \frac{1}{2}(Tx + T^2x) = \frac{1}{2}(Tx + x) = m$ so $m \in M$, and $Tn = \frac{1}{2}(Tx - T^2x) = \frac{1}{2}(Tx - x) = -n$ so $n \in N$. Thus $x = m + n \in M + N$.

Suppose $v \in M \cap N$. Then $Tv = v$ and $Tv = -v$, which implies $v = 0$. Thus $M \cap N = \{0\}$. It follows that $V = M \oplus N$. ∎

Problem 17

Let $V = \mathcal{F}(\mathbb{R})$ be the real vector space of all functions $x : \mathbb{R} \longrightarrow \mathbb{R}$ (1.3.4). For $x \in V$ define $Tx \in V$ by the formula $(Tx)(t) = x(-t)$. Analyze T in the light of Exercise 16. [Remember 1.6, Exercise 13]

Solution: The set of functions for which $x(t) = x(-t)$, i.e., the even functions, are in M. The functions for which $x(t) = -x(-t)$, i.e., the odd functions, are in N. Every function $x \in V$ can be expressed as $x = \frac{1}{2}(x + Tx) + \frac{1}{2}(x - Tx) \in M \oplus N$.

Problem 18

Let $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, prove that $(ST)' = T'S'$. (cf. Exercise 4).

Proof. Let $f \in W'$ be a linear form on W. Then $(ST)'(f) = f \circ (ST) = (f \circ S) \circ T = T'(S'(f))$. ∎

Problem 19

Let $S, T \in \mathcal{L}(V, W)$ and let $M = \{x \in V \mid Sx = Tx\}$. Prove that M is a linear subspace of V. [Hint: Consider $Ker(S - T)$.]

Proof. Note that $M = \{x \in V \mid Sx - Tx = 0\} = \{x \in V \mid (S - T)x = 0\} = \mathbf{ker}(S - T)$. Since the kernel of a linear map is always a linear subspace, M is a linear subspace of V. ∎

Problem 20

If $T : V \longrightarrow W, S : W \longrightarrow V$ are linear mappings such that $ST = I$, prove that $KerT = \{0\}$ and $ImS = V$.

Proof. Let $x \in KerT$ such that $Tx = 0$. Then $STx = Tx = 0$. Since $ST = I$, we have $x = 0$ thus $KernT = \{0\}$.

Next, let $v \in V$. Then $v = Iv = STv = S(Tv)$. Thus $v \in ImS$. Therefore $ImS = V$. ∎

Problem 21

If $V = \{\theta\}$ or $W = \{\theta\}$ then $\mathcal{L}(V, W) = \{0\}$. If $V \neq \{\theta\}$ then $I_v \neq 0$.

Proof. Suppose $V = \{\theta\}$ or $W = \{\theta\}$. Clearly there is only one linear mapping between V and W which is for all $x \in V, T(x) = 0 \in W$.

Suppose $V \neq \{\theta\}$. V has at least two elements, one of which maps to 0. The other must map to a nonzero element under I_V. ∎

Problem 22

A lightning proof of Theorem 2.3.11 can be based on an earlier exercise: Since $0 \in \mathcal{L}(V, W)$, Lemma 2.3.10 shows that $\mathcal{V}, \mathcal{W}$ is a linear subspace of the vector space $\mathcal{F}(V, W)$ defined as in 1.3, Exercise 1, therefore $\mathcal{L}(V, W)$ is also a vector space (1.6.2).

Solution: OK.

Problem 23

If $T : U \longrightarrow V$ and $S : V \longrightarrow W$ are linear mappings, prove that $Ker(ST) = T^{-1}(KerS)$.

Proof. Let $x \in U$. Then

$$x \in Ker(ST) \iff STx = 0 \iff S(Tx) = 0 \iff Tx \in KerS \iff x \in T^{-1}(KerS).$$

Therefore $Ker(ST) = T^{-1}(KerS)$. ∎

Problem 24

Let $V = \mathcal{D}$ be the vector space of real polynomial functions, $D : V \longrightarrow V$ the differentiation mapping $Dp = p'$ (2.1.4). Let u be the monomial $u(t) = t$ and define another linear mapping $M : V \longrightarrow V$ by the formula $Mp = up$. (So to speak, M is multiplication by t.) Prove that $DM - MD = I$ (the identity mapping). [Hint: The claim is that $(up)' - up' = p$. Remember the 'product rule'?]

Proof. By the product rule,

$$(up)' = u'p + up'.$$

Since $u(t) = t$ we know $u' = 1$. Thus

$$(up)' = p + up'.$$

∎

Problem 25

Let $D : \mathcal{D} \longrightarrow \mathcal{D}$ be the differentiating map of Exercise 24 and let $S : \mathcal{D} \longrightarrow \mathcal{D}$ be the linear mapping such that $(Sp)' = p$ and $(Sp)(0) = 0$ (2.1.5). Prove:

1. If $R : \mathcal{D} \longrightarrow \mathcal{D}$ is any linear mapping such that $(Rp)' = p$ for all $p \in \mathcal{D}$, then $DR = I$.
2. If $T : \mathcal{D} \longrightarrow \mathcal{D}$ is a linear mapping such that $DT = 0$, then there exists a linear form f on $\mathcal{D}$ such that (in the notation of Exercise 2.3.4) $T = f \otimes 1$, where 1 is the constant function 1. [Hint: $ImT \subset KerD$.]
3. With R as in (i), $R = S + f \otimes 1$ for a suitable linear form f on $\mathcal{D}$. [Hint: Consider $T = R - S$.] Remember 2.1, Exercise 8?

Proof. Suppose $R : \mathcal{D} \longrightarrow \mathcal{D}$ is a linear mapping such that $(Rp)' = p$ for all $p \in \mathcal{D}$. Let $p \in \mathcal{D}$. Then $(DR)(p) = D(Rp) = (Rp)' = p$. Thus $DR = I$. ∎

Proof. If $T : \mathcal{D} \longrightarrow \mathcal{D}$ is a linear mapping such that $DT = 0$, then for all $p \in \mathcal{D}$, Tp is a constant function. Thus $T = T \otimes 1$, where 1 is the constant function 1. ∎

Proof. Let $R : \mathcal{D} \longrightarrow \mathcal{D}$ and let $T = R - S$. Then for any $p \in \mathcal{D}$, $D(Tp) = D((R - S)p) = DR(p) - DS(p) = I(p) - I(p) = 0$. Thus $DT = 0$. By part (ii), there exists a linear form f on $\mathcal{D}$ such that $T = f \otimes 1$. Therefore $R = S + T = S + f \otimes 1$. ∎

Problem 26

Let $S, T \in \mathcal{L}(V)$. If $ST - I$ is injective then so is $TS - I$. [Hint: $S(TS - I) = (ST - I)S$.]

Proof. Suppose $ST - I$ is injective. Thus $\ker(ST - I) = \{0\}$. Suppose $x \in V$ such that $(TS - I)(x) = 0$. Then applying S we have $S(TS - I)(x) = 0$. By the hint we have $((ST - I)S)(x) = 0 \iff (ST - I)(S(x)) = 0$. Since $ST - I$ is injective we have $S(x) = 0$. Then $(TS - I)(x) = TS(x) - x = T0 - x = -x = 0$. Thus $x = 0$. ∎

2.4 Isomorphic Vector Spaces

Problem 2

Regard $\mathbb{C}^n$ as a real vector space in the natural way (sums as usual, multiplication only by real scalars); then $\mathbb{C}^n \cong \mathbb{R}^{2n}$ as real vector spaces.

Proof. Let $r_1, r_2, \dots, r_{2n-1}, r_{2n} \in \mathbb{R}$. Consider the mapping

$$(r_1, r_2, \dots, r_{2n-1}, r_{2n}) \mapsto (r_1 + r_2 i, \dots, r_{2n-1} + r_{2n} i).$$

We know that the map $(r_1, r_2) \mapsto r_1 + r_2 i$ is linear and bijective over $\mathbb{R}$. Our mapping is simply an extension of this map to n components and is therefore linear and bijective. ∎

Problem 3

Let X be a set, V a vector space, $f : X \longrightarrow V$ a bijection. There exists a unique vector space structure on X for which f is a linear mapping (hence a vector space isomorphism). [Hint: Define sums and scalar multiples in X by the formulas $x + y = f^{-1}(f(x) + f(y)), cx = f^{-1}(cx)$. This trick is called 'transport of structure'.]

Proof. Clearly X is a vector space, since after applying f each vector space axiom becomes the corresponding axiom in V, and injectivity of f implies the axioms hold in X. ∎

Problem 4

Let V be the set of n-ples $x = (x_1, \dots, x_n)$ of real numbers > 0, that is,

$$V = \{(x_1, \dots, x_n) \mid x_i \in (0, +\infty \text{ for } i = 1, \dots, n)\}.$$

For $x = (x_1, \dots, x_n), y = (y_1, \dots, y_n)$ in V and $c \in \mathbb{R}$, define

$$x \oplus y = (x_1 y_1, \dots, x_n y_n), c \cdot x = (x_1^c, \dots, x_n^c),$$

(here $x_i y_i$ and c_i^c are the usual product and power), Define a mapping $T : V \longrightarrow \mathbb{R}^n$ by the formula $T(x_1, \dots, x_n) = (\log x_1, \dots, \log x_n)$.

1. Show that $T(x \oplus y) = Tx + Ty$ and $T(c \cdot x) = c(Tx)$ for all x, y in V and all $c \in \mathbb{R}$.
2. True or false (explain): V is a real vector space for the operations $\cdot$.

Proof. Let $x, y \in V$ and c be a scalar. Then

$$T(x \oplus y) = T((x_1 y_1, \dots, x_n y_n)) = (\log(x_1 y_1), \dots, \log(x_n y_n)) = (\log x_1 + \log y_1, \dots, \log x_n + \log y_n)$$
$$= (\log x_1, \dots, \log x_n) + (\log y_1, \dots, \log y_n) = Tx + Ty.$$

Also,

$$T(c \cdot x) = T((x_1^c, \dots, x_n^c)) = (\log(x_1^c), \dots, \log(x_n^c)) = (c \log x_1, \dots, c \log x_n) = c(Tx).$$

∎

Solution: False since scalar multiplication does not satisfy the vector space axioms.

Problem 5

Let V be a vector space, $T \in \mathcal{L}(V)$ bijective, and c a nonzero scalar. Prove that cT is bijective and that $(cT)^{-1} = c^{-1}T^{-1}$.

Proof. Let $v, v' \in V$ and suppose $cT(v) = cT(v')$. Then $cT(v) = cT(v') \iff cT(v) - cT(v') = 0 \iff c(T(v) - T(v')) = 0$. Since $c \neq 0$, we must have $T(v) = T(v')$. Since T is injective, it follows that $v = v'$. Thus cT is injective.

Let $v \in V$. Since $c \neq 0$, the scalar $\frac{1}{c}$ exists. Since V is closed under scalar multiplication, $\frac{1}{c} \cdot v = \frac{v}{c} \in V$. Since T is surjective, there exists $x \in V$ such that $T(x) = \frac{v}{c}$. Then $cT(x) = c\left(\frac{v}{c}\right) = v$. Thus cT is surjective.

Let $v \in V$, and suppose $(cT)^{-1}(v) = x$. Then $cT(x) = v \implies T(x) = \frac{v}{c} \implies x = T^{-1}\left(\frac{v}{c}\right) = T^{-1}(c^{-1}v)$. Thus $(cT)^{-1} = c^{-1}T^{-1}$. ∎

Problem 6

Let $T : V \longrightarrow W$ be a bijective linear mapping. For each $S \in \mathcal{L}(V)$, define $\rho : \mathcal{L}(V) \longrightarrow TST^{-1}$ (note that the product is defined). Prove that $\rho : \mathcal{L}(V) \longrightarrow \mathcal{L}(W)$ is a bijective linear mapping such that $\rho(RS) = \rho(R)\rho(S)$ for all $R, S \in \mathcal{L}(V)$.

Proof. Since the composition of linear mappings is linear, TST^{-1} is a linear mapping. Let $R, S \in \mathcal{L}(V)$. Then

$$\rho(RS) = T(RS)T^{-1} = (TRT^{-1})(TST^{-1}) = \rho(R)\rho(S).$$

Suppose $\rho(S)(v) = 0$ for some $v \in W$. Then $\rho(S)(v) = 0 \iff TST^{-1}(v) = 0$. Now $v = T(x)$ for some $x \in V$ thus $TST^{-1}(T(x)) = TS(x) = 0$. Then since T is injective $S(x) = 0$ thus ρ is injective. It is surjective because for any $U \in \mathcal{L}(W)$, setting $S = T^{-1}UT$ gives $\rho(S) = U$. ∎

Problem 7

Let V be a vector space, $T \in \mathcal{L}(V)$. Prove:

1. If $T^2 = 0$ then $I - T$ is bijective.
2. If $T^n = 0$ for some positive integer n, then $I - T$ is bijective.

Proof. Suppose $T^2 = 0$. Let $v \in V$ and suppose $(I - T)(v) = 0 \implies T(v) - T^2(v) = 0 \implies T(v) = 0$. But $I(v) - T(v) = 0 \implies TI(v) - T^2(v) = T(0) \implies v - T(v) = 0 \implies v = T(v)$. Thus $v = 0$. It follows that $I - T$ is injective.

Let $y \in V$. We want to find $x \in V$ such that $(I - T)(x) = y$. Let $x = y + T(y)$. Then $(I - T)(x) = (I - T)(y + T(y)) = I(y + T(y)) - T(y + T(y)) = I(y) + IT(y) - T(y) - T^2(y) = y + T(y) - T(y) = y$. ∎

Proof. We proceed via induction on n. The previous proof shows the base case $n = 2$. Suppose for some $n - 1$, $T^{n-1} = 0$ implies $I - T$ is bijective. Now suppose $T^n = 0$. Let $v \in V$ and suppose $(I - T)(v) = 0$. Composing with T^{n-1}, we get $T^{n-1}(v) - T^n(v) = T^{n-1}(v) = 0$. By the induction hypothesis, this implies $v = 0$, so $I - T$ is injective.

For surjectivity, let $y \in V$ and set $x = y + T(y) + \cdots + T^{n-1}(y)$. Then $(I - T)(x) = y$, so $I - T$ is bijective. ∎

Problem 8

Let U, V, W be vector spaces over the same field. Form the product vector spaces $U \times V, V \times W$ (Definition 1.3.11), then the product vector spaces $(U \times V) \times W$ and $U \times (V \times W)$. Prove that $(U \times V) \times W \cong U \times (V \times W)$.

Proof. Consider the mapping

$$\rho : (U \times V) \times W \longrightarrow U \times (V \times W), \quad \rho((x_1, x_2), x_3) = (x_1, (x_2, x_3)).$$

Suppose $\rho((x_1, x_2), x_3) = \rho((y_1, y_2), y_3)$. Then $(x_1, (x_2, x_3)) = (y_1, (y_2, y_3))$, so $x_1 = y_1$, $x_2 = y_2$, $x_3 = y_3$. Thus $((x_1, x_2), x_3) = ((y_1, y_2), y_3)$, and ρ is injective.

Let $(u, (v, w)) \in U \times (V \times W)$. Then $\rho((u, v), w) = (u, (v, w))$. Thus ρ is surjective. ∎

Problem 9

Prove that $\mathbb{R}^2 \times \mathbb{R}^3 \cong \mathbb{R}^5$.

Proof. Consider the mapping

$$\rho : \mathbb{R}^2 \times \mathbb{R}^3 \longrightarrow \mathbb{R}^5, \quad \rho(((x_1, x_2), (x_3, x_4, x_5))) = (x_1, x_2, x_3, x_4, x_5)$$

Suppose $\rho(((x_1, x_2), (x_3, x_4, x_5))) = \rho(((y_1, y_2), (y_3, y_4, y_5)))$. Then $(x_1, x_2) = (y_1, y_2)$ and $(x_3, x_4, x_5) = (y_3, y_4, y_5)$. Thus $((x_1, x_2), (x_3, x_4, x_5)) = ((y_1, y_2), (y_3, y_4, y_5))$, and ρ is injective.

Let $(x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5$. Then $\rho(((x_2, x_2), (x_3, x_4, x_5))) = (x_1, x_2, x_3, x_4, x_5)$. Thus ρ is surjective. ∎

Problem 10

With notations as in Exercie 8, prove that $V \times W \cong W \times V$.

Proof. Consider the mapping

$$\rho : V \times W \longrightarrow W \times V, \quad \rho((v, w)) = (w, v).$$

Suppose $\rho((v_1, w_1)) = \rho((v_2, w_2))$. Then $(w_1, v_1) = (w_2, v_2)$, so $v_1 = v_2$ and $w_1 = w_2$. Thus $(v_1, w_1) = (v_2, w_2)$, and ρ is injective.

Let $(w, v) \in W \times V$. Then $\rho((v, w)) = (w, v)$. Thus ρ is surjective. ∎

Problem 11

With notations as in Example 2.4.2, prove that $(x_1, x_2) \mapsto (x_1, x_2 + x_2, 0)$ is an isomorphism $\mathbb{R}^2 \mapsto W$.

Proof. Consider the mapping

$$\rho : \mathbb{R}^2 \longrightarrow W, \quad \rho((x_1, x_2)) = (x_1, x_2, 0).$$

Suppose $\rho((x_1, x_2)) = \rho((y_1, y_2))$. Then $(x_1, x_2, 0) = (y_1, y_2, 0)$, so $x_1 = y_1$ and $x_2 = y_2$. Thus $(x_1, x_2) = (y_1, y_2)$, and ρ is injective.

Let $(x_1, x_2, 0) \in W$. Then $\rho((x_1, x_2)) = (x_1, x_2, 0)$. Thus ρ is surjective. ∎

2.5 Equivalence Relations and Quotient Sets

Problem 2

Let X be the set of all nonzero vectors in $\mathbb{R}^n$, that is, $X = \mathbb{R}^n - \{\theta\}$. For x, y in X, write $x \sim y$ if x is proportional to y, that is, if there exists a scalar c (necessarily nonzero) such that $x = cy$; this is an equivalence relation in X.

Proof. Notice that $x = 1x$, thus $x \sim x$.

Suppose $x \sim y$. Then $x = cy$ for some scalar c. Since $c \neq 0$, we have $\frac{1}{c}x = y$. Thus $y \sim x$.

Suppose $x \sim y$ and $y \sim z$. Then $x = c_1 y$ and $y = c_2 z$ for some scalars c_1, c_2. Then $x = c_1 y = c_1(c_2 z) = (c_1 c_2)z$. Thus $x \sim z$. ∎

Problem 4

If λ is a nonempty set of vector spaces over the field F, then the relation of isomorphism (2.4.1) is an equivalence relation in λ.

Proof. Let $x, y, z \in \lambda$. Notice that $x \sim x$ through the identity mapping.

Suppose $x \sim y$. Then there exists an isomorphism ρ such that $\rho(x) = y$. But then $x = \rho^{-1}(y)$, thus $y \sim x$.

Suppose $x \sim y$ and $y \sim z$. Then there exist ρ_1, ρ_2 such that $\rho_1(x) = y$ and $\rho_2(y) = z$. Then $z = \rho_2(\rho_1(x))$ Thus $x \sim z$. ∎

Problem 9

Let $f : X \longrightarrow Y$ be any mapping, $x \sim x'$ the equivalence relation defined by $f(x) = f(x')$ (2.5.1), $q : X \longrightarrow X/\sim$ the quotient mapping (2.5.17).
Define a mapping $g : X/\sim \longrightarrow f(X)$ as follows : if $u \in X/\sim$ define $g(u) = f(x)$, where x is any element of X such that $u = q(x)$. [If also $u = q(x')$ then $x \sim x'$, so $f(x) = f(x')$; thus $g(u)$ depends only on u not on the particular x chosen to represent the class u.] Prove (cf. Fig. 10):

1. g is bijective;
2. $f = i \circ g \circ q$, where $i : f(X) \longrightarrow Y$ is the insertion mapping (Appendix A.3.8).

In particular, if $f : X \longrightarrow Y$ is surjective, then $g : X/\sim \longrightarrow Y$ is bijective.

Proof. Suppose $u, u' \in X/\sim$ such that $g(u) = g(u')$. By definition, $g(u) = f(x)$ and $g(u') = f(x')$ for some $x, x' \in X$ with $u = q(x)$ and $u' = q(x')$. Then $f(x) = f(x')$, so $x \sim x'$. Thus $q(x) = q(x')$, and therefore $u = u'$. Thus g is injective.

Let y be an arbitrary element of $f(X)$. Then $y = f(x)$ for some $x \in X$. Let $u = q(x) \in X/\sim$. By definition of g, we have $g(u) = f(x) = y$. Thus g is surjective. ∎

Proof. Let $x \in X$. Then $q(x) \in X/\sim$ and $g(q(x)) = f(x)$ by definition of g. Since $i : f(X) \to Y$ is the insertion mapping, $i(f(x)) = f(x)$. Thus $(i \circ g \circ q)(x) = f(x)$ so $f = i \circ g \circ q$.

Suppose $f : X \longrightarrow Y$ is surjective. Then $f(X) = Y$, so $g : X/\sim \longrightarrow Y$.

Suppose $u, u' \in X/\sim$ such that $g(u) = g(u')$. Then $g(u) = f(x)$ and $g(u') = f(x')$ for some $x, x' \in X$. Thus $f(x) = f(x')$, so $x \sim x'$ and thus $u = u'$. Thus g is injective.

Let $y \in Y$. Since f is surjective there exists $x \in X$ such that $f(x) = y$. Let $u = q(x) \in X/\sim$. Then $g(u) = f(x) = y$. Thus g is surjective. ∎

Problem 10

For x, y in $\mathbb{R}$, write $x \sim y$ if $x \sim y$ is an integer (that is, $x - y \in \mathbb{Z}$); this is an equivalence relation in $\mathbb{R}$. The quotient set is denoted $\mathbb{R}/\mathbb{Z}$; the equivalence class $[x]$ of $x \in \mathbb{R}$ is the set $x + \mathbb{Z} = \{x + n \mid n \in \mathbb{Z}\}$.

1. There is a connection between functions defined on $\mathbb{R}/\mathbb{Z}$ and functions defined on $\mathbb{R}$ that are periodic of period 1; can you make it precise?
2. There exists a bijection $g : \mathbb{R}/\mathbb{Z} \longrightarrow U$, where $U = \{(a, b) \in \mathbb{R}^2 \mid a^2 + b^2 = 1\}$ is the 'unit circle' in $\mathbb{R}^2$. [Hint: Apply Exercise 9 to the function $f : \mathbb{R} \longrightarrow U$ defined by $f(t) = (\cos 2\pi t, \sin 2\pi t)$.]

Solution: Consider a periodic function of period 1, say $f : \mathbb{R} \longrightarrow \mathbb{R}$. Then for all $x \in \mathbb{R}$ and all $n \in \mathbb{Z}$,

$$f(x + n) = f(x).$$

Thus f takes the same value on all elements of the equivalence class

$$[x] = x + \mathbb{Z}.$$

Conversely, any function defined on $\mathbb{R}/\mathbb{Z}$ determines a function on $\mathbb{R}$ that is periodic of period 1 by composition with the quotient map.

Proof. Consider the function

$$f : \mathbb{R} \longrightarrow U, \qquad f(t) = (\cos 2\pi t, \sin 2\pi t).$$

Notice that $f(t + n) = f(t)$ for all $n \in \mathbb{Z}$, so f is constant on equivalence classes of $\mathbb{R}/\mathbb{Z}$. By Exercise 9 there is a map

$$g : \mathbb{R}/\mathbb{Z} \longrightarrow U, \qquad g([t]) = f(t).$$

Suppose $g([t]) = g([s])$. Then $f(t) = f(s)$ so $t - s \in \mathbb{Z}$. Thus $[t] = [s]$. Suppose $(a, b) \in U$. There exists $t \in [0, 1)$ such that $(\cos 2\pi t, \sin 2\pi t) = (a, b)$, so $(a, b) = g([t])$. ∎

Problem 11

For x, y in $\mathbb{R}$, write $x \sim y$ if $x - y$ is an integral multiple of 2π (that is, $x - y \in 2\pi\mathbb{Z}$). With an eye on Exercise 10, discuss periodic functions of period 2π and describe a bijection of $\mathbb{R}/2\pi\mathbb{Z}$ onto the unit circle U.

Solution: The equivalence class of $x \in \mathbb{R}$ in $\mathbb{R}/2\pi\mathbb{Z}$ is $[x] = \{x+2\pi n \mid n \in \mathbb{Z}\}$. A function $f : \mathbb{R} \longrightarrow \mathbb{R}$ is periodic of period 2π if $f(x+2\pi) = f(x)$ for all $x \in \mathbb{R}$. A bijection of $\mathbb{R}/2\pi\mathbb{Z}$ onto the unit circle $U = \{(a,b) \in \mathbb{R}^2 \mid a^2+b^2 = 1\}$ can be defined as

$$g : \mathbb{R}/2\pi\mathbb{Z} \longrightarrow U, \qquad g([x]) = (\cos x, \sin x),$$

2.6 Quotient Vector Spaces

Problem 1

Prove that if $V = M \oplus N$ (1.6, Exercise 12) then $V/M \cong N$. [Hint: Restrict the quotient mapping $V \longrightarrow V/M$ to N (Appendix $A.3.8$, and calculate the kernel and range of the restricted mapping.)]

Proof. Suppose $V = M \oplus N$. Let $q : V \longrightarrow V/M$ be the quotient mapping. Then consider $(q \mid N) : N \longrightarrow V/M$, which is the restriction of q to N. It was shown in the proof of Theorem 2.6.1 that q is linear. Suppose $x \in N$ such that $(q \mid N)(x) = [0]$. By definition, this means $x \in M$. But $x \in N$ and $V = M \oplus N$ thus $x = 0$. Thus $(q \mid N)$ is injective. Let $[y] \in V/M$. Since $V = M \oplus N$ then $y = m+n$ with $m \in M$ and $n \in N$. Then $(q \mid N)(n) = [n] = [m+n] = [y]$. Thus $(q \mid N)$ is surjective. Therefore $q \mid N$ is bijective, and it follows that $V/M \cong N$. ∎

Problem 2

Let M and N be linear subspaces of the vector spaces V and W, respectively. Form the product vector space $V \times W$ (Definition 1.3.11). Then $M \times N$ is a linear subspace of $V \times W$ and

$$(V \times W)/(M \times N) \cong (V/M) \times (W/N).$$

Prove this by showing that

$$(x,y) + M \times N \longrightarrow (x+M, y+N),$$

defines a function $(V \times W)/(M \times N) \longrightarrow (V/M) \times (W/N)$ and that this function is linear and bijective. [The essential first step to show that if $u = (x,y) + M \times N = (x',y') + M \times N$, then $(x+M, y+N) = (x'+M, y'+N)$, that is, the proposed functional value at u depends only on u and not on the particular ordered pair $(x,y) \in u$ chosen to represent it (cf. the proof of Theorem 2.6.1).]

Proof. Let $q : (V \times W)/(M \times N) \longrightarrow (V/M) \times (W/N)$ be a mapping defined as

$$q((x,y) + M \times N) = (x+M, y+N).$$

We first show that q is well defined. Suppose $u = (x,y) + M \times N = (x',y') + M \times N$. Then

$$(x,y) - (x',y') \in M \times N.$$

Thus $(x-x', y-y') = (m,n)$ for some $m \in M$ and $n \in N$. Then $x-x' \in M$ and $y-y' \in N$ thus $x+M = x'+M$ and $y+N = y'+N$. Thus $(x+M, y+N) = (x'+M, y'+N)$.

We now show q is additive. Let $(x,y), (x',y')$ be elements in $V \times W$. Then

$$q((x,y)+(x',y')) = q((x+x', y+y')) = (x+x'+M, y+y'+N) = (x+M, y+N)+(x'+M, y'+N) = q((x,y))+q((x',y')).$$

We now show q is homogeneous.

$$q(c(x,y)) = q((cx,cy)) = (cx+M, cy+N) = c(x+M, y+N) = cq(x,y).$$

We now show q is injective. Suppose $q((x,y)) = (0+M, 0+N)$. Then $x \in M$ and $y \in N$. Thus $(x,y) \in M \times N$ so $(x,y) + M \times N = M \times N$. Therefore $(x,y) = 0$ in $(V \times W)/(M \times N)$, and q is injective.

We now show q is surjective. Let $(x+M, y+N)$ be an element in $(V/M)\times(W/N)$. Then $q((x,y)) = (x+M, y+N)$, thus q is surjective. ∎

Problem 3

Let $f : \mathbb{R}^3 \longrightarrow \mathbb{R}$ be the linear form $f(x) = 2x_1 - 3x_2 + x_3$ and let M be its kernel. As in Example 2.5.9, think of V/M as the set of planes consisting of M and the planes parallel to it. If u and v are two of these planes, their sum $u + v$ in V/M can be visualized as follows: choose a point P on the plane u and a point Q on the plane v; add the arrows OP and OQ using the parallelogram rule, obtaining an arrow OR; the plane through R parallel to M is $u + v$.

Solution: OK.

Problem 4

Let M be a linear subspace of V, $u = x + M$ and $v = y + M$ two elements of V/M.

1. Let $A = \{s + t \mid s \in u, t \in v\}$ be the sum of the sets $x + M$ and $y + M$ in the sense of Definition 1.6.9. Then A is also a coset, namely $A = x + y + M$. [This offers a definition of $u + v$ that is obviously independent of the vectors x and y chosen to represent u and v.]
2. If c is a nonzero scalar, then the set $B = \{cs \mid s \in u\}$ is a coset, namely $B = cx + M$. What if $c = 0$?

Proof. Let a be an arbitrary element in A such that $a = s + t$. Suppose $s = x + m$ and $t = y + m'$ where $m, m' \in M$. Then taking sums we have $s + t = x + y + m + m'$. Now $m + m' \in M$ since M is closed under addition. Thus $x + y + m + m' \in x + y + M$.

Conversely, suppose $t \in x + y + M$ such that $t = x + 0 + y + m$. Then $0, m \in M$ thus $x + 0 \in x + M$ and $y + m \in y + M$ thus $t \in A$. ∎

Proof. Suppose c is a nonzero scalar. Let b be an arbitrary element in B such that $b = cs$ where $s \in u$. Suppose $s = x + m$ where $m \in M$. Then $b = cs = c(x + m) = cx + cm$. Now $cm \in M$ thus $b = cx + cm \in cx + M$.

Conversely, suppose t is an arbitrary element in $cx + M$ such that $t = cx + m$ where $m \in M$. Since $m \in M$ and $c \neq 0$ we have $\frac{1}{c} \cdot m = \frac{m}{c} \in M$, so $t = cx + m \iff t = c(x + \frac{m}{c})$. Now $x + \frac{m}{c} \in u$ so $t \in B$ as required. ∎

Solution: If $c = 0$ then B is the set $\{0\}$.

Problem 5

Let V be a vector space, M and N linear subspaces of V, and x, y vectors in V. Prove:

1. $x + M \subset y + N$ if and only if $M \subset N$ and $x - y \in N$.
2. $(x + M) \cap (y + N) \neq \emptyset$ if and only if $x - y \in M + N$.
3. If $z \in (x + M) \cap (y + N)$ then $(x + M) \cap (y + N) = z + M \cap N$.

Proof. ($\longrightarrow$) Suppose $x + M \subset y + N$. Let m be an arbitrary element in M. Since $x + m \in x + M$ and $x + M \subset y + N$ we have $x + m = y + n$ for some $n \in N$. Now for arbitrary $m \in M$, we have $x + m = y + n$ for some $n \in N$, so $m = n - (x - y)$. Since $n \in N$ and $x - y \in N$ we have $m \in N$. Thus $M \subset N$ as required.

($\longleftarrow$) Suppose $M \subset N$ and $x - y \in N$. Let $t \in x + M$ such that $t = x + m$ for some $m \in M$. Since $M \subset N, x - y \in N$, we have $m \in N$ thus $t = y + ((x - y) + m) \in y + N$. Therefore $x + M \subset y + N$. ∎

Proof. ($\longrightarrow$) Suppose $(x + M) \cap (y + N) \neq \emptyset$. Let $t \in (x + M) \cap (y + N)$ such that $t = x + m$ and $t = y + n$ for $m \in M$ and $n \in N$. Then $x + m = y + n \iff x - y = n - m \in N - M = M + N$.

($\longleftarrow$) Suppose $x - y \in M + N$. Then $x - y = m + n$ for some $m \in M$ and $n \in N$. Then $x - n = y + m$. Then $x - n \in x + N$ and $y + m \in y + M$ thus $x + N \cap y + M \neq \emptyset$. ∎

Proof. Suppose $z \in (x+M) \cap (y+N)$. Then $z = x+m$ and $z = y+n$ for $m \in M$ and $n \in N$. Let t be an arbitrary element in $(x+M) \cap (y+N)$. Then $t = x+m'$ and $t = y+n'$ for some $m' \in M$ and $n' \in N$. Then

$$t - z = (x+m') - (x+m) \in M,$$

and

$$t - z = (y+n') - (y+n) \in N.$$

Thus $t - z \in M \cap N$. Thus $t \in z + M \cap N$. Conversely, let $t \in z + M \cap N$. Then $t = z + m$ where $m \in M \cap N$. Thus $t \in x + M$ and $t \in x + N$ and therefore $t \in (x+M) \cap (y+N)$. ∎

Problem 6

Let X be a nonempty set. Let R be a subset of the cartesian product $X \times X$, such that (i) R contains the 'diagnol' $X \times X$, that is, $(x,x) \in R$ for all $x \in X$; (ii) R is 'symmetric in the diagnol', that is, if $(x,y) \in R$ then $(y,x) \in R$; and (iii) if $(x,y) \in R$ and $(y,z) \in R$ then $(x,z) \in R$. Does this suggest a way of defining an equivalence relation $x \sim y$ in X.

Solution:

$$x \sim y \iff (x,y) \in R$$

Problem 7

Let $T : V \longrightarrow W$ be a linear mapping, M a linear subspace of V such that $M \subset KerT$, and $Q : V \longrightarrow V/M$ the quotient mapping. Prove:

1. There exists a (unique) linear mapping $S : V/M \longrightarrow W$ such that $T = SQ$ (Fig. 11).
2. S is injective if and only if $M = KerT$.

Proof. Consider $S : V/M \longrightarrow W$ defined

$$S(x+M) = T(x).$$

We first show S is well defined. Suppose $x + M = y + M$. Therefore $x - y \in M$. Since $M \subset KerT$, we have $T(x-y) = 0 \implies T(x) = T(y)$. Then let $x+M, y+M$ be arbitrary elements in V/M. Then

$$S((x+M)+(y+M)) = S(x+y+M) = T(x+y) = T(x) + T(y) = S(x+M) + S(y+M).$$

Let c be a scalar. Then

$$S(c(x+M)) = S(cx+M) = T(cx) = cT(x).$$

Thus S is a linear mapping. Now let v be an arbitrary element in V. Then $(SQ)(v) = S(Q(v)) = S(v+M) = T(v)$.

Suppose there exists S' such that $S' : V/M \longrightarrow W$ and $T = S'Q$. For all $v \in V$ we have $T(v) = S'(Q(v)) = S(Q(v))$. Thus S, S' agree at all cosets and therefore $S' = S$ and S is unique. ∎

Proof. Suppose S is injective. We know $M \subset KerT$ so let t be an arbitrary element in $KerT$. Then $T(t) = (SQ)(t) = S(t+M) = 0$. Since S is injective we have $t + M = 0 + M$ thus $t \in M$. Thus $M = KerT$.

Suppose $M = KerT$. Let $x + M$ be an arbitrary element in V/M. Suppose $S(x+M) = 0$. Then $T(x) = 0$ thus $x \in M$ and S is injective. ∎

Problem 8

Let M be a linear subspace of $V, Q : V \longrightarrow V/M$ the quotient mapping, $T : V \longrightarrow V$ a linear mapping such that $T(M) \subset M$. Prove that there exists a (unique) linear mapping: $S : V/M \longrightarrow V/M$ such that $SQ = QT$ (Fig. 12). [Hint: Apply Exercise 7 to the linear mapping $QT : V \longrightarrow V/M$.]

Proof. Let v be an arbitrary element in M. Since $T(M) \subset M$, we have $T(v) \in M$. Then $QT(v) = Q(T(v)) = 0$. Thus $M \subset Ker(QT)$. We can then apply Exercise 7 to find a linear mapping $S : V/M \longrightarrow V/M$ such that $SQ = QT$. ∎

2.7 The First Isomorphism Theorem

Problem 1

When there's a first, there's a second. The following results is called the Second isomorphism theorem. Let V be a vector space, M and N linear subpsaces of V, $M \cap N$ their intersection and $M + N$ their sum (1.6.9). Then $M \cap N$ is a linear subspace of M, N is a linear subspace of $M + N$, and

$$M/(M \cap N) \cong (M + N)/N.$$

[Hint: Let $Q : V \longrightarrow V/N$ be the quotient mapping and let $T = Q \mid M$ be the restriction of Q to M (Appendix 3.8), that is, $T : M \longrightarrow V/N$ and $Tx = x + N$ for all $x \in M$. Show that T has range $(M + N)/N$ and kernel $M \cap N$.]

Proof. Let $Q : V \longrightarrow V/N$ be the quotient mapping and let $T = Q \mid M$ be the restriction of Q to M, that is, $T : M \longrightarrow V/N$ and $Tx = x + N$ for all $x \in M$.

We first show that $Ran(T) = (M + N)/N$. Notice

$$Ran(T) = T(M) = \{m + N : m \in M\}.$$

$\underline{Ran(T) \subset (M + N)/N}$: Let $x \in Ran(T)$. Then $x = m + N$ for some $m \in M$. Since $m \in M \subset M + N$, we have $m + N \in (M + N)/N$. Thus $Ran(T) \subset (M + N)/N$.

$\underline{(M + N)/N \subset Ran(T)}$: Let $x \in (M+N)/N$. Then $x = (m+n)+N$ for some $m \in M, n \in N$. But $(m+n)+N = m + N = Tm$, so $x \in Ran(T)$.

Thus

$$Ran(T) = (M + N)/N.$$

We now show that $Ker(T) = M \cap N$.

$\underline{Ker(T) \subset M \cap N}$: Suppose $x \in Ker(T)$. Then $Tx = x + N = N$, so $x \in N$. Since $x \in M$, we have $x \in M \cap N$.

$\underline{M \cap N \subset Ker(T)}$: Suppose $x \in M \cap N$. Then $x \in N$, so $Tx = x + N = N$. Thus $x \in Ker(T)$.

Thus $Ker(T) = M \cap N$.

Therefore, by the First Isomorphism Theorem,

$$M/(M \cap N) \cong Ran(T) = (M + N)/N.$$

∎

Problem 2

Let M, N be linear subspaces of the vector spaces V, W and let

$$T : V \times W \longrightarrow (V/M) \times (W/N),$$

be the linear mapping defined by $T(x, y) = (x + M, y + N)$. Determine the kernel of T and deduce another proof of 2.6, Exercise 2.

2.6 Problem 2

Let M and N be linear subspaces of the vector spaces V and W, respectively. Form the product vector space $V \times W$ (Definition 1.3.11). Then $M \times N$ is a linear subspace of $V \times W$ and

$$(V \times W)/(M \times N) \cong (V/M) \times (W/N).$$

Prove this by showing that

$$(x, y) + M \times N \longrightarrow (x + M, y + N),$$

defines a function $(V \times W)/(M \times N) \longrightarrow (V/M) \times (W/N)$ and that this function is linear and bijective. [The essential first step to show that if $u = (x, y) + M \times N = (x', y') + M \times N$, then $(x + M, y + N) = (x' + M, y' + N)$, that is, the proposed functional value at u depends only on u and not on the particular ordered pair $(x, y) \in u$ chosen to represent it (cf. the proof of Theorem 2.6.1).]

Proof. We must find $(v, w) \in V \times W$ such that

$$T(v, w) = (0 + M, 0 + N) \in (V/M) \times (W/N).$$

This holds iff $v + M = M$ and $w + N = N$ so $v \in M$ and $w \in N$. Thus

$$\textbf{ker}(T) = M \times N.$$

By the First Isomorphism Theorem

$$(V \times W)/(M \times N) \cong \textbf{Ran}(T) = (V/M) \times (W/N).$$

∎

Problem 3

Let V be a vector space over F, $f : V \longrightarrow F$ a linear form on V (Definition 2.1.11), N the kernel of f. Prove that if f is not identically zero, then $V/N \cong F$. [Hint: If $f(z) \neq 0$ and c is any scalar, evaluate f at the vector $(c/f(z))z$.]

Proof. Suppose f is not identically zero. Thus there exists $v \in V$ such that $f(v) \neq 0$. Suppose $f(v) = t$ where $t \in F$ and let c be an arbitrary element in F. We first show f is surjective. Let $c' = \frac{c}{t}$. Then

$$f(c'v) = c'f(v) = \frac{c}{t} \cdot t = c.$$

Thus f is surjective. Now $\textbf{ker}(f) = N$ by definition. Thus, by the First Isomorphism Theorem

$$V/N \cong Ran(f) = F.$$

∎

Problem 4

Let V be a vector space, $V \times V$ the product vector space (1.3.11), and $\nabla = \{(x, x) \mid x \in V\}$ (called the diagnol of $V \times V$). Prove that ∇ is a linear subspace of $V \times V$ and that $(V \times V)/\nabla \cong V$. [Hint: 2.2, Exercise 4.]

Proof. Firstly, $(0, 0) \in \nabla$. Let $(x, x), (y, y) \in \nabla$ and c a scalar. Then $(x, x) + (y, y) = (x + y, x + y) \in \nabla$. Also, $c(x, x) = (cx, cx) \in \nabla$. Thus ∇ is a linear subspace of $V \times V$. Define $f : V \times V \to V$ by $f(x, y) = x - y$. Then $\textbf{ker}(f) = \{(x, x) \mid x \in V\} = \nabla$. By Exercise 4 of 2.2, f is surjective. Thus by the First Isomorphism Theorem

$$(V \times V)/\nabla \cong V.$$

∎

Problem 5

Let N be the set of vectors in $\mathbb{R}^5$ whose last two coordinates are zero:

$$N = \{x_1, x_2, x_3, 0, 0\} \mid x_1, x_2, x_3 \in \mathbb{R}.$$

Prove that N is a linear subspace of $\mathbb{R}^5$, $\mathbb{R}^3 \cong N$, and $\mathbb{R}^5/N \cong \mathbb{R}^2$. [Taking some liberties with the notation, $\mathbb{R}^5/\mathbb{R}^3 = \mathbb{R}^2$.]

Proof. Firstly, $(0,0,0,0,0) \in N$. Let $(x_1, x_2, x_3, 0, 0), (y_1, y_2, y_3, 0, 0) \in N$ and c be a scalar. Then $(x_1, x_2, x_3, 0, 0) + (y_1, y_2, y_3, 0, 0) = (x_1 + y_1, x_2 + y_2, x_3 + y_3, 0, 0) \in N$. Also $c(x_1, x_2, x_3, 0, 0) = (cx_1, cx_2, cx_3, 0, 0) \in N$. Thus N is a linear subspace of $\mathbb{R}^5$.

Consider the mapping $f : \mathbb{R}^3 \longrightarrow N$ defined

$$f(x_1, x_2, x_3) = (x_1, x_2, x_3, 0, 0).$$

Let $(x_1, x_2, x_3, 0, 0) \in N$. Then $f(x_1, x_2, x_3) = (x_1, x_2, x_3, 0, 0)$, so f is surjective. Also, if $f(x_1, x_2, x_3) = (0,0,0,0,0)$ then $(x_1, x_2, x_3) = (0,0,0)$, so f is injective. Thus f is bijective, and $\mathbb{R}^3 \cong N$.

Consider the linear mapping $g : \mathbb{R}^5 \to \mathbb{R}^2$ defined by

$$g(x_1, x_2, x_3, x_4, x_5) = (x_4, x_5).$$

Then $\ker(g) = N$, and g is surjective. Thus by the First Isomorphism Theorem

$$\mathbb{R}^5/N \cong \mathbb{R}^2.$$

∎

Problem 6

Let T be a set, A a subset of T, $V = \mathcal{F}(T, \mathbb{R})$ and $W = \mathcal{F}(A, \mathbb{R})$ the vector spaces of functions constructed in 1.3.4. Let N be the set of all functions $x \in V$ such that $x(t) = 0$ for all $t \in A$. Then N is a linear subspace of V and $V/N \cong W$. [Hint: Consider the mapping $x \mapsto x \mid A, (x \in V)$, where $x \mid A$ is the restriction of the function x to A (Appendix A.3.8).]

Proof. Firstly, the zero function $0 \in V$ satisfies $0(t) = 0$ for all $t \in A$, so $0 \in N$. Let $x, y \in N$ and c be a scalar. Then for all $t \in A$

$$(x + y)(t) = x(t) + y(t) = 0 + 0 = 0,$$

and

$$(cx)(t) = cx(t) = c \cdot 0 = 0.$$

Thus $x + y \in N$ and $cx \in N$, so N is a linear subspace of V. Consider the mapping $f : V \to W$ defined by

$$f(x) = x|_A.$$

Then f is linear. Also

$$\ker(f) = \{x \in V \mid x(t) = 0 \text{ for all } t \in A\} = N.$$

Since every function in W is the restriction of some function in V, f is surjective. Thus by the First Isomorphism Theorem

$$V/N \cong W.$$

∎

3 Structure of Vector Spaces

3.1 Linear Subspace Generated by a Subset

Problem 1

If $T : V \longrightarrow W$ is linear and A is a subset of V such that $T(A)$ is generating for W, then T is surjective.

Proof. Let w be an arbitrary element in W. Since $T(A)$ generates W, w is a linear combination of elements in $T(A)$. Suppose

$$w = c_1 a_1 + \cdots + c_n a_n \text{ such that } a_1, \ldots, a_n \in T(A).$$

But $a_i = T(v_i)$ for some $v_i \in V$, so

$$c_1 a_1 + \cdots + c_n a_n = c_1 T(v_1) + \cdots + c_n T(v_n) = T(c_1 v_1) + \cdots + T(c_n v_n) = T(c_1 v_1 + \cdots + c_n v_n).$$

The last two equalities hold since T is linear. Thus T is surjective. ∎

Problem 2

If M and N are linear subspaces of V such that $V = M + N$ (1.6.9) and $T : V \longrightarrow V/N$ is the quotient mapping, then $T(M) = V/N$ but M need not equal V; thus we cannot conclude in Exercise 1 that A is a generating for V. Can you exhibit an example of such V, M and N?

Proof. Consider $V = \mathbb{R}$, $M = \{0\}$, and $N = \mathbb{R}$. Then $V = M + N$, but $M \neq V$. Let $T : V \to V/N$ be the quotient map. Since $V/N = \{0\}$, we have $T(M) = \{0\} = V/N$. ∎

Problem 3

(Substitution Principle) Suppose $x, x_1, \ldots, x_n$ are vectors such that x is a linear combination of $x_1, \ldots, x_n$ but not of $x_1, \ldots, x_{n-1}$. Prove:

$$[\{x_1, \ldots, x_n\}] = [\{x_1, \ldots, x_{n-1}, x\}].$$

Proof. Since x is a linear combination of $x_1, \ldots, x_n$ suppose

$$x = c_1 x_1 + \cdots + c_n x_n.$$

Since x is not a linear combination of $x_1, \ldots, x_{n-1}$, we have $c_n \neq 0$, so

$$x_n = \frac{1}{c_n}(x - c_1 x_1 - \cdots - c_{n-1} x_{n-1}).$$

Let $v \in [\{x_1, \ldots, x_n\}]$ and suppose

$$v = d_1 x_1 + \cdots + d_n x_n.$$

Substituting for x_n, we get

$$v = d_1 x_1 + \cdots + d_{n-1} x_{n-1} + d_n \frac{1}{c_n}(x - c_1 x_1 - \cdots - c_{n-1} x_{n-1}) \in [\{x_1, \ldots, x_{n-1}, x\}].$$

Conversely, let $v \in [\{x_1, \ldots, x_{n-1}, x\}]$ and suppose

$$v = d_1 x_1 + \cdots + d_{n-1} x_{n-1} + dx.$$

Then

$$x = c_1 x_1 + \cdots + c_n x_n \implies dx = (dc_1) x_1 + \cdots + (dc_n) x_n.$$

Thus

$$v = d_1 x_1 + \cdots + d_{n-1} x_{n-1} + (dc_1) x_1 + \cdots + (dc_n) x_n \in [\{x_1, \ldots, x_n\}].$$

∎

Problem 4

Let $S : V \longrightarrow W$ and $T : V \longrightarrow W$ be linear mappings, and let A be a subset of V such that $V = [A]$. Prove that if $Sx = Tx$ for all $x \in A$, then $S = T$. (Informally, a lienar mapping is determined by its values on a generating set.)

Proof. Suppose $Sx = Tx$ for all $x \in A$. Let $x' \in V$ and since $V = [A]$ so

$$x' = c_1 a_1 + \cdots + c_n a_n.$$

Then

$$\begin{aligned} T(x') &= T(c_1 a_1 + \cdots + c_n a_n) \\ &= T(c_1 a_1) + \cdots + T(c_n a_n) \\ &= c_1 T(a_1) + \cdots + c_n T(a_n) \\ &= c_1 S(a_1) + \cdots + c_n S(a_n) \\ &= S(c_1 a_1) + \cdots + S(c_n a_n) \\ &= S(c_1 a_1 + \cdots + c_n a_n) \\ &= S(x'). \end{aligned}$$

Thus $Sx' = Tx'$ for all $x' \in V$, so $S = T$. ∎

Problem 5

Let V be a vector space, A a nonempty subset of V. For each finite subset $\{x_1, \dots, x_n\}$ of A, let

$$M(x_1, \dots, x_n) = \{a_1 x_1 + \cdots + a_n x_n \mid a_1, \dots, a_n \text{ scalars.}\};$$

as noted in the proof of Corollary $3.1.5$, this is a linear subspace of V. Let $\mathcal{N}$ be the set of all linear subspaces of V obtained in this way (by considering all possible finite subsets of A); by definition, $[A] = \bigcup \mathcal{N}$ (the union of the subspaces belonging to $\mathcal{N}$). Prove that $[A]$ is a linear subspace of V by verifying criterion 1.6, Exercise 9.

Proof. We must show that if $M, K \in \mathcal{N}$ then there exists $P \in \mathcal{N}$ such that $M \subset P$ and $K \subset P$. Therefore suppose $M, K \in \mathcal{N}$. Then there exist finite subsets $\{x_1, \dots, x_n\} \subseteq A$ and $\{y_1, \dots, y_m\} \subseteq A$ such that

$$M = M(x_1, \dots, x_n), \quad K = M(y_1, \dots, y_m).$$

Let

$$P = M(x_1, \dots, x_n, y_1, \dots, y_m).$$

Then $P \in \mathcal{N}$, and clearly $M \subset P$ and $K \subset P$. From 1.6 Exercise 9 it follows that $\bigcup \mathcal{N}$ is a linear subspace of V. ∎

Problem 6

Let V and W be vector spaces, $V \times W$ the product vector space $(1.3.11)$; let $A \subset V$ and $B \subset W$. Prove:

1. $[A \times B] \subset [A] \times [B]$,
2. In general, $[A \times B] \neq [A] \times [B]$, Hint: Consider $A = \emptyset, B = W$; or $V = W = R$ and $A = B = \{1\}$.
3. $[(A \times \{\emptyset\}) \cup (\{\emptyset\} \times B)] = [A] \times [B]$. Hint

$$\left(\sum_{i=1}^{m} c_i (x_i, 0) + \sum_{j=1}^{n} d_j (0, y_j) \right).$$

Proof. Let (x, y) be an arbitrary element in $[A \times B]$. Then

$$(x, y) = c_1(a_1, b_1) + \cdots + c_n(a_n, b_n)$$

for some $(a_i, b_i) \in A \times B$. But then

$$(x, y) = (c_1 a_1 + \cdots + c_n a_n,\ c_1 b_1 + \cdots + c_n b_n).$$

Thus

$$x = c_1 a_1 + \cdots + c_n a_n \in [A], \quad y = c_1 b_1 + \cdots + c_n b_n \in [B],$$

so $(x, y) \in [A] \times [B]$. ∎

Proof. Let $V = W = \mathbb{R}$ and $A = B = \{1\}$. Then

$$A \times B = \{(1, 1)\},$$

so

$$[A \times B] = \{c(1, 1) \mid c \in \mathbb{R}\} = \{(c, c) \mid c \in \mathbb{R}\}.$$

Conversely

$$[A] = \mathbb{R}, \quad [B] = \mathbb{R},$$

so

$$[A] \times [B] = \mathbb{R} \times \mathbb{R}.$$

Since

$$\{(c, c) \mid c \in \mathbb{R}\} \neq \mathbb{R} \times \mathbb{R},$$

it follows that

$$[A \times B] \neq [A] \times [B].$$

∎

Proof. Let $(x, y) \in [(A \times \{0\}) \cup (\{0\} \times B)]$. Then (x, y) is a linear combination of elements of the form $(a, 0)$ and $(0, b)$ so

$$(x, y) = \sum_{i=1}^{m} c_i(x_i, 0) + \sum_{j=1}^{n} d_j(0, y_j),$$

where $x_i \in A$ and $y_j \in B$. Then

$$(x, y) = \left(\sum_{i=1}^{m} c_i x_i,\ \sum_{j=1}^{n} d_j y_j \right),$$

so $x \in [A]$ and $y \in [B]$. Thus

$$[(A \times \{0\}) \cup (\{0\} \times B)] \subset [A] \times [B].$$

Conversely let $(x, y) \in [A] \times [B]$. Then

$$x = \sum_{i=1}^{m} c_i x_i, \quad y = \sum_{j=1}^{n} d_j y_j,$$

with $x_i \in A$, $y_j \in B$. Then

$$(x, y) = \sum_{i=1}^{m} c_i(x_i, 0) + \sum_{j=1}^{n} d_j(0, y_j),$$

which shows $(x, y) \in [(A \times \{0\}) \cup (\{0\} \times B)]$. Thus

$$[(A \times \{0\}) \cup (\{0\} \times B)] = [A] \times [B].$$

∎

3.2 Linear Dependence

Problem 1

Let $V = \mathcal{F}(\mathbb{R})$ be the vector space of all real-valued functions of a real variable (Example 1.3.4). Define $f : \mathbb{R} \longrightarrow \mathbb{R}$ by $f(t) = \sin(t + \pi/6)$. Prove that the function f, sin, cos are linearly dependent.

Proof. Simply note

$$\begin{aligned} f(t) &= \sin(t + \pi/6) \\ &= \sin(t)\cos(\pi/6) + \cos(t)\sin(\pi/6) \\ &= \frac{\sqrt{3}}{2}\sin(t) + \frac{1}{2}\cos(t). \end{aligned}$$

∎

Problem 2

In the vector space of real polynomial function (Example 1.3.5), show that the function p_1, p_2, p_3, p_4 defined by $p_1(t) = t^2 + t + 1$, $p_2(t) = t^2 + 5$, $p_3(t) = 2t + 1$, $p_4(t) = 3t^2 + 5t$ are linearly dependent (cf. 1.5.3).

Proof. Simply note

$$\begin{aligned} -35p_1(t) + 8p_2(t) - 5p_3(t) + 9p_4(t) &= -35(t^2 + t + 1) + 8(t^2 + 5) - 5(2t + 1) + 9(3t^2 + 5t) \\ &= (-35t^2 - 35t - 35) + (8t^2 + 40) + (-10t - 5) + (27t^2 + 45t) \\ &= 0. \end{aligned}$$

∎

Problem 3

Show that in $\mathbb{R}^3$ the vectors e_1, e_2, e_3, $(3, -2, 4)$ are linearly dependent (cf. 1.5.3).

Proof. Simply note

$$(3, -2, 4) = 3e_1 + -2e_2 + 4e_3.$$

∎

Problem 5

If $T : V \longrightarrow W$ is a linear mapping and $x_1, \dots, x_n$ are linearly dependent vectors in V, then the vectors $Tx_1, \dots, Tx_n$ of W are linearly dependent.

Proof. Since $x_1, \dots, x_n$ are linearly dependent, there exist scalars $c_1, \dots, c_n$, not all zero, such that

$$c_1x_1 + \cdots + c_nx_n = 0 \implies T(c_1x_1 + \cdots + c_nx_n) = T(0) = c_1T(x_1) + \cdots + c_nT(x_n) = 0.$$

∎

Problem 6

A finite list of vectors $x_1, \dots, x_n$ is linearly dependent if and only if there exists an index j such that $x_j \in [\{x_i \mid i \neq j\}]$ (cf. 3.1.5).

Proof. (⟶) Suppose a finite list of vectors $x_1, \dots, x_n$ is linearly dependent. We have $c_1, \dots, c_n$ not all zero such that

$$c_1 x_1 + \cdots + c_n x_n = 0.$$

Wlog suppose $c_j \neq 0$ for some $1 \leq j \leq n$. Then

$$x_j = \frac{-1}{c_j}(c_1 x_1 + \cdots + c_{j-1} x_{j-1} + c_{j+1} x_{j+1} + \cdots + c_n x_n) \in [\{x_i \mid i \neq j\}].$$

(⟵) Suppose there exists an index j such that $x_j \in [\{x_i \mid i \neq j\}]$. Then we have

$$x_j = c_1 x_1 + \cdots + c_{j-1} x_{j-1} + c_{j+1} x_{j+1} + \cdots + c_n x_n.$$

Thus

$$-x_j + c_1 x_1 + \cdots + c_{j-1} x_{j-1} + c_{j+1} x_{j+1} + \cdots + c_n x_n = 0.$$

Thus $x_1, \dots, x_n$ is linearly dependent. ∎

Problem 7

If $T : V \longrightarrow W$ is an injective linear mapping and if $x_1, \dots, x_n$ are vectors in V such that $Tx_1, \dots, Tx_n$ are linearly dependent in W, then $x_1, \dots, x_n$ are linearly dependent in V.

Proof. Suppose $T : V \longrightarrow W$ is an injective linear mapping and $Tx_1, \dots, Tx_n$ are linearly dependent in W. We have scalars $c_1, \dots, c_n$ not all zero such that

$$c_1 T x_1 + \cdots + c_n T x_n = T(c_1 x_1 + \cdots + c_n x_n) = 0.$$

Since T is injective we have

$$c_1 x_1 + \cdots + c_n x_n = 0.$$

Thus $x_1, \dots, x_n$ are linearly dependent. ∎

Problem 8

Let V be a vector space. A subset A of V is said to be linearly dependent if there exists a finite list of distinct vectors $x_1, \dots, x_n$ in A that is linearly dependent in the sense of Definition 3.2.1. Prove:

1. A subset of A of V is linearly dependent if and only if some vector in A is the linear span of the remaining vectors of A, that is, there exists $x \in A$ such that $x \in [A - \{x\}]$.
2. If $A \subset B \subset V$ and A is linearly dependent, then B is also linearly dependent.

Proof. This just follows from Problem 6. ∎

Proof. Since B is linearly dependent there exists scalars $c_1, \dots, c_n$ not all zero such that

$$c_1 x_1 + \cdots + c_n x_n = 0.$$

Since $A \subset B$ we have $x_1, \dots, x_n \in B$ thus B is linearly dependent. ∎

3.3 Linear Independence

Problem 1

Show that the vectors $(2,-1,1),(-1,2,1),(1,1,1)$ in $\mathbb{R}^3$ are linearly independent.

Proof. Suppose there exist scalars not all zero c_1, c_2, c_3 such that

$$c_1(2,-1,1)+c_2(-1,2,1)+c_3(1,1,1)=0.$$

Wlog suppose $c_1 \neq 0$ so we have

$$(2,-1,1)=-\frac{c_2}{c_1}(-1,2,1)-\frac{c_3}{c_1}(1,1,1).$$

From this we get the following equations

1. $2=\frac{c_2}{c_1}-\frac{c_3}{c_1}$
2. $-1=-2\frac{c_2}{c_1}-\frac{c_3}{c_1}$
3. $1=-\frac{c_2}{c_1}-\frac{c_3}{c_1}$

Let $a=\frac{c_2}{c_1}$ and $b=\frac{c_3}{c_1}$. Then the system becomes

$$\begin{aligned} 2&=a-b\\ -1&=-2a-b\\ 1&=-a-b. \end{aligned}$$

From the first equation, $a=2+b$. Substituting into the third gives

$$1=-(2+b)-b=-2-2b,$$

so $b=-\frac{3}{2}$ and thus $a=\frac{1}{2}$. Substituting into the second equation

$$-1=-2\left(\frac{1}{2}\right)-\left(-\frac{3}{2}\right)=-1+\frac{3}{2}=\frac{1}{2},$$

a contradiction. ∎

Problem 2

If $x_1,\dots,x_n,x_{n+1}$ are linearly independent, then so are $x_1,\dots,x_n$.

Proof. Suppose $x_1,\dots,x_n,x_{n+1}$ are linearly independent. For contradiction, suppose $x_1,\dots,x_n$ are dependent. Then for scalars $c_1,\dots,c_n$ not all zero we have

$$\sum_{i=1}^{n} c_i x_i = 0 \text{ so } \sum_{i=1}^{n} c_i x_i + 0\cdot x_{n+1} = 0.$$

Contradicting that $x_1,\dots,x_n,x_{n+1}$ are linearly independent. ∎

Problem 3

If $x_1,\dots,x_n$ are linearly independent, then the following implication is true: (*) $a_1x_1+\cdots+a_nx_n=b_1x_1+\cdots+b_nx_n \implies a_i=b_i$ for all i. Conversely, the validity of (*) implies that $x_1,\dots,x_n$ are lienarly independent.

Proof. ($\underline{\longrightarrow}$) Suppose $x_1, \ldots, x_n$ are linearly independent. Suppose

$$\sum a_i x_i = \sum b_i x_i.$$

Then

$$\sum (a_i - b_i) x_i = 0.$$

Since the vectors are linearly independent it follows that

$$a_i - b_i = 0 \text{ for all } i,$$

thus $a_i = b_i$ for all i.

($\underline{\longleftarrow}$) Suppose (*) holds. Now suppose

$$\sum a_i x_i = 0.$$

Then

$$\sum a_i x_i = \sum 0 \cdot x_i.$$

By (*) it follows that $a_i = 0$ for all i. Thus $x_1, \ldots, x_n$ are linearly independent. ∎

Problem 6

If $T : V \longrightarrow W$ is a linear mapping and $x_1, \ldots, x_n$ are vectors in V such that $Tx_1, \ldots, Tx_n$ are linearly independent, then $x_1, \ldots, x_n$ are linearly independent.

Proof. This is exactly Problem 5 in the previous section written in its contrapositive form. ∎

Problem 9

In the vector space $\mathcal{F}(\mathbb{R})$ of all functions $\mathbb{R} \longrightarrow \mathbb{R}$ (Example 1.3.4), sin and cos are linearly independent; so are the functions sin, cos, p where p is any nonzero polynomial function (Hint: Differentiate.); so are sin, cos, p, exp (the base e exponential function).

Proof. Suppose sin and cos are linearly dependent thus there exists scalars c_1, c_2 not both zero such that

$$c_1 \cos(x) + c_2 \sin(x) = 0.$$

At $x = 0$ we have

$$c_1 \cos(0) + c_2 \sin(0) = 0 \implies c_1 = 0.$$

But then at $x = \pi/2$ we have

$$c_1 \cos(\pi/2) + c_2 \sin(\pi/2) = 0 \implies c_2 = 0.$$

Thus contradicting that c_1, c_2 are not both zero.

Suppose sin, cos, and p are linearly dependent where p is any nonzero polynomial function. There exists scalars c_1, c_2, c_3 not all zero such that

$$c_1 \cos(x) + c_2 \sin(x) + c_3 p(x) = 0.$$

If $c_3 = 0$ then we have the previous case. Therefore suppose $c_3 \neq 0$ and we find

$$p(x) = -\frac{c_1}{c_3} \cos(x) - \frac{c_2}{c_3} \sin(x).$$

If p is not constant, the rhs is bounded, while a nonzero polynomial is unbounded, giving a contradiction. If p is constant, then the rhs would have to be constant, but a nontrivial linear combination of sin and cos is not constant, also giving a contradiction.

Suppose sin, cos, exp and p are linearly dependent where p is any nonzero polynomial function. There exists scalars c_1, c_2, c_3, c_4 not all zero such that

$$c_1 \cos(x) + c_2 \sin(x) + c_3 p(x) + c_4 e^x = 0.$$

If $c_4 = 0$ we have the previous case. Therefore suppose $c_4 \neq 0$. Let p have degree n. Differentiating $n+1$ times gives

$$A \cos(x) + B \sin(x) + C e^x = 0$$

for some constants A, B, C with $C \neq 0$. Solving for e^x we obtain

$$e^x = -\frac{A}{C} \cos(x) - \frac{B}{C} \sin(x).$$

The rhs is bounded, while e^x is unbounded as $x \to \infty$, giving a contradiction. ∎

Problem 12

Let $V = \mathbb{C}^2$ and regard V as a real vector space by restriction of scalars (1.3, Exercise 3). Let $x_1 = (1, i), x_2 = (i, -1)$. Show that x_1, x_2 are linearly independent in $V_{\mathbb{R}}$ but linearly dependent in $V_{\mathbb{C}}$.

Proof. Let $c_1 = -i$ and $c_2 = 1$. Then

$$c_1 x_1 + c_2 x_2 = (-i)(1, i) + 1(i, -1) = (-i + i,\, -i^2 - 1) = (0, 0).$$

Thus x_1, x_2 are linearly dependent in $V_{\mathbb{C}}$.

Now suppose $a_1 x_1 + a_2 x_2 = 0$ with $a_1, a_2 \in \mathbb{R}$. Then

$$a_1(1, i) + a_2(i, -1) = (a_1 + i a_2,\, i a_1 - a_2) = 0$$

so

$$a_1 + i a_2 = 0, \quad i a_1 - a_2 = 0.$$

From $a_1 + i a_2 = 0$, we get $a_1 = -i a_2$. Since $a_1 \in \mathbb{R}$, we have $a_2 = 0$ so $a_1 = 0$ which is a contradiction. Therefore x_1, x_2 are linearly independent in $V_{\mathbb{R}}$. ∎

Problem 14

With $V = \mathcal{D}$ as in Exercise 13, let $D \in \mathcal{L}(V)$ be the linear mapping defined by $Dp = p'$ (the derivative of p). Prove that for every positive integer n, the elements $I, D, D^2, \dots, D^n$ of $\mathcal{L}(V)$ are linearly independent.

Proof. Let p be a polynomial of degree n. Suppose $I, D, D^2, \dots, D^n$ are linearly dependent. To simplify notation, let $I = D^0$. Let $c_0, c_1, \dots, c_n$ be scalars not all zero such that

$$\sum c_i D^i = 0.$$

Let j be the largest value i such that $c_i \neq 0$. Then

$$c_j D^j(p) = -\sum_{i<j} c_i D^i(p).$$

Now $\deg(D^j p) = \deg(p) - j$, while for $i < j$ we have $\deg(D^i p) = \deg(p) - i > \deg(p) - j$. Thus $I, D, D^2, \dots, D^n$ are linearly independent in $\mathcal{L}(V)$. ∎

Problem 15

A set A of vectors is said to be linearly independent if every finite list $x_1, \dots, x_n$ of distinct vectors in A is linearly independent (in other words, A is not linearly dependent in the sense of 3.2, Exercise 8)

1. A is linearly independent if and only if no vector A is linear combination of remaining vectors of A, that is, for every vector $x \in A$, $x \neq [A - \{x\}]$.
2. The empty set $\emptyset$ is linearly independent 'by default'. Hint: Independence of A means $x \in A \implies x \notin [A - \{x\}]$ Remember 'vacuous implication' (Appendix A.1.5)?
3. If A is linearly independent and $B \subset A$, then B is linearly independent.
4. If $T : V \longrightarrow W$ is an injective linear mapping and A is linearly independent subset of V, then $T(A)$ is linearly independent subset of W.

Proof. (1)
($\longrightarrow$) Suppose A is linearly independent. Let $x \in A$. If $x \in [A - \{x\}]$, then

$$x = \sum c_i x_i \quad (x_i \in A \setminus \{x\}),$$

so

$$x - \sum c_i x_i = 0,$$

a linear dependence in A which is a contradiction. Thus $x \notin [A - \{x\}]$.

($\longleftarrow$) Suppose no $x \in A$ satisfies $x \in [A - \{x\}]$. If

$$\sum c_i x_i = 0$$

with some $c_j \neq 0$, then

$$x_j = -\sum_{i \neq j} \frac{c_i}{c_j} x_i \in [A - \{x_j\}],$$

a contradiction. Thus A is linearly independent.

(2)
The condition is vacuously true for $A = \emptyset$, so $\emptyset$ is linearly independent.

(3)
Let $B \subset A$. Any finite list in B is also a finite list in A, so it is linearly independent since A is. Thus B is linearly independent.

(4)
Let $T : V \to W$ be injective and let A be linearly independent. If

$$\sum c_i T(x_i) = 0,$$

then

$$T\left(\sum c_i x_i\right) = 0.$$

Since T is injective, $\sum c_i x_i = 0$, so all $c_i = 0$. Thus $T(A)$ is linearly independent. ∎

Problem 16

Let V be a vector space over F, f a nonzero linear form on V (2.1.11), N the kernel of f, and y a vector such that $f(y) \neq 0$. Prove;

1. $V = N \oplus Fy$ (in the sense of 1.6 Exercise 12). Hint: 2.2, Exercise 2.
2. A linear form g on V is proportional to f if and only if $g(x) = 0$ for all $x \in N$. Hint: Assuming $g = 0$ on N, the problem is to find a scalar c such that $g = cf$. Try $c = g(y)/f(y)$.

Proof. By 1.6 Exercise 12 every vector $v \in V$ can be written

$$v = n + cy \in N + Fy \text{ with } n \in N, c \in F.$$

Suppose $v \in N \cap Fy$. Then $v = cy$ and $v = n$ for some $c \in F, n \in N$. Since $v \in N$, we have

$$0 = f(v) = f(cy) = cf(y).$$

Because $f(y) \neq 0$, it follows that $c = 0$ thus $v = 0$. Thus $V = N \oplus Fy$. ∎

Proof. ($\longrightarrow$) Suppose a linear form g on V is proportional to f. Let $x \in N$, since $x \in \ker F$ we have

$$cf(x) = g(x) = 0.$$

($\longleftarrow$) Suppose $g(x) = 0$ for all $x \in N$. Since $f(y) \neq 0$, define

$$c = \frac{g(y)}{f(y)}.$$

Let $x \in V$. Write $x = n + ay$ with $n \in N$. Then

$$g(x) = g(n + ay) = g(n) + ag(y) = ag(y),$$

since $g(n) = 0$, and

$$f(x) = f(n + ay) = f(n) + af(y) = af(y),$$

since $f(n) = 0$. Thus

$$cf(x) = \frac{g(y)}{f(y)} \cdot af(y) = ag(y) = g(x).$$

∎

Problem 18

Let V and W be vector spaces over F and let M, N be linear subspaces of V such that $V = M \oplus N$ (1.6, Exercise 2). Prove:

1. If $R : M \longrightarrow W$ and $S : N \longrightarrow W$ are linear mappings, there exists a unique linear mapping $T : V \longrightarrow W$ such that T agrees with R on M and with S on N.
2. Every linear mapping $M \longrightarrow W$ may be extended to a linear mapping $V \longrightarrow W$ (if we like, with the same range).

Proof. Suppose $R : M \longrightarrow W$ and $S : N \longrightarrow W$ are linear mappings. Consider the mapping

$$T : V \longrightarrow W \text{ defined by } T(v) = R(m) + S(n),$$

where $v = m + n$ with $m \in M, n \in N$.

Write $v_1, v_2 \in V = M \oplus N$ such that $v_1 = m_1 + n_1$, $v_2 = m_2 + n_2$. Let c be a scalar in F. Then

$$\begin{aligned} T(v_1 + v_2) &= T((m_1 + m_2) + (n_1 + n_2)) \\ &= R(m_1 + m_2) + S(n_1 + n_2) \\ &= R(m_1) + R(m_2) + S(n_1) + S(n_2) \\ &= T(v_1) + T(v_2), \end{aligned}$$

and

$$\begin{aligned} T(cv_1) &= T(cm_1 + cn_1) \\ &= R(cm_1) + S(cn_1) \\ &= cR(m_1) + cS(n_1) \\ &= cT(v_1). \end{aligned}$$

Thus T is linear.

For uniqueness, suppose $T' : V \to W$ is linear with $T'|_M = R$ and $T'|_N = S$. Let $v \in V$ such that $v = m + n$ then

$$T'(v) = T'(m+n) = T'(m) + T'(n) = R(m) + S(n) = T(v),$$

so $T' = T$. ∎

Proof. Let $R : M \longrightarrow W$ be linear and write $V = M \oplus N$. Define $T : V \longrightarrow W$ by

$$T(m+n) = R(m) \text{ such that } m \in M,\ n \in N.$$

Then for $v_1 = m_1 + n_1$, $v_2 = m_2 + n_2$ and scalar c,

$$\begin{aligned} T(v_1 + v_2) &= T((m_1 + m_2) + (n_1 + n_2)) \\ &= R(m_1 + m_2) \\ &= R(m_1) + R(m_2) \\ &= T(v_1) + T(v_2), \end{aligned}$$

and

$$\begin{aligned} T(cv_1) &= T(cm_1 + cn_1) \\ &= R(cm_1) \\ &= cR(m_1) \\ &= cT(v_1). \end{aligned}$$

Thus T is linear. Let $m \in M$ then

$$T(m) = T(m+0) = R(m),$$

so T extends R. ∎

Problem 19

Let $M_1, \dots, M_n$ be linear subspaces of a vector space V and let $M_1 \times \cdots \times M_n$ be the product vector space (1.3.11). The mapping $T : M_1 \times \cdots \times M_n \longrightarrow V$ defined by $T(x_1, \dots, x_n) = x_1 + \cdots + x_n$ is linear. If T is injective, the linear subspaces $M_1, \dots, M_n$ are said to be (linearly independent).

1. M_1, M_2 are linearly independent independent if and only if $M_1 \cap M_2 = \{0\}$.
2. M_1, M_2, M_3 are independent if and only if $M \cap (M_2 + M_3) = M_2 \cap (M_1 + M_3) = M_3 \cap (M_1 + M_2) = \{\emptyset\}$.
3. Generalize the statement (ii).
4. If $x_1, \dots, x_n$ are nonzero vectors and M_i is the set of all scalar multiples of x_i, then the linear subspaces of $M_1, \dots, M_n$ are independent if and only if the vectors $x_1, \dots, x_n$ are linearly independent.

Proof. ($\longrightarrow$) Suppose M_1, M_2 are linearly independent. For contradiction, suppose $M_1 \cap M_2 \neq \{0\}$. Then let $x \in M_1 \cap M_2$ such that $x \neq 0$. Now we know $-x \in M_2$, and then

$$T(x, -x) = x + (-x) = 0.$$

But $(x, -x) \neq (0, 0)$, contradicting that T is injective.

($\longleftarrow$) Suppose $M_1 \cap M_2 = \{0\}$. Now suppose for some $(x_1, x_2) \in M_1 \times M_2$ we have

$$T(x_1, x_2) = 0.$$

Thus $x_1 + x_2 = 0$. Therefore $x_1 = -x_2$, so $x_1 = x_2 \in M_1 \cap M_2$. Thus $x_1 = x_2 = 0$. Therefore T is injective, and it follows that M_1, M_2 are linearly independent. ∎

Proof. ($\longrightarrow$) Suppose M_1, M_2, M_3 are linearly independent. We show $M_1 \cap (M_2 + M_3) = \{0\}$ the others are similar. Let $x \in M_1 \cap (M_2 + M_3)$. Then $x \in M_1$ and $x = m_2 + m_3$ for some $m_2 \in M_2, m_3 \in M_3$. Then

$$T(x, -m_2, -m_3) = x - m_2 - m_3 = 0.$$

Since T is injective, it follows that

$$(x, -m_2, -m_3) = (0, 0, 0).$$

Thus $x = 0$, so $M_1 \cap (M_2 + M_3) = \{0\}$. Similarly,

$$M_2 \cap (M_1 + M_3) = \{0\}, \quad M_3 \cap (M_1 + M_2) = \{0\}.$$

($\longleftarrow$) Suppose

$$M_1 \cap (M_2 + M_3) = M_2 \cap (M_1 + M_3) = M_3 \cap (M_1 + M_2) = \{0\}.$$

Let $(x_1, x_2, x_3) \in M_1 \times M_2 \times M_3$ such that

$$T(x_1, x_2, x_3) = 0.$$

Then

$$x_1 + x_2 + x_3 = 0.$$

So

$$x_1 = -(x_2 + x_3) \in M_1 \cap (M_2 + M_3),$$

therefore $x_1 = 0$. Similarly,

$$x_2 = x_3 = 0.$$

Thus $(x_1, x_2, x_3) = (0, 0, 0)$, so T is injective. ∎

Proof. We proceed via induction on n.

Base Case: $n = 2$ and $n = 3$ are exactly parts (i) and (ii).

Inductive Step: Assume the statement holds for $n - 1$ subspaces. Let $M_1, \dots, M_n$ be subspaces of V, and let

$$T(x_1, \dots, x_n) = x_1 + \cdots + x_n.$$

($\longrightarrow$) Suppose $M_1, \dots, M_n$ are linearly independent. Fix k. Let

$$x \in M_k \cap \sum_{i \neq k} M_i.$$

Then $x = \sum_{i \neq k} m_i$ with $m_i \in M_i$. So

$$T(0, \dots, 0, x, 0, \dots, 0, -m_1, \dots, -m_{k-1}, -m_{k+1}, \dots, -m_n) = 0.$$

By injectivity, all components are zero, so $x = 0$. Thus

$$M_k \cap \sum_{i \neq k} M_i = \{0\}.$$

($\longleftarrow$) Suppose for each k

$$M_k \cap \sum_{i \neq k} M_i = \{0\}.$$

Let $(x_1, \dots, x_n)$ such that $T(x_1, \dots, x_n) = 0$. Then

$$x_1 = -(x_2 + \cdots + x_n) \in M_1 \cap \sum_{i \neq 1} M_i,$$

so $x_1 = 0$. Repeating this argument gives $x_k = 0$ for all k. Thus T is injective. ∎

Proof. Suppose $x_1, \dots, x_n$ are nonzero vectors and M_i is the set of all scalar multiples of x_i.

($\longrightarrow$) Suppose the linear subspaces $M_1, \dots, M_n$ are independent. If $x_1, \dots, x_n$ are linearly dependent then there exist scalars $c_1, \dots, c_n$, not all zero, such that

$$\sum c_i x_i = 0.$$

But then each $c_i x_i \in M_i$, contradicting that $M_1, \dots, M_n$ are independent.

($\longleftarrow$) Suppose the vectors $x_1, \dots, x_n$ are linearly independent. If $M_1, \dots, M_n$ are linearly dependent then there exist vectors $v_i \in M_i$, not all zero, such that

$$\sum v_i = 0.$$

Since each $v_i = c_i x_i$ for some scalars c_i, we get

$$\sum c_i x_i = 0,$$

with not all c_i zero, contradicting that $x_1, \dots, x_n$ are linearly independent. ∎

Problem 20

With notations as in the preceeding exercise, if $M_1, \dots, M_n$ are independent and $M_1 + \cdots + M_n = V$, then V is said to be the direct sum (or 'internal direct sum') of the linear subspaces $(M_1, \dots, M_n)$, written $V = M_1 \oplus \cdots \oplus M_n$. (The case $n = 2$ is covered by 1.6, Exercise 12.) This is equivalent to saying that T is bijective, and it implies that $M_1 \times \cdots \times M_n \cong M_1 \oplus \cdots \oplus M_n = V$.

Alrighty then.

Problem 21

Let $V_1, \dots, V_n$ be vector spaces over the same field of scalars and let $V = V_1 \times \cdots \times V_n$ be the product vector space (Definition 1.3.11). For each index j, let M_j be the linear subspace of V consisting of all vectors $(x_1, \dots, x_n) \in V$ such that $x_i = 0$ for all $i \neq j$. For example, M_2 is the set of all vectors $(0, x_2, 0, \dots, 0)$ with $x_2 \in V_2$. Prove:

1. $M_j \cong V_j$ for all $j = 1, \dots, n$.
2. $V = M_1 \oplus \cdots \oplus M_n$ in the sense of Exercise 20. For this reason $V_1 \times \cdots \times V_n$ is also called the 'direct sum' of $V_1, \dots, V_n$ (or 'external direct sum', to distinguish it from the situation in Exercise 20).

Proof. Consider the bijection $\rho : M_j \longrightarrow V_j$ defined

$$\rho(0, \dots, x_j, \dots, 0) = x_j.$$

∎

Proof. Let $x \in V$. Then $x = (x_1, \dots, x_n)$ such that

$$x = (x_1, 0, \dots, 0) + \cdots + (0, \dots, 0, x_n),$$

where each $(0, \dots, x_j, \dots, 0) \in M_j$. It is clear from the definition that $M_i \cap M_j = \{0\}$ for $i \neq j$ thus $M_1 \cap \cdots \cap M_n = \{0\}$. Thus $x \in M_1 + \cdots + M_n$.

Now let $x \in M_1 \oplus \cdots \oplus M_n$. Then x can be written as

$$x = m_1 + \cdots + m_n \text{ such that } m_j \in M_j.$$

Each m_j has the form $(0, \dots, x_j, \dots, 0)$, so their sum is

$$(x_1, \dots, x_n) \in V.$$

Thus $V = M_1 \oplus \cdots \oplus M_n$. ∎

Problem 22

Suppose $V = M_1 \oplus \cdots \oplus M_n$ as in Exercise 20. Prove:

1. $V = (M_1 \oplus \cdots \oplus M_{n-1}) \oplus M_n$.
2. $V/M_n \cong M_1 \oplus \cdots \oplus M_{n-1}$.

Proof. Let $x \in V$. Then

$$x = m_1 + \cdots + m_n = [(x_1, \dots, 0_n) + \cdots + (0_1, \dots, x_{n-1})] + (0_1, \dots, x_n)$$
$$= (m_1 + \cdots + m_{n-1}) + m_n \in (M_1 \oplus \cdots \oplus M_{n-1}) + M_n.$$

Thus $x \in (M_1 \oplus \cdots \oplus M_{n-1}) + M_n$.

Conversely, let $x \in (M_1 \oplus \cdots \oplus M_{n-1}) \oplus M_n$. Then $x = y + m_n$ where $y \in M_1 \oplus \cdots \oplus M_{n-1}$ and $m_n \in M_n$. Write $y = m_1 + \cdots + m_{n-1}$ with $m_i \in M_i$. Then

$$x = m_1 + \cdots + m_n \in V.$$

Thus $V = (M_1 \oplus \cdots \oplus M_{n-1}) \oplus M_n$. ∎

Proof. Consider the linear mapping $\rho : V \longrightarrow M_1 \oplus \cdots \oplus M_{n-1}$ defined by

$$\rho(x_1, \dots, x_n) = (x_1, \dots, x_{n-1}).$$

Clearly $\text{ker}(\rho) = M_n$ and $\text{ran}(\rho) = M_1 \oplus \cdots \oplus M_{n-1}$. By the first isomorphism theorem we have

$$V/M_n \cong M_1 \oplus \cdots \oplus M_{n-1}.$$

∎

Problem 23

Let $M_1, \dots, M_n$ be linear subspaces of V. Prove that the following conditions are equivalent:

1. $M_1, \dots, M_n$ are independent (in the sense of Exercise 19);
2. $(M_1 + \cdots + M_i) \cap M_{i+1} = \{\emptyset\}$ for $i = 1, \dots, n-1$.

Proof. This is exactly problem 19. ∎

Problem 24

Let V be a vector space over F and let $x_1, \dots, x_n$ be vectors in V. For $k = 1, \dots, n$ let M_k be the linear subspace of V generated by $\{x_1, \dots, x_k\}$ (3.1.4):

$$\begin{aligned} M_1 &= [\{x_1\}] = Fx_1 \\ M_2 &= [\{x_1, x_2\}] \\ &\dots \\ M_k &= [\{x_1, \dots, x_k\}] \\ &\dots \\ M_n &= [\{x_1, \dots, x_n\}]. \end{aligned}$$

Define $M_0 = \{0\}$. Thus

$$\{0\} = M_0 \subset M_1 \subset M_2 \dots \subset M_n.$$

Prove the following conditions are equivalent:

1. $x_1, \dots, x_n$ are linearly independent;

2. $M_{k-1} \subset M_k$ properly, for $k = 1, \dots, n$.

Proof. Suppose $x_1, \dots, x_n$ are linearly independent. If $M_i = M_{i+1}$ then

$$[\{x_1, \dots, x_i\}] = [\{x_1, \dots, x_{i+1}\}].$$

This would imply $x_{i+1} \in M_i$, so

$$x_{i+1} = \sum_{j=1}^{i} c_j x_j$$

for some scalars c_j. Then

$$\sum_{j=1}^{i} (-c_j) x_j + 1 \cdot x_{i+1} = 0,$$

contradicting linear independence of $x_1, \dots, x_n$. Thus $M_{k-1} \subset M_k$ properly for all k.

Conversely, suppose $M_{k-1} \subset M_k$ properly for $k = 1, \dots, n$. Assume $x_1, \dots, x_n$ are not linearly independent. Then there exist scalars $c_1, \dots, c_n$, not all zero, such that

$$\sum_{i=1}^{n} c_i x_i = 0.$$

Let k be the largest index such that $c_k \neq 0$. Then

$$x_k = -\sum_{i=1}^{k-1} \frac{c_i}{c_k} x_i,$$

so $x_k \in M_{k-1}$. But then $M_k = M_{k-1}$, contradicting the assumption that the inclusion is proper. Therefore $x_1, \dots, x_n$ are linearly independent. ∎

3.4 Finitely Generated Vector Spaces

Problem 1

True or false (explain): The vectors $x_1 = (1, 2, -3), x_2 = (2, -1, 2), x_3 = (5, 10, -15), x_4 = (6, -3, 6)$ generate $\mathbb{R}^3$.

Solution: False because $5(1, 2, -3) = (5, 10, -5)$ and $3(2, -1, 2) = (6, -3, 6)$. Thus, these vectors cover a plane in $\mathbb{R}^3$.

Problem 3

Let V and W be vector spaces over F, $V \times W$ the product vector space (1.3.11). Prove: $V \times W$ is finitely generated if and only if both V and W are finitely generated. Hint: Remark 2.3.8 3.4.12 and 3.1, Exercise 6.

Proof. ($\longrightarrow$) Suppose $V \times W$ is finitely generated. So there is a finite list of vectors $(v_1, w_1), \dots, (v_n, w_n) \in V \times W$ such that

$$V \times W = [\{(v_1, w_1), \dots, (v_n, w_n)\}].$$

Let $v \in V$. Then $(v, 0) \in V \times W$ so $(v, 0)$ is a linear combination of $(v_1, w_1), \dots, (v_n, w_n)$. Thus v is a linear combination of $v_1, \dots, v_n$ and thus

$$V = [\{v_1, \dots, v_n\}].$$

Similarly

$$W = [\{w_1, \dots, w_n\}].$$

Thus both V and W are finitely generated.

($\longleftarrow$) Suppose both V and W are finitely generated. Then

$$V = [\{v_1, \dots, v_n\}] \text{ and } W = [\{w_1, \dots, w_m\}].$$

We have

$$V \times W = [\{v_1, \dots, v_n\}] \times [\{w_1, \dots, w_m\}].$$

Then

$$V \times W = [\{(v_1, 0), \dots, (v_n, 0), (0, w_1), \dots, (0, w_m)\}],$$

thus $V \times W$ is finitely generated. ∎

Problem 4

Let V be a vector space of all function $\mathbb{R} \longrightarrow \mathbb{R}$ having derivatives of all orders (see Example 2.3.8) and let

$$M = \{f \in V : f'' = f\},$$

where f'' is the second derivative of f. Prove that M is the linear subspace of V generated by the two functions $t \longrightarrow e^t$ and $t \longrightarrow e^{-t}$. Hint: if $f'' = f$, look at $f = \frac{1}{2}(f + f') + \frac{1}{2}(f - f')$.

Proof. It is clear that $Ae^t + Be^{-t} \in M$. Let f be an arbitrary element in M. Thus

$$f'' = f.$$

Thus

$$f = \frac{1}{2}(f + f') + \frac{1}{2}(f - f').$$

Let $g = \frac{1}{2}(f + f')$ and $h = \frac{1}{2}(f - f')$. Then $f = g + h$. Then

$$g' = \frac{1}{2}(f' + f'') = \frac{1}{2}(f' + f) = g,$$

so $g = Ae^t$ for some constant A. Similarly

$$h' = \frac{1}{2}(f' - f'') = \frac{1}{2}(f' - f) = -h,$$

so $h = Be^{-t}$ for some constant B. Thus $f = Ae^t + Be^{-t}$. ∎

Problem 5

With notations as in Exercise 4, M is also generated by the function $sinh$, $cosh$.

Proof. Simply note

$$\text{sinh}(t) = \frac{e^t - e^{-t}}{2} \text{ and } \text{cosh}(t) = \frac{e^t + e^{-t}}{2}.$$

∎

Problem 6

With V as in Exercise 4, let

$$N = \{f \in V \mid f'' = -f\}.$$

Prove that N is generated by the function sin, cos. Hint: If $f'' = -f$, calculate the derivative of $f \sin + f' \cos$

and of $f\cos - f'\sin$.

Proof. It is clear that $A\sin t + B\cos t \in N$. Let f be an arbitrary element in N. Thus $f'' = -f$. Now

$$\frac{d}{dt}(f\sin t + f'\cos t) = f'\sin t + f\cos t + f''\cos t - f'\sin t = f\cos t + f''\cos t = 0.$$

Thus $f\sin t + f'\cos t = A$ for some constant A. Then

$$\frac{d}{dt}(f\cos t - f'\sin t) = f'\cos t - f\sin t - f''\sin t - f'\cos t = -f\sin t - f''\sin t = 0.$$

Thus $f\cos t - f'\sin t = B$ for some constant B. Solving these two equations gives $f = A\sin t + B\cos t$. ∎

Problem 7

If V is a vector space and M is linear subspace such that both M and V/M are finitely generated, then V is finitely generated. More precisely, if $x_1, \dots, x_m$ generate M and if $Qy_1, \dots, Qy_r$ generate V/M (where $Q : V \longrightarrow V/M$ is the quotient mapping), then V is generated by $x_1, \dots, x_m, y_1, \dots, y_r$.

Proof. Suppose $x_1, \dots, x_m$ generate M and $Qy_1, \dots, Qy_r$ generate V/M. Let v be an arbitrary element in V. Since $Qy_1, \dots, Qy_r$ generate V/M, there exist scalars $c_1, \dots, c_r$ such that

$$Qv = \sum c_i Qy_i.$$

Thus

$$Qv = Q\left(\sum c_i y_i\right) \implies v - \sum c_i y_i \in M.$$

Thus

$$\sum c_i y_i = v + m \text{ where } m \in M.$$

Since M is finitely generated there exist $x_1, \dots, x_m$ and scalars $c'_1, \dots, c'_m$ such that

$$m = \sum c'_i x_i.$$

So

$$\sum c_i y_i = v + \sum c'_i x_i \implies v = \sum c_i y_i - \sum c'_i x_i.$$

∎

3.5 Basis, Dimension

Problem 1

Prove that $(1,2,3), (2,1,0), (-3,2,0)$ is a basis of $\mathbb{R}^3$. Then find a shorter proof.

Proof. Let $(a,b,c) \in \mathbb{R}^3$. We require c_1, c_2, c_3 such that

$$c_1(1,2,3) + c_2(2,1,0) + c_3(-3,2,0) = (a,b,c).$$

Thus

$$(c_1, 2c_1, 3c_1) + (2c_2, c_2, 0) + (-3c_3, 2c_3, 0) = (a,b,c).$$

From this we get three equations

$$c_1 + 2c_2 - 3c_3 = a, \quad 2c_1 + c_2 + 2c_3 = b, \quad 3c_1 = c.$$

So we have

$$c_1 = \frac{c}{3}, \quad c_2 = \frac{2a + 3b - 8c}{21}, \quad c_3 = \frac{2b - a - c}{7}.$$

Verifying,

$$c_1(1,2,3) + c_2(2,1,0) + c_3(-3,2,0) = \frac{c}{3}(1,2,3) + \frac{2a + 3b - 8c}{21}(2,1,0) + \frac{2b - a - c}{7}(-3,2,0) = (a,b,c).$$

Therefore, $(1,2,3), (2,1,0), (-3,2,0)$ generate $\mathbb{R}^3$. Now suppose there exists c_1, c_2, c_3 not all zero such that

$$c_1(1,2,3) + c_2(2,1,0) + c_3(-3,2,0) = 0.$$

We get three equations

$$c_1 + 2c_2 - 3c_3 = 0, \quad 2c_1 + c_2 + 2c_3 = 0, \quad 3c_1 = 0.$$

From $3c_1 = 0$ we have $c_1 = 0$. Substituting into the first two equations gives

$$2c_2 - 3c_3 = 0, \quad c_2 + 2c_3 = 0.$$

From the second equation $c_2 = -2c_3$. Substituting into the first gives

$$2(-2c_3) - 3c_3 = -7c_3 = 0,$$

so $c_3 = 0$ thus $c_2 = 0$. Therefore

$$c_1 = c_2 = c_3 = 0.$$

Thus the vectors are linearly independent. Since $(1,2,3), (2,1,0), (-3,2,0)$ are generating and linearly independent they form a basis of $\mathbb{R}^3$. ∎

Proof. Let $(a,b,c) \in \mathbb{R}^3$ and

$$c_1 = \frac{c}{3}, \quad c_2 = \frac{2a + 3b - 8c}{21}, \quad c_3 = \frac{2b - a - c}{7}.$$

Then,

$$c_1(1,2,3) + c_2(2,1,0) + c_3(-3,2,0) = (a,b,c).$$

Thus $(1,2,3), (2,1,0), (-3,2,0)$ generate $\mathbb{R}^3$. Since $\hat{x}, \hat{y}, \hat{z}$ are linearly independent, and

$$|\{(1,2,3), (2,1,0), (-3,2,0)\}| = |\{\hat{x}, \hat{y}, \hat{z}\}|,$$

by Theorem 3.5.8 both lists are bases of $\mathbb{R}^3$. ∎

Problem 2

Let F be any field. Explain why $dim_F F^n = n$.

Proof. We know that $\{e_1, \dots, e_n\}$ forms a basis of F^n and $|\{e_1, \dots, e_n\}| = n$. By Corollary 3.5.11, every basis has n elements. ∎

Problem 3

Prove that $F^m \cong F^n$ if and only if $m = n$.

Proof. $(\underline{\longrightarrow})$ Suppose $F^m \cong F^n$. There exists a bijective linear mapping $L : F^m \longrightarrow F^n$. Then $L(\{e_1, \dots, e_m\}) = \{Le_1, \dots, Le_m\}$. Since L is surjective, $\{Le_1, \dots, Le_m\}$ generates F^n, and since L is injective, these vectors are linearly independent. Thus $m = n$.

$(\underline{\longleftarrow})$ Suppose $m = n$. Consider the bijective linear mapping $L : F^n \longrightarrow F^n$ defined by $L(e_i) = e_i$. Then L is an isomorphism, so $F^m \cong F^n$. ∎

Problem 4

Let V be a vector space, $V \neq \{0\}$. Prove that V is finitely generated if and only if $V \cong F^n$ for some positive integer n; if $V \cong F^n$ then $n = dim_F V$ (in particular, n is unique).

Proof. ($\longrightarrow$) Suppose V is finitely generated. Say $\{x_1, \dots, x_n\}$ is a basis for V. Consider the bijective linear mapping $L : F^n \longrightarrow V$ defined

$$L(a_1, \dots, a_n) = \sum a_i x_i.$$

Thus $V \cong F^n$.

($\longleftarrow$) Suppose $V \cong F^n$ for some positive integer n. Thus there exists a bijective linear mapping $L : F^n \longrightarrow V$. Let $\{e_1, \dots, e_n\}$ be the standard basis of F^n. Then $\{L(e_1), \dots, L(e_n)\}$ spans V and is linearly independent thus is a basis of V. Therefore V is finitely generated. ∎

Problem 5

Let α be a real number. Prove that the vectors

$$u = (\cos\alpha, \sin\alpha), v = (-\sin\alpha, \cos\alpha),$$

are a basis of $\mathbb{R}^2$

Proof. Suppose there exists c such that

$$(\cos\alpha, \sin\alpha) = c(-\sin\alpha, \cos\alpha).$$

We get two equations

$$\cos\alpha = -c\sin\alpha \text{ and } \sin\alpha = c\cos\alpha.$$

Multiplying these equations gives

$$\cos\alpha\sin\alpha = -c^2\sin\alpha\cos\alpha.$$

Thus

$$(1 + c^2)\sin\alpha\cos\alpha = 0.$$

Since $1 + c^2 \neq 0$ we have $\sin\alpha = 0$ or $\cos\alpha = 0$. In either case both give $\sin\alpha = 0$ and $\cos\alpha = 0$, a contradiction. Since $\dim\mathbb{R}^2 = 2$ and $\{u, v\}$ contains two linearly independent vectors in $\mathbb{R}^2$ it forms a basis of $\mathbb{R}^2$. ∎

Problem 7

Let $x_1, \dots, x_n$ be a basis of V. For each $x \in V$, there exist unique scalars $a_1, \dots, a_n$ such that $x = a_1x_1 + \dots + a_nx_n$ (Theorem 3.5.5); one calls $a_1, \dots, a_n$ the coordinates of x with respect to the basis $x_1, \dots, x_n$.

1. In $\mathbb{R}^2$, find the coordinates of the vector $(2, 3)$ with respect to the basis $(\frac{1}{2}\sqrt{3}, \frac{1}{2}), (-\frac{1}{2}, \frac{1}{2}\sqrt{3})$.
2. In $\mathbb{R}^n$, find the coordinates of the vector $(a_1, \dots, a_n)$ with respect to the canonical basis $e_1, \dots, e_n$ (3.5.2).

Proof. We must find a_1, a_2 such that

$$(2, 3) = a_1\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right) + a_2\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right).$$

This gives the system

$$\frac{\sqrt{3}}{2}a_1 - \frac{1}{2}a_2 = 2, \quad \frac{1}{2}a_1 + \frac{\sqrt{3}}{2}a_2 = 3.$$

Thus

$$a_1 = \sqrt{3} + 1, \quad a_2 = 3 - \sqrt{3}.$$

∎

Proof. Clearly the coordinates of $(a_1, \dots, a_n)$ with respect to the canonical basis $e_1, \dots, e_n$ are $a_1, \dots, a_n$. ∎

Problem 9

Let V be a finite-dimensional vector space and let M be a linear subspace of V. Prove:

1. V/M is finite-dimensional, and $dim(V/M) \leq dimV$;
2. $dim(V/M) = dimV \iff M = \{0\}$;
3. $dim(V/M) = 0 \iff M = V$.

Proof. By 3.5.14, M is finite-dimensional. Furthermore, by 3.5.16 we can write

$$[\{x_1, \dots, x_r, x_{r+1}, \dots, x_n\}] = V \text{ with } [\{x_{r+1}, \dots, x_n\}] = M.$$

By 3.4.8, $Qx_1, \dots, Qx_r$ generate V/M, so V/M is finite-dimensional with $\dim(V/M) \leq r \leq n = \dim V$. ∎

Proof. $\underline{(\longrightarrow)}$ Suppose $\dim(V/M) = \dim V = n$. Suppose $Qx_1, \dots, Qx_r$ and $x_{r+1}, \dots, x_n$ generate V/M and M respectively. We know that $x_1, \dots, x_r, x_{r+1}, \dots, x_n$ generate V by Problem 7. Then $\dim V = r + \dim M = n$, thus $\dim M = n - r$. But $\dim(V/M) = r = n$, so $\dim M = 0 \implies M = \{0\}$.

$\underline{(\longleftarrow)}$ Suppose $M = \{0\}$. Suppose $Qx_1, \dots, Qx_r$ and $x_{r+1}, \dots, x_n$ generate V/M and M respectively. We know that $x_1, \dots, x_r, x_{r+1}, \dots, x_n$ generate V by Problem 7. But $\dim M = 0$, so there are no $x_{r+1}, \dots, x_n$, thus $\dim(V/M) = r = n = \dim V$. ∎

Proof. $\underline{(\longrightarrow)}$ Suppose $\dim(V/M) = 0$. Suppose $Qx_1, \dots, Qx_r$ and $x_{r+1}, \dots, x_n$ generate V/M and M respectively. Let $v \in V$. Then

$$v = \sum_{i=1}^{r} a_i x_i + \sum_{i=r+1}^{n} b_i x_i.$$

But since $\dim(V/M) = 0$ there are no terms in $\sum_{i=1}^{r} a_i x_i$. Thus $v = \sum_{i=r+1}^{n} b_i x_i$ where each $x_i \in M$, thus $v \in M$. So $V \subset M$. We know $M \subset V$, thus $V = M$.

$\underline{(\longleftarrow)}$ Suppose $M = V$. Suppose $Qx_1, \dots, Qx_r$ and $x_{r+1}, \dots, x_n$ generate V/M and M respectively. But since $M = V$, every $x_i \in M$, so $Qx_i = 0$ for all i. Thus V/M is generated by $\{0\}$, thus $\dim(V/M) = 0$. ∎

Problem 11

Let $x_1, \dots, x_n$ be a basis of V and let $1 \leq m < n$. Let M be the linear subspace of V generated by $x_1, \dots, x_m$, and N the linear subspace generated by $x_{m+1}, \dots, x_n$. Prove that $V = M \oplus N$ (in the sense of 1.6, Exercise 12).

Proof. Let $x \in V$. We can write

$$x = \sum_{i=1}^{n} c_i x_i,$$

for suitable $c_i \in F$. So

$$x = \sum_{i=1}^{m} c_i x_i + \sum_{i=m+1}^{n} c_i x_i \in M + N.$$

Thus $V = M + N$.

It remains to show that $M \cap N = \{0\}$. Suppose $x \in M \cap N$ such that

$$x = \sum_{i=1}^{m} a_i x_i$$

and

$$x = \sum_{i=m+1}^{n} b_i x_i.$$

Then

$$\sum_{i=1}^{m} a_i x_i - \sum_{i=m+1}^{n} b_i x_i = 0.$$

Since $x_1, \ldots, x_n$ is a basis, it is linearly independent, so all coefficients are zero. Thus $a_i = 0$ and $b_i = 0$ for all i, and thus $x = 0$. Therefore $M \cap N = \{0\}$, so $V = M \oplus N$. ∎

Problem 12

Let V be a vector space over F, let $x_1, \ldots, x_n$ be nonzero vectors in V, and let $M_i = Fx_i$ for $i = 1, \ldots, n$. Prove that the following two conditions are equivalent:

1. $x_1, \ldots, x_n$ is a basis of V;
2. $V = M_1 \oplus \cdots \oplus M_n$ (in the sense of 3.3, Exercise 20).

Proof. ($\longrightarrow$) Suppose $x_1, \ldots, x_n$ is a basis of V. Let $M_i = [\{x_i\}]$. We have mapping $T : M_1 \times \cdots \times M_n \longrightarrow V$ defined

$$T(v_1, \ldots, v_n) = \sum v_i.$$

Suppose $v \in V$ we have

$$v = \sum c_i x_i,$$

for appropriate $c_1, \ldots, c_n \in F$, so $c_i x_i \in M_i$. Since $x_1, \ldots, x_n$ are linearly independent we have $\ker T = \{0\}$. Thus $M_1, \ldots, M_n$ are linearly independent. Thus $v \in M_1 \oplus \cdots \oplus M_n$.

Since $M_i \subseteq V$ for each i, we have $M_1 + \cdots + M_n \subseteq V$.

Thus $V = M_1 \oplus \cdots \oplus M_n$.

($\longleftarrow$) Suppose $V = M_1 \oplus \cdots \oplus M_n$. Now, suppose $v \in V$. Then $v = v_1 + \cdots + v_n$ with $v_i \in M_i$. Then for scalars $c_1, \ldots, c_n \in F$ we have $v_i = c_i x_i$. Clearly the $x_1, \ldots, x_n$ are generating. By definition of $\oplus$ the x_i are linearly independent. Thus $x_1, \ldots, x_n$ is a basis of V. ∎

Problem 16

Let V be any vector space (not necessarily finitely generated). A subset B of V is said to be a basis of V if it has the following two properties:

1. B is generating, that is $[B] = V$ (3.1.4);
2. B is linearly independent, that is, for every vector $x \in B$, $x \notin [B - \{x\}]$ (3.3, Exercise 15).

Solution: Alrighty then.

Problem 17

Prove that if B is a basis of V (in the sense of Exercise 16), then B is 'minimal-generating' and 'maximal-independent' in the following sense:

1. If A is a proper subset of B (that is, $A \subset B$ and $A \neq B$) then A is not generating;
2. If $B \subset C \subset V$ and B is a proper subset of C, then C is not linearly independent.

Proof. Suppose B is a basis of V. Now, suppose A is a proper subset of B. For contradiction, suppose $[A] = V$. Let $x \in B - A$. Since $x \in V$ we have $x = \sum c_i a_i$ for some $c_i \in F$ and $a_i \in A$. But since $A \subset B$ we have $x = \sum c_i b_i$ where $b_i \in B$. This contradicts that B is linearly independent.

Now, suppose $B \subset C \subset V$ and B is a proper subset of C. Let $x \in C - B$. Since $x \in V$ we have $x = \sum c_i b_i$ where $b_i \in B$. But since $B \subset C$ we have $b_i \in C$, so x is a linear combination of elements of C. Thus C is not linearly independent. ∎

Problem 18

Let $\mathcal{D}$ be the vector space of real polynomial functions (Example 1.3.5). For $n = 0, 1, 2, \ldots$ let p_n be the monomial function $p_n(t) = t^n$, and let $B = \{p_0, p_1, p_2, \ldots\}$. Prove that B is a basis of $\mathcal{D}$ in the sense of Exercise 16.

Proof. Let f be an arbitrary polynomial in $\mathcal{D}$. Let deg$(f) = m$. By definition f can be written

$$f = \sum_{i=0}^{m} c_i t^i = \sum_{i=0}^{m} c_i p_i(t) \in [B].$$

Thus B is generating.

Now suppose $C \subset B$ and $C \neq B$. Let $g \in B - C$. Then $g = p_k$ for some k, and $p_k \notin C$. Since every element of $[C]$ is a linear combination of elements of C, it cannot equal p_k. Thus $[C] \neq [B] = \mathcal{D}$, so C is not generating.

It follows that B is a basis of $\mathcal{D}$. ∎

Problem 19

Let V be the vector space of finitely nonzero sequences of real numbers (1.3.9). Find a basis B of V (in the sense of Exercise 16).

Proof. For $n = 0, 1, 2, \ldots$ let p_n be the sequence $p_n = (0, \ldots, 0, 1, 0, \ldots)$ where the 1 is in the n-th position, and let $B = \{p_0, p_1, p_2, \ldots\}$.

Let x be an arbitrary element of V. Since x is a finitely nonzero sequence, we have

$$x = (c_0, c_1, \ldots, c_m, 0, 0, \ldots).$$

Then

$$x = \sum_{i=0}^{m} c_i p_i \in [B].$$

Thus B is generating.

Suppose $C \subset B$ and $C \neq B$. Let $g \in B - C$. Then $g = p_k$ for some k. Since every element of $[C]$ has zero in the k-th position, it cannot equal p_k. Thus $[C] \neq V$, so C is not generating.

It follows that B is a basis of V. ∎

Problem 20

Let V be a finite-dimensional complex vector space. Regard V also as a real vector space by restriction of scalars (1.3, Exercise 3). Prove that $dim_{\mathbb{R}} V = 2 \cdot dim_{\mathbb{C}} V$.

Proof. Let $\{v_1, \ldots, v_n\}$ be a basis of V over $\mathbb{C}$. Let $v \in V$ such that

$$v = \sum_{k=1}^{n} z_k v_k$$

where $z_k \in \mathbb{C}$. We can write $z_k = a_k + b_k i$ for $a_k, b_k \in \mathbb{R}$. Then

$$v = \sum_{k=1}^{n} (a_k + b_k i) v_k = \sum_{k=1}^{n} a_k v_k + \sum_{k=1}^{n} b_k (i v_k).$$

Consider the set

$$B = \{v_1, iv_1, \dots, v_n, iv_n\}.$$

Now, consider the bijection $T : \mathbb{R}^{2n} \to V$ defined

$$T(a_1, b_1, \dots, a_n, b_n) = \sum_{k=1}^{n} a_k v_k + \sum_{k=1}^{n} b_k (i v_k).$$

It follows that B is a basis of V over $\mathbb{R}$. Therefore

$$\dim_{\mathbb{R}} V = 2 \cdot \dim_{\mathbb{C}} V.$$

∎

Problem 21

If V is a finite-dimensional real vector space and W is its complexification (1.3, Exercise 4), prove that $dim_C W = dim_R V$.

Proof. Let $\{v_1, \dots, v_n\}$ be a basis of V over $\mathbb{R}$. Since W is the complexification of V, every element of W can be written as

$$w = \sum_{i=1}^{n} c_i v_i$$

where $c_1, \dots, c_n \in \mathbb{C}$. Consider the map $T : \mathbb{C}^n \longrightarrow W$ defined by

$$T(c_1, \dots, c_n) = \sum_{i=1}^{n} c_i v_i.$$

T is a bijection so $\{v_1, \dots, v_n\}$ is a basis of W over $\mathbb{C}$. Therefore

$$\dim_{\mathbb{C}} W = n = \dim_{\mathbb{R}} V.$$

∎

Problem 22

Let $x_1, \dots, x_n$ be vectors in $\mathbb{R}^n$. Prove that $x_1, \dots, x_n$ is a basis of the real vector space $\mathbb{R}^n$ if and only if it is a basis of the complex vector space $\mathbb{C}^n$.

Proof. ($\longrightarrow$) Suppose $x_1, \dots, x_n$ is a basis of $\mathbb{R}^n$. Let $v \in \mathbb{C}^n$. Then $v = u + iw$ where $u, w \in \mathbb{R}^n$. Since $x_1, \dots, x_n$ span $\mathbb{R}^n$ we can write $u = \sum a_i x_i$ and $w = \sum b_i x_i$ with $a_i, b_i \in \mathbb{R}$, so

$$v = \sum a_i x_i + i \sum b_i x_i = \sum (a_i + b_i i) x_i,$$

thus $x_1, \dots, x_n$ span $\mathbb{C}^n$. Now let $\sum_{i=1}^{n} c_i x_i = 0$ with $c_i \in \mathbb{C}$. Writing $c_i = a_i + b_i i$,

$$\sum_{i=1}^{n} a_i x_i + i \sum_{i=1}^{n} b_i x_i = 0 \implies \sum_{i=1}^{n} a_i x_i = 0 \text{ and } \sum_{i=1}^{n} b_i x_i = 0.$$

Since $x_1, \dots, x_n$ are linearly independent over $\mathbb{R}$, all $a_i = b_i = 0$, thus all $c_i = 0$. Thus $x_1, \dots, x_n$ is a basis of $\mathbb{C}^n$.

($\Longleftarrow$) Suppose $x_1, \dots, x_n$ is a basis of $\mathbb{C}^n$. Let $\sum_{i=1}^n a_i x_i = 0$ with $a_i \in \mathbb{R} \subset \mathbb{C}$. Since $x_1, \dots, x_n$ are linearly independent over $\mathbb{C}$, all $a_i = 0$. Since $x_1, \dots, x_n$ generate $\mathbb{C}^n$ over $\mathbb{C}$ and each $x_i \in \mathbb{R}^n$, for any $v \in \mathbb{R}^n \subset \mathbb{C}^n$ we have $v = \sum_{i=1}^n c_i x_i$ with $c_i \in \mathbb{C}$. Taking real parts gives $v = \sum_{i=1}^n a_i x_i$ with $a_i \in \mathbb{R}$, so $x_1, \dots, x_n$ generate $\mathbb{R}^n$. Thus $x_1, \dots, x_n$ is a basis of $\mathbb{R}^n$. ∎

Problem 23

Let V and W be finite-dimensional vector spaces, $V \times W$ the product vector space (1.3.11). Prove:

$$dim(V \times W) = dimV + dimW.$$

Hint: If $x_1, \dots, x_n$ is a basis of V and $y_1, \dots, y_m$ is a basis of W, show that the vectors

$$(x_1, 0), \dots, (x_n, 0), (0, y_1), \dots, (0, y_m),$$

are a basis of $V \times W$, cf. 3.1, Exercise 6.

Proof. Suppose $x_1, \dots, x_n$ is a basis of V and $y_1, \dots, y_m$ is a basis of W. Consider the set

$$(x_1, 0), \dots, (x_n, 0), (0, y_1), \dots, (0, y_m).$$

Let $(v, w) \in V \times W$. Then

$$v = \sum a_i x_i \text{ and } w = \sum b_j y_j.$$

Thus

$$(v, w) = \sum a_i (x_i, 0) + \sum b_j (0, y_j),$$

so the set is generating. Now suppose

$$\sum a_i (x_i, 0) + \sum b_j (0, y_j) = (0, 0).$$

Then

$$(\sum a_i x_i, \sum b_j y_j) = (0, 0).$$

Thus $\sum a_i x_i = 0$ and $\sum b_j y_j = 0$. Since $x_1, \dots, x_n$ and $y_1, \dots, y_m$ are bases, all $a_i, b_j = 0$. Thus the set is linearly independent. Therefore it is a basis of $V \times W$, so

$$dim(V \times W) = dimV + dimW.$$

∎

Problem 24

If V is a finite-dimensional vector space and M is a linear subspace of V, then there exists a linear subspace N of V such that $V = M \oplus N$ in the sense of 1.6 Exercise 12. Hint: Corollary 3.5.16.

Proof. Let $x_1, \dots, x_r$ be a basis for M. By Corollary 3.5.16, $x_1, \dots, x_r$ can be expanded to be a basis of V. Suppose the expanded basis is

$$x_1, \dots, x_r, x_{r+1}, \dots, x_n.$$

Let $N = \text{span}(x_{r+1}, \dots, x_n)$. Now, let $v \in V$. Then for chosen $c_1, \dots, c_n$ we have

$$v = \sum_{i=1}^n c_i x_i = \sum_{i=1}^r c_i x_i + \sum_{i=r+1}^n c_i x_i \in M + N.$$

Now suppose $v \in M \cap N$. Then

$$v = \sum_{i=1}^r c_i x_i \text{ and } v = \sum_{i=r+1}^n c_i x_i \implies \sum_{i=1}^r c_i x_i - \sum_{i=r+1}^n c_i x_i = 0.$$

Since $x_1, \dots, x_n$ are linearly independent we must have $c_1 = \dots = c_n = 0$. Thus $v = 0$, so $M \cap N = \{0\}$. It follows that $V = M \oplus N$. ∎

Problem 25

As in Theorem 3.4.8, let V be a vector space, $x_1, \dots, x_r, x_{r+1}, \dots, x_n \in V$, $M = [\{x_{r+1}, \dots, x_n\}]$ and $Q : V \longrightarrow V/M$ the quotient mapping. The following conditions are equivalent:

1. ⓐ $x_1, \dots, x_n$ is a basis of V;
2. ⓑ $Qx_1, \dots, Qx_n$ is a basis of V/M and $x_{r+1}, \dots, x_n$ is a basis of M.

Proof. (ⓐ⟶ⓑ) Suppose $x_1, \dots, x_n$ is a basis of V. By 3.4.8, $Qx_1, \dots, Qx_r$ generate V/M. For contradiction suppose for some i with $1 \le i \le r$ we have $Qx_i = \sum_{j \neq i} c_j Qx_j$. Then

$$Qx_i - \sum_{j \neq i} c_j Qx_j = 0 \implies Q\left(x_i - \sum_{j \neq i} c_j x_j \right) = 0 \implies x_i - \sum_{j \neq i} c_j x_j \in \ker Q = M.$$

So $x_i - \sum_{j \neq i} c_j x_j = \sum_{k=r+1}^{n} d_k x_k$ for some scalars d_k, which contradicts that $x_1, \dots, x_n$ are linearly independent. It follows that $Qx_1, \dots, Qx_r$ is a basis of V/M. Since $x_1, \dots, x_n$ is a basis of V and $M = [\{x_{r+1}, \dots, x_n\}]$, it is clear that $x_{r+1}, \dots, x_n$ is a basis of M.

(ⓐ⟵ⓑ) Suppose $Qx_1, \dots, Qx_r$ is a basis of V/M and $x_{r+1}, \dots, x_n$ is a basis of M. Let $v \in V$. Since $Qx_1, \dots, Qx_r$ generate V/M, we have

$$Qv = \sum_{i=1}^{r} c_i Qx_i = Q\left(\sum_{i=1}^{r} c_i x_i \right),$$

so $v - \sum_{i=1}^{r} c_i x_i \in \ker Q = M$. Since $x_{r+1}, \dots, x_n$ generate M, we have $v - \sum_{i=1}^{r} c_i x_i = \sum_{i=r+1}^{n} c_i x_i$, thus $v = \sum_{i=1}^{n} c_i x_i$ and $x_1, \dots, x_n$ span V. Now suppose $\sum_{i=1}^{n} c_i x_i = 0$. Then

$$Q\left(\sum_{i=1}^{n} c_i x_i \right) = 0 \implies \sum_{i=1}^{r} c_i Qx_i = 0.$$

Since $Qx_1, \dots, Qx_r$ are linearly independent we must have $c_1 = \dots = c_r = 0$. Thus $\sum_{i=r+1}^{n} c_i x_i = 0$, and since $x_{r+1}, \dots, x_n$ are linearly independent we must have $c_{r+1} = \dots = c_n = 0$. Thus $x_1, \dots, x_n$ are linearly independent and are thus a basis of V. ∎

Problem 26

Let V be a finite-dimensional vector space, M and N linear subspace of V such that $dim M + dim N = dim V$. Prove: $M + N = V \iff M \cap N = \{0\}$. Hint: Corollary 3.5.14, (ii) and Theorem 3.5.13 (2).

Proof. Note that $M + N$ is a subspace of V with $\dim(M + N) \le \dim V$.

(⟶) Suppose $M + N = V$. Let $x_1, \dots, x_r$ be a basis of $M \cap N$. By 3.5.16, extend to a basis $x_1, \dots, x_r, m_1, \dots, m_s$ of M and a basis $x_1, \dots, x_r, n_1, \dots, n_t$ of N. Then $x_1, \dots, x_r, m_1, \dots, m_s, n_1, \dots, n_t$ generate $M + N = V$, so $r + s + t \ge \dim V = \dim M + \dim N = (r + s) + (r + t)$. Thus $r \le 0$ thus $r = 0$ and $M \cap N = \{0\}$.

(⟵) Suppose $M \cap N = \{0\}$. Let $m_1, \dots, m_s$ and $n_1, \dots, n_t$ be bases of M and N respectively. Then $s + t = \dim V$. Then $m_1, \dots, m_s, n_1, \dots, n_t$ generate $M + N$. These vectors are linearly independent so suppose $\sum a_i m_i + \sum b_j n_j = 0$ then $\sum a_i m_i = -\sum b_j n_j \in M \cap N = \{0\}$, so all $a_i = b_i = 0$. Thus $m_1, \dots, m_s, n_1, \dots, n_t$ is a basis of $M + N$ of size $\dim V$, so by 3.5.14 (ii) $M + N = V$. ∎

Problem 27

Let V be a finite-dimensional vector space over a field F, $V \times V$ the product vector space (1.3.11), Prove: If $dim(V \times V) = (dimV)^2$, then either $V = \{0\}$ or $V \cong F^2$.

Proof. Suppose $\dim(V \times V) = (\dim V)^2$. There are two cases. If $V = \{0\}$ then this is trivial. Thus suppose $V \neq \{0\}$. By Problem 4 $V \cong F^n$ for some integer n so $\dim V = n$. By Problem 23 $\dim(V \times V) = \dim V + \dim V$, we have

$$\dim(V \times V) = 2n.$$

Thus the hypothesis gives

$$2n = n^2.$$

Thus $n = 2$. Therefore $V \cong F^2$. ∎

3.6 Rank + Nullity = Dimension

Problem 1

Does there exist a linear mapping $T : \mathbb{R}^7 \longrightarrow \mathbb{R}^3$ whose kernel is 3-dimensional?

Proof. Here, $R = 3$ and $D = 7$ so $N = 4$ so no. ∎

Problem 2

Let V be a finite-dimensional vector space, M and N linear subspaces of V. Prove that $dimM + dimN = dim(M + N) + dim(M \cap N)$.

Proof. Consider the mapping $T : M \longrightarrow (M + N)/N$ defined

$$Tx = x + N \text{ for all } x \in M.$$

By 2.7 Problem 1 we know $ranT = (M + N)/N$ and $\ker T = M \cap N$. Clearly $domT = M$. Applying the rank-nullity theorem

$$\dim((M + N)/N) + \dim(M \cap N) = \dim(M).$$

Then since $\dim((M + N)/N) = \dim(M + N) - \dim(N)$ we have

$$\dim(M + N) - \dim(N) + \dim(M \cap N) = \dim(M),$$

so

$$\dim(M + N) + \dim(M \cap N) = \dim(M) + \dim(N).$$

∎

Problem 3

Let V be a vector space over F, $f : V \longrightarrow F$ a linear form on V, N the kernel of f. Prove that if f is not identically zero, then $V/N \cong F$.

Proof. Suppose f is not identically zero, so there exists $z \in V$ with $f(z) \neq 0$. Let $c = f(z) \in F$. Then $f(c^{-1}z) = c^{-1}f(z) = 1$, so f is surjective. By the First Isomorphism Theorem

$$V/\ker f = V/N \cong ranf = F.$$

∎

Problem 4

Let V be a finite-dimensional vector space, M and N linear subspaces of V such that the dimensions of

$$(M+N)/N, (M+N)/M, M \cap N,$$

are $2, 3, 4$, respectively. Find the dimensions of M, N and $M+N$.

Proof. From the given dimensions we have the relations

$$\dim(M+N) - \dim(N) = 2, \quad \dim(M+N) - \dim(M) = 3, \quad \dim(M \cap N) = 4.$$

By Problem 2,

$$\dim(M+N) + \dim(M \cap N) = \dim(M) + \dim(N),$$

and substituting $\dim(M+N) = \dim(N) + 2$ and $\dim(M \cap N) = 4$ gives

$$(\dim(N) + 2) + 4 = \dim(M) + \dim(N),$$

so $\dim(M) = 6$. Substituting back,

$$\dim(M+N) = \dim(M) + 3 = 9, \qquad \dim(N) = \dim(M+N) - 2 = 7.$$ ∎

Problem 6

The converse of Theorem 3.6.4 is true: If a product space $V \times W$ is finite-dimensional, then so are the factor spaces V and W.

Proof. Suppose $V \times W$ is finite-dimensional. Since $V \times \{0\}$ and $\{0\} \times W$ are subspaces of $V \times W$, they are also finite-dimensional. Since $V \cong V \times \{0\}$ and $W \cong \{0\} \times W$, it follows that V and W are finite-dimensional. ∎

Problem 7

Let V be an n-dimensional vector space and let f be a nonzero linear form on V, that is, a nonzero element of the dual space V' of V. Prove that the kernel of f is $(n-1)$-dimensional. Conversely, every $(n-1)$-dimensional linear subspace of V is the kernel of a linear form.

Proof. Since f is a nonzero linear form, $\operatorname{ran} f = F$ and so $\dim \operatorname{ran} f = 1$. By the First Isomorphism Theorem,

$$V/\ker f \cong \operatorname{ran} f \implies \dim V - \dim \ker f = \dim \operatorname{ran} f = 1 \implies \dim \ker f = n-1.$$

Conversely, let $N \subseteq V$ be an $(n-1)$-dimensional subspace. Let $\{e_1, \dots, e_{n-1}\}$ be a basis for N and extend it to a basis $\{e_1, \dots, e_{n-1}, e_n\}$ of V. Define $f : V \to F$ by

$$f\left(\sum_{i=1}^{n} a_i e_i\right) = a_n.$$

Then f is a nonzero linear form with $\ker f = N$. ∎

Problem 8

If V is an n-dimensional vector space and if f, g are linear forms on V that are not proportional (neither is a multiple of the other), prove that $(\ker f) \cap (\ker g)$ is $(n-2)$-dimensional.

Proof. Suppose V is an n-dimensional vector space and f, g are linear forms on V that are not proportional. By Problem 7, we know that $\dim\ker f = \dim\ker g = n-1$. By Problem 16, g is proportional to f if and only if $g(x) = 0$ for all $x \in \ker f$. Thus there exists $x \in \ker f$ such that $g(x) \neq 0$, so the restriction $g|_{\ker f} : \ker f \to F$ is a nonzero linear form on the $(n-1)$-dimensional space $\ker f$. By Problem 7,

$$\dim\ker(g|_{\ker f}) = (n-1) - 1 = n-2.$$

But $\ker(g|_{\ker f}) = \ker f \cap \ker g$, so $\dim(\ker f \cap \ker g) = n-2$. ∎

Problem 9

Let V be a vector space and suppose $f_1, \dots, f_n$ are linear forms on V such that

$$(\ker f_1) \cap \dots \cap (\ker f_n) = \{0\}.$$

Prove: V is a finite-dimensional and $\dim V \leq n$.

Proof. Define $T : V \to F^n$ by $Tx = (f_1(x), \dots, f_n(x))$. Then T is linear and $\ker T = (\ker f_1) \cap \dots \cap (\ker f_n) = \{0\}$, so T is injective. Thus $V \cong \operatorname{ran} T \subseteq F^n$, so V is finite-dimensional and $\dim V \leq n$. ∎

Problem 14

Let $T : V \longrightarrow W$ and $S : W \longrightarrow U$ be linear mappings, with V finite-dimensional.

1. If S is injective, then $\ker ST = \ker T$ and $\rho(ST) = \rho(T)$.
2. If T is surjective, then $\operatorname{ran} ST = \operatorname{ran} S$ and $v(ST) - v(S) = \dim V - \dim W$.

Proof. Suppose S is injective. Let $v \in \ker ST$, then $(ST)(v) = S(T(v)) = 0$. Since S is injective, $T(v) = 0$, so $v \in \ker T$. Conversely, if $v \in \ker T$ then $(ST)(v) = S(T(v)) = S(0) = 0$, so $v \in \ker ST$. Thus $\ker ST = \ker T$, and by rank-nullity $\rho(ST) = \rho(T)$.

Suppose T is surjective. For any $u \in \operatorname{ran} S$, there exists $w \in W$ such that $u = S(w)$. Since T is surjective, there exists $v \in V$ such that $T(v) = w$, so $u = S(T(v)) = (ST)(v) \in \operatorname{ran} ST$. Thus $\operatorname{ran} ST = \operatorname{ran} S$. By rank-nullity applied to ST and S,

$$v(ST) = \dim V - \rho(ST) = \dim V - \rho(S) = \dim V - (\dim W - v(S)),$$

so $v(ST) - v(S) = \dim V - \dim W$. ∎

Problem 15

Let V be a finite-dimensional vector space, $T \in \mathcal{L}(V)$. The following are equivalent:

1. ⓐ $V = \ker T + \operatorname{ran} T$;
2. ⓑ $\ker T \cap \operatorname{ran} T = \{0\}$;
3. ⓒ $V = \ker T \oplus \operatorname{ran} T$.

Proof. (ⓐ$\longrightarrow$ⓑ) Suppose $V = \ker T + \operatorname{ran} T$.

$$\dim V = \dim\ker T + \dim\operatorname{ran} T,$$

and since $V = \ker T + \operatorname{ran} T$ by the dimension formula from Problem 2

$$\dim V = \dim\ker T + \dim\operatorname{ran} T - \dim(\ker T \cap \operatorname{ran} T),$$

so $\dim(\ker T \cap \operatorname{ran} T) = 0$ thus $\ker T \cap \operatorname{ran} T = \{0\}$.

(ⓑ$\longrightarrow$ⓒ) Suppose $\ker T \cap \operatorname{ran} T = \{0\}$. By rank-nullity

$$\dim(\ker T + \operatorname{ran} T) = \dim \ker T + \dim \operatorname{ran} T - \dim(\ker T \cap \operatorname{ran} T) = \dim V - 0 = \dim V,$$

so $\ker T + \operatorname{ran} T = V$. Since $\ker T \cap \operatorname{ran} T = \{0\}$ we have $V = \ker T \oplus \operatorname{ran} T$.

(ⓒ$\longrightarrow$ⓐ) This is immediate since $V = \ker T \oplus \operatorname{ran} T$ implies $V = \ker T + \operatorname{ran} T$ by definition of the direct sum. ∎

3.7 Applications of $R + N = D$

Problem 1

Suppose that the homogeneous system

$$(*) \quad \begin{cases} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n = 0 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n = 0 \\ \quad \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \ldots + a_{nn}x_n = 0 \end{cases}$$

(n equations in n unknowns) has only the trivial solution. Prove that for any given scalars $c_1, \ldots, c_n$, the system

$$(**) \quad \begin{cases} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n = c_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n = c_2 \\ \quad \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \ldots + a_{nn}x_n = c_n \end{cases}$$

has a unique solution. (Hint: Theorem 3.7.4.)

Proof. Define $T : F^n \to F^n$ by

$$T(x_1, \ldots, x_n) = \left(\sum_{j=1}^{n} a_{1j}x_j, \ldots, \sum_{j=1}^{n} a_{nj}x_j \right).$$

The system $(*)$ having only the trivial solution means $\ker T = \{0\}$ so T is injective. By Theorem 3.7.4 T is bijective. Thus for any $(c_1, \ldots, c_n) \in F^n$, the equation $Tx = (c_1, \ldots, c_n)$ has a unique solution, which is precisely the system $(**)$. ∎

Problem 2

Prove that for any scalars a, b, c, the system

$$\begin{aligned} x_1 + 2x_2 - 3x_3 &= a \\ 2x_1 + x_2 + 2x_3 &= b \\ 3x_1 \qquad\qquad &= c \end{aligned}$$

has a unique solution. (Hint: §3.5, Exercise 1.)

Proof. Define $T : F^3 \to F^3$ by

$$T(x_1, x_2, x_3) = (x_1 + 2x_2 - 3x_3,\ 2x_1 + x_2 + 2x_3,\ 3x_1).$$

Suppose $T(x_1, x_2, x_3) = (0, 0, 0)$. Clearly $\ker T = \{0\}$ so the homogeneous system has only the trivial solution. By Problem 1, the system has a unique solution for any scalars a, b, c. ∎

Problem 3

Let V be a vector space and let $S, T \in \mathcal{L}(V)$. Prove:

(i) $\rho(S+T) \leq \rho(S) + \rho(T)$. (Hint: §3.6, Exercise 2.)

(ii) $\rho(ST) \leq \rho(S)$ and $\rho(ST) \leq \rho(T)$.

(iii) $\nu(S+T) \geq \nu(S) + \nu(T) - \dim V$.

Proof. We have

$$\rho(S) + \rho(T) = \dim(\operatorname{ran} S + \operatorname{ran} T) + \dim(\operatorname{ran} S \cap \operatorname{ran} T).$$

So $\rho(S+T) = \rho(S) + \rho(T) - \dim(\operatorname{ran} S \cap \operatorname{ran} T)$. Then

$$\rho(S) + \rho(T) - \rho(S+T) = \rho(S) + \rho(T) - [\rho(S) + \rho(T) - \dim(\operatorname{ran} S \cap \operatorname{ran} T)] = \dim(\operatorname{ran} S \cap \operatorname{ran} T) \geq 0.$$

Thus $\rho(S+T) \leq \rho(S) + \rho(T)$. ∎

Proof. Since $(ST)(x) = S(T(x)) \in \operatorname{ran} S$ for any $x \in V$, we have $\operatorname{ran}(ST) \subseteq \operatorname{ran} S$, so $\rho(ST) \leq \rho(S)$. Since $\operatorname{ran}(ST) = S(\operatorname{ran} T)$ and $S|_{\operatorname{ran} T} : \operatorname{ran} T \to V$ is linear, we have $\rho(ST) = \dim S(\operatorname{ran} T) \leq \dim \operatorname{ran} T = \rho(T)$. ∎

Proof. By rank-nullity $\nu(S) = \dim V - \rho(S)$ and $\nu(T) = \dim V - \rho(T)$. Then

$$\nu(S+T) = \dim V - \rho(S+T) \geq \dim V - \rho(S) - \rho(T) = \nu(S) + \nu(T) - \dim V.$$

∎

Problem 4

Let V be a finite-dimensional vector space, $T \in \mathcal{L}(V)$. Prove:

(i) If T and T^2 have the same rank, then $V = (\ker T) \oplus T(V)$ in the sense of §1.6, Exercise 12. (Hint: T and T^2 also have the same nullity; use $\ker T \subseteq \ker T^2$ in the light of Corollary 3.5.14.)

(ii) Prove that if k is a nonnegative integer such that T^k and T^{k+1} have the same rank, then $V = (\ker T^k) \oplus T^k(V)$.

(iii) Review §2.3, Exercise 13.

Proof. Suppose T and T^2 have the same rank. ∎

Proof. Suppose k is a nonnegative integer such that T^k and T^{k+1} have the same rank. ∎

Solution: OK.

Problem 5

Let V be a vector space of infinite sequences of real numbers. Define linear mappings $T : V \longrightarrow V$ and $S : V \longrightarrow V$ as follows: if $x = (a_1, a_2, a_3, ...) \in V$, then $Tx = (0, a_1, a_2, a_3, ...)$ and $Sx = (a_2, a_3, a_4, ...) \in V$. Show that $ST = I$ but $TS \neq I$. (T is called the shift operator on V.)

Proof. Suppose $x = (a_1, a_2, a_3, ...) \in V$. Then

$$(ST)x = S(0, a_1, a_2, a_3, ...) = (a_1, a_2, a_3, ...) = Ix,$$

so $ST = I$. To see that $TS \neq I$, note that

$$(TS)x = T(a_2, a_3, a_4, ...) = (0, a_2, a_3, a_4, ...) \neq (a_1, a_2, a_3, ...) = Ix.$$

∎

Problem 6

Let $\mathcal{D}$ be the vector space of all real polynomial functions (Example 1.3.5). As in Examples 2.1.4 and 2.1.5, let $S, T \in \mathcal{L}(V)$ be the linear mappings such that $Tp = p'$ and Sp is the prederivative of p with constant term 0. Then $TS = I$. Is ST bijective?

Proof. ST is not bijective. Consider any nonzero constant polynomial $f = c_0 \neq 0$. Then $T(f) = f' = 0$, so $ST(f) = S(0) = 0$. Thus $f \neq 0$ but $ST(f) = 0$, so ST is not injective and thus not bijective. ∎

Problem 7

Let V be any vector space (not necessarily finite-dimensional) and suppose $R, S, T \in \mathcal{L}(V)$ are such that $ST = I$ and $TR = I$. Prove that T is bijective and $R = S = T^{-1}$. (Hint: Look at $S(TR)$.)

Proof. We have

$$TR = I \implies S(TR) = S \implies (ST)R = S \implies IR = S \implies R = S.$$

Since $R = S$ and $TR = I$, we have $TS = I$. Combined with $ST = I$, we have $ST = TS = I$, so T is bijective and $R = S = T^{-1}$. ∎

Problem 8

With V as in Exercise 5, define $M(a_1, a_2, a_3, ...) = (a_1, 2a_2, 3a_3, ...)$. Show that M is bijective and find the formula for M^{-1}.

Proof. Consider the mapping $T \in \mathcal{L}(V)$ defined

$$T(a_1, a_2, a_3, ...) = (a_1, a_2/2, a_3/3, ...).$$

Consider $(a_1, a_2, a_3, ...) \in V$ then

$$MT(a_1, a_2, a_3, ...) = M(a_1, a_2/2, a_3/3, ...) = (a_1, a_2, a_3, ...)$$

also

$$TM(a_1, a_2, a_3, ...) = T(a_1, 2a_2, 3a_3, ...) = (a_1, a_2, a_3, ...).$$

Thus $MT = TM = I$, so $T = M^{-1}$ and M is bijective. ∎

Problem 9

Let V be a finite-dimensional vector space and let $S, T \in \mathcal{L}(V)$. Prove:

(i) If there exists a nonzero vector x such that $STx = \theta$, then there exists a nonzero vector y such that $TSy = \theta$.

(ii) ST is bijective $\Leftrightarrow$ S and T are bijective $\Leftrightarrow$ TS is bijective.

(Hint: Exercise 3.)

Proof. Suppose there exists a nonzero vector x such that $STx = \theta$. There are two cases. If $Tx = \theta$, then $x \neq \theta$ implies $\nu(T) \geq 1$, and by Exercise 3, $\nu(TS) \geq \nu(T) \geq 1$. If $Tx \neq \theta$, then since $S(Tx) = \theta$ with $Tx \neq \theta$, we have $\nu(S) \geq 1$, and by Exercise 3, $\nu(TS) \geq \nu(S) \geq 1$. ∎

Proof. Suppose ST is bijective. If T is not injective, there exist $x_1 \neq x_2$ with $T(x_1) = T(x_2)$, so $ST(x_1) = ST(x_2)$, contradicting injectivity of ST. If S is not surjective, there exists $y \in V$ with $y \notin$ ran S, so $y \notin$ ran(ST), contradicting surjectivity of ST. Thus T is injective and S is surjective. Since V is finite-dimensional, both S and T are bijective.

Conversely, suppose S and T are bijective. Then $(ST)^{-1} = T^{-1}S^{-1}$, so ST is bijective.

Thus TS is bijective $\Leftrightarrow$ T and S are bijective $\Leftrightarrow$ ST is bijective. ∎

3.8 Dimensions of $\mathcal{L}(V, W)$

Exercise 1

Hiding in the proof of Corollary 3.8.3 is the proof that if V is 1-dimensional then every $S \in \mathcal{L}(V)$ is a scalar multiple of the identity. Can you find it?

Solution: When dim $V = 1$, every two nonzero vectors are linearly dependent, so the independence of x and Sx assumed for contradiction in the proof doesn't occur. Thus we conclude that $Sx = cx$ for all $x \in V$ with $ST = TS$ not needed.

Exercise 2

True or false (explain): $\mathcal{L}(\mathbb{R}^2, \mathbb{R}^3) \cong \mathbb{R}^6$.

Proof. True: The dimensions are equal. Note that dim $\mathcal{L}(\mathbb{R}^2, \mathbb{R}^3) = 2 \cdot 3 = 6 =$ dim $\mathbb{R}^6$. We know there is a linear mapping sending the basis of $\mathcal{L}(\mathbb{R}^2, \mathbb{R}^3)$ to the basis of $\mathbb{R}^6$. Thus they are isomorphic. ∎

Exercise 3

If V and W are finite-dimensional vector spaces, prove that $\mathcal{L}(V, W) \cong \mathcal{L}(W, V)$. (Hint: Don't look for a mapping $\mathcal{L}(V, W) \to \mathcal{L}(W, V)$.)

Proof. The dimensions are equal. Note that $\dim \mathcal{L}(V, W) = (\dim V)(\dim W) = (\dim W)(\dim V) = \dim \mathcal{L}(W, V)$. We know there is a linear mapping sending the basis of $\mathcal{L}(V, W)$ to the basis of $\mathcal{L}(W, V)$. Thus they are isomorphic. ∎

Exercise 4

Let V be a finite-dimensional vector space, V' its dual space (§2.3, Exercise 5). Prove: V' is also finite-dimensional and $\dim V' = \dim V$. (Hint: Corollary 3.8.2.)

Proof. Let $x_1, \ldots, x_n$ be a basis of V. Consider the mapping $T : V' \longrightarrow F^n$ defined by

$$T(f) = (f(x_1), \ldots, f(x_n)).$$

T is an isomorphism, so $V' \cong F^n$. Thus V' is finite-dimensional and $\dim V' = n = \dim V$. ∎

Exercise 5

Here is an alternate approach to Exercise 4: If F is the field of scalars and $n = \dim V$, prove that $V' \cong F^n$. (Hint: If $x_1, \ldots, x_n$ is a basis of V, then the mapping $V' \to F^n$ defined by $f \mapsto (f(x_1), \ldots, f(x_n))$ is surjective by Theorem 3.8.1.)

Proof. Suppose $x_1, \ldots, x_n$ is a basis of V. Consider the mapping $T : V' \longrightarrow F^n$ defined by

$$T(f) = (f(x_1), \ldots, f(x_n)).$$

T is surjective by Theorem 3.8.1. Since $\dim V' = \dim V = n = \dim F^n$ and T is surjective, T is an isomorphism by Theorem 3.7.4. Thus $V' \cong F^n$. ∎

Exercise 6

Here is a sketch of an alternate proof of Corollary 3.8.2, based on constructing an explicit basis for $\mathcal{L}(V, W)$. Let $x_1, \ldots, x_n$ be a basis of V and $y_1, \ldots, y_m$ a basis of W. For each $i \in \{1, \ldots, m\}$ and each $j \in \{1, \ldots, n\}$, there exists (by Theorem 3.8.1) a linear mapping $E_{ij} : V \to W$ such that

$$E_{ij}x_j = y_i \quad \text{and} \quad E_{ij}x_k = \theta \quad \text{for } k \neq j.$$

Show that the mn linear mappings constructed in this way are a basis of $\mathcal{L}(V, W)$. (Verify that the E_{ij} are linearly independent and that every $T \in \mathcal{L}(V, W)$ is a linear combination of them.)

Proof. Let $x_1, \ldots, x_n$ be a basis of V and $y_1, \ldots, y_m$ a basis of W. For each $i \in \{1, \ldots, m\}$ and each $j \in \{1, \ldots, n\}$, there exists (by Theorem 3.8.1) a linear mapping $E_{ij} : V \to W$ such that

$$E_{ij}x_j = y_i \quad \text{and} \quad E_{ij}x_k = \theta \quad \text{for } k \neq j.$$

We first show linear independence. Suppose, for contradiction, there exist scalars c_{ij}, not all zero, such that

$$\sum_{i,j} c_{ij}E_{ij} = \sum_{i=1}^{m}\sum_{j=1}^{n} c_{ij}E_{ij} = 0.$$

Let x_l be a basis element of V. Applying this transformation to x_l gives

$$\sum_{i=1}^{m}\sum_{j=1}^{n} c_{ij}E_{ij}(x_l) = \theta.$$

Since $E_{ij}(x_l) = \theta$ for $j \neq l$, all terms with $j \neq l$ vanish, so we can simplify to

$$\sum_{i=1}^{m} c_{il}E_{il}(x_l) = \sum_{i=1}^{m} c_{il}\, y_i = \theta.$$

Since $y_1, \ldots, y_m$ is a basis of W, they are linearly independent, so $c_{il} = 0$ for all i. Since l was arbitrary, $c_{ij} = 0$ for all i, j, a contradiction.

We now show generation. Let $T \in \mathcal{L}(V, W)$. Now, T is completely determined by where it sends $x_1, \ldots, x_n$. For each basis element x_j, since $Tx_j \in W$, we can write

$$Tx_j = \sum_{i=1}^{m} a_{ij}\, y_i$$

for some scalars a_{ij}. Consider the map $\sum_{i,j} a_{ij}E_{ij}$. Applying it to any basis element x_l gives

$$\sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}E_{ij}(x_l) = \sum_{i=1}^{m} a_{il}\, y_i = Tx_l.$$

Since T and $\sum_{i,j} a_{ij}E_{ij}$ agree on every basis element of V they are equal. ∎

Exercise 7

Prove: If V is a finite-dimensional vector space and $T \in \mathcal{L}(V)$, then there exist scalars $a_0, a_1, \ldots, a_t$, not all 0, such that

$$a_0t + a_1Tx + a_2T^2x + \ldots + a_tT^tx = 0$$

for all $x \in V$. (Hint: The powers of T are defined following 2.3.13. Consider $I, T, T^2, T^3, \ldots$ in the finite-dimensional vector space $\mathcal{L}(V)$.)

Proof. Suppose V is a finite-dimensional vector space and $T \in \mathcal{L}(V)$. Since V is finite-dimensional $\mathcal{L}(V)$ is also finite-dimensional. Thus the infinite sequence $I, T, T^2, T^3, \ldots$ in $\mathcal{L}(V)$ must be linearly dependent, so there exist scalars $a_0, a_1, \ldots, a_t$, not all zero, such that

$$a_0 I + a_1 T + a_2 T^2 + \ldots + a_t T^t = 0.$$

$\blacksquare$

Exercise 8

Let $x_1, \dots, x_n$ be a basis of V and, for each pair of indices $i, j \in \{1, \dots, n\}$, let $E_{ij} \in \mathcal{L}(V)$ be the linear mapping such that $E_{ij}x_j = x_i$ and $E_{ij}x_k = 0$ when $j \neq k$ (cf. Exercise 6). Show that (1) $E_{ij}E_{hk} = \delta_{jh}E_{ik}$, (2) $E_{ij}E_{jk} = E_{ik}$, and (3) $E_{11} + E_{22} + \dots + E_{nn} = I$.

Exercise 9

Let $T : V \to W$ be a linear mapping such that W is finite-dimensional with $\dim W = m$, and let $y_1, \dots, y_m$ be a basis of W. For each $x \in V$, Tx is uniquely a linear combination of $y_1, \dots, y_m$; denote the coefficient of y_j by $f_j(x)$, thus

$$Tx = \sum_{i=1}^{m} f_i(x)y_i \quad (x \in V).$$

Prove: The f_i are linear forms on V and, in the notation of Example 2.3.4, $T = \sum_{i=1}^{m} f_i \otimes y_i$.

Exercise 10

If V and W are finite-dimensional vector spaces such that $\dim \mathcal{L}(V, W) = 7$, then $\mathcal{L}(V, W)$ is isomorphic to either V or W.

Exercise 11

If V and W are finite-dimensional vector spaces over the same field, then there exists a natural isomorphism

$$\mathcal{L}(U, W) \times \mathcal{L}(V, W) \cong \mathcal{L}(U \times V, W).$$

(Hint: If $R : U \to W$ and $S : V \to W$ are linear mappings, consider the mapping $(x, y) \mapsto Rx + Sy$.)

Exercise 12

If V is a finite-dimensional vector space and x is a nonzero vector in V, then there exists a linear form f on V such that $f(x) \neq 0$. (Hint: Expand x to a basis of V (3.5.15) and apply Theorem 3.8.1. Alternate solution: If $\dim V = n$, compose an isomorphism $T : V \to F^n$ with an appropriate coordinate function $F^n \to F$.)

Exercise 13

Let V be a finite-dimensional vector space, M a linear subspace of V, and x a vector in V such that $x \notin M$. Prove that there exists a linear form f on V such that $f(x) \neq 0$ and $f(y) = 0$ for all $y \in M$. (Hint: Adjoin x to a basis of M, expand to a basis of V (3.5.15), then apply Theorem 3.8.1. Alternate solution: apply Exercise 12 in the quotient space V/M and compose the linear form so obtained with the quotient mapping $V \to V/M$.)

3.9 Duality in Vector Spaces

Exercise 1

Simplify the proof of Theorem 3.9.6 by citing §2.3, Exercise 2.

Exercise 2

If V is a finite-dimensional vector space, M is a linear subspace of V, and $Q : V \to V/M$ is the quotient mapping, then $Q' : (V/M)' \to V'$ is an injective linear mapping with range M°. In particular, $(V/M)' \cong M^\circ$. (Hint: By Corollary 3.9.13, Q' is injective. Show that $\operatorname{Im} Q'$ and M° have the same dimension (cf.

Theorems 3.9.12, 3.9.11) and that Im $Q' \subset M^\circ$, therefore Im $Q' = M^\circ$.}

Exercise 3

If V and W are finite-dimensional vector spaces and $T : V \to W$ is a linear mapping, prove that T and T' have the same nullity if and only if dim V = dim W.

Exercise 4

If V and W are finite-dimensional vector spaces, then $(V \times W)' \cong V' \times W'$ on grounds of dimensionality (Theorems 3.9.2 and 3.6.4). There is also a 'natural' isomorphism $T : V' \times W' \to (V \times W)'$, defined as follows: if $f \in V'$ and $g \in W'$, let $T(f, g)$ be the linear form $(x, y) \mapsto f(x) + g(y)$ on $V \times W$. {Shortcut: §3.8, Exercise 11.}

Exercise 5

Let $V = \mathcal{P}$ be the vector space of real polynomial functions (1.3.5). Every nonnegative integer k defines a linear form φ_k on V by the formula $\varphi_k(p) = p^{(k)}(0)$, where $p^{(0)} = p$, $p^{(1)} = p'$, $p^{(2)} = p''$, etc. If $D : V \to V$ is the differentiation mapping $Dp = p'$ (2.1.4), show that $D'\varphi_k = \varphi_{k+1}$. {Hint: $\varphi_k(p) = (D^k p)(0)$.}

Exercise 6

Let V be a vector space of dimension n and let $f_1, \dots, f_n$ be a basis of V' (Theorem 3.9.2). Prove that the mapping $x \mapsto (f_1(x), \dots, f_n(x))$ is a vector space isomorphism $V \to F^n$. {Hint: §3.8, Exercise 12.}

Exercise 7

If V and W are finite-dimensional vector spaces, $T : V \to W$ is a linear mapping, and $T' : W' \to V'$ is the transpose of T, then T is bijective if and only if T' is bijective. {Hint: Corollary 3.9.13.}

Exercise 8

It is an exercise in concentration to calculate the transpose of a linear form $f : V \to F$.

(i) For each $a \in F$, the formula $a'(c) = ac$ defines a linear form a' on F, and $a \mapsto a'$ is an isomorphism $F \to F'$.

(ii) $a'f$ (the composite of a' and f) is equal to af (the scalar multiple of f by the scalar a).

(iii) $f'a' = af$ for all $a \in F$.

Exercise 9

Let $x_1, \dots, x_n$ be a basis of V and let $f_1, \dots, f_n$ be the basis of V' dual to $x_1, \dots, x_n$. Show that $x_1'', \dots, x_n''$ is the basis of V'' dual to $f_1, \dots, f_n$. {See Definition 3.9.15 for the notations.}

Exercise 10

Let V and W be finite-dimensional vector spaces. Show that the mapping $\mathcal{L}(V, W) \to \mathcal{L}(W', V')$ defined by $T \mapsto T'$ is linear and bijective.

Exercise 11

If V and W are finite-dimensional, then the mapping $\mathcal{L}(V, W) \to \mathcal{L}(V'', W'')$ defined by $T \mapsto T''$ is linear and bijective.

Exercise 12

Let V be a finite-dimensional vector space and let $f_1, \dots, f_n$ be a basis of V' (3.9.2). Prove that there exists a basis $x_1, \dots, x_n$ of V such that $f_1, \dots, f_n$ is the basis of V' dual to $x_1, \dots, x_n$. {Hint: Apply 3.9.3 with V and V' replaced by V' and V'', then cite 3.9.17. Alternate proof: cf. Exercise 6 and the proof of 3.9.3.}

Exercise 13

Let V be a finite-dimensional vector space, M a linear subspace of V, M° the annihilator of M in V' (3.9.10). Prove:

(i) $M^\circ = \{0\} \Leftrightarrow M = V$.

(ii) $M^\circ = V' \Leftrightarrow M = \{\theta\}$.

Exercise 14

Let V be a finite-dimensional vector space, V' its dual, N a linear subspace of V'. Define

$$^\circ N = \{x \in V : f(x) = 0 \text{ for all } f \in N\},$$

that is, $^\circ N$ is the intersection of the kernels of all the $f \in N$. {This is a sort of 'backward annihilator', analogous to the 'forward annihilator' M° of Definition 3.9.10; the effect on subspaces is that $M \mapsto M^\circ$ moves us forward in the list $V, V', V'', V''', \dots$ and $N \mapsto {}^\circ N$ moves us backward.} Prove that $^\circ(M^\circ) = M$ for every linear subspace M of V. {Hint: The inclusion $M \subset {}^\circ(M^\circ)$ is trivial; for the reverse inclusion, see §3.8, Exercise 13. Alternate proof: With $S : V \to V''$ as in 3.9.15, let $(M^\circ)^\circ$ be the annihilator of M° in V''; show that $S(M) \subset (M^\circ)^\circ$ then argue that $S(M) = (M^\circ)^\circ$ on the grounds of dimensionality (3.9.11), and finally observe that $^\circ(M^\circ) = S^{-1}((M^\circ)^\circ)$.}

Exercise 15

Let V be a finite-dimensional vector space, M and N linear subspaces of V, and M°, N° their annihilators in V' (3.9.10). Prove:

(i) If $M^\circ = N^\circ$ then $M = N$. {Hint: $S(M) = S(N)$ by the hint for Exercise 14.}

(ii) $(M + N)^\circ = M^\circ \cap N^\circ$.

(iii) $(M \cap N)^\circ = M^\circ + N^\circ$. {Hint: $S(M \cap N) = S(M) \cap S(N)$, that is, $((M \cap N)^\circ)^\circ = (M^\circ)^\circ \cap (N^\circ)^\circ$; cite (ii) to conclude that $((M \cap N)^\circ)^\circ = (M^\circ + N^\circ)^\circ$, then cite (i).}

Exercise 16

Let V be a finite-dimensional vector space, N a linear subspace of V'. The 'backward annihilator' $^\circ N$ of N is a linear subspace of V (Exercise 14); the 'forward annihilator' N° of N is a linear subspace of V'' (3.9.10). Prove:

(i) $S(^\circ N) = N^\circ$.

(ii) $(^\circ N)^\circ = N$. {Hint: If $M = {}^\circ N$ then $N^\circ = S(M) = (M^\circ)^\circ$; cite (i) of Exercise 15.}

Exercise 17

Let V be a finite-dimensional vector space and let f and $f_1, \dots, f_n$ be linear forms on V. The following conditions are equivalent:

(a) f is a linear combination of $f_1, \dots, f_n$;

(b) f vanishes on the intersection of the null spaces of the f_i, that is, if x is a vector in V such that $f_i(x) = 0$ for $i = 1, \dots, n$ then $f(x) = 0$.

{Hint: Let N be the linear span of $f_1, \dots, f_n$ and look at Exercise 16.}

Exercise 18

Let $T : V \to W$ be a linear mapping, M a linear subspace of V, N a linear subspace of W. Prove:

(i) If $T(M) \subset N$ then $T'(N^\circ) \subset M^\circ$.

(ii) $T'(W') \subset (\mathbf{ker}\, T)^\circ$. {Hint: In (i), consider $M = \mathbf{ker}\, T$, $N = \{\theta\}$.}

(iii) If V and W are finite-dimensional, then $T'(W') = (\mathbf{ker}\, T)^\circ$. {Hint: In (ii), calculate dimensions.}

Exercise 19

Let V be a finite-dimensional vector space, M a linear subspace of V, and $T : M \to V$ the insertion mapping $Tx = x$ ($x \in M$). Show that $T' : V' \to M'$ is the mapping R in the proof of Theorem 3.9.11.

Exercise 20

If V is a vector space and $f_1, \dots, f_n$ are linearly independent linear forms on V such that

$$(\mathbf{ker}\, f_1) \cap \dots \cap (\mathbf{ker}\, f_n) = \{\theta\}$$

then V is n-dimensional. {Hint: Cite §3.6, Exercise 9, then argue that dim $V \geq n$. Shortcut: Exercise 17.}

Exercise 21

Let V and W be finite-dimensional vector spaces, $T : V \to W$ a linear mapping. If W is 12-dimensional and the kernel of T' is 4-dimensional, find the rank of T.